

Debiased Machine Learning U-Statistics*

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Abstract

We propose a method for debiasing estimators based on U-statistics involving Machine Learning (ML) first steps. Novel applications include inference on Inequality of Opportunity (IOp) based on the Gini of ML predictions of income, inference on gravity models in international trade, and inference on algorithmic fairness. In those applications, standard and commonly used plug-in methods lead to large regularization and model selection biases, as well as invalid inferences. Those biases are greatly reduced by our debiasing procedures. Our methods deliver new, robust and simple estimators with provably asymptotics and good performance in finite samples. In our empirical application, we report the first unbiased measures of income IOp in Europe. Plug-in estimators of IOp are very sensitive to the ML algorithm, for some countries even up to 60 percentage points apart, while debiased IOp estimators are robust.

JEL Classification: C13; C14; C21; D31; D63

Keywords: local robustness, orthogonal moments, machine learning, U-statistics, Inequality of Opportunity, gravity models, fairness.

R package: <https://joelters.github.io/home/code/>

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1 Introduction

Many parameters of interest in economics can be estimated as functionals of U-statistics which depend on a first step estimation of possibly infinite-dimensional nuisance parameters (henceforth referred to as Machine Learning (ML) first steps). We call estimators based on these two-step U-statistics ML U-estimators (in short, MLU-estimators). There is a recent and growing empirical literature employing ML methods to estimate nuisance parameters. In this paper, we cover a non-exhaustive list of example applications of MLU-estimators, including inequality of opportunity, dyadic regressions, semiparametric gravity models, and algorithmic fairness through Integral Probability Metrics. In these and other applications, standard plug-in MLU-estimators have large regularization and model selection biases. We propose novel Debiased MLU-estimators (in short, DMLU-estimators) for these and other applications and the first valid inferences in these settings.

The main econometric problem that ties all these examples together relates to two-step estimation in a U-statistic setting. In all cases, an infinite-dimensional object is estimated with ML or other nonparametric procedures. Then, the first step is used to estimate a finite-dimensional parameter of interest which is identified as the solution to a U-statistic sample equation. The first step estimation typically involves a bias-variance trade-off which is optimized using regularization and/or model selection. Bias will be introduced in the first-step estimation if it entails a variance reduction that improves the prediction. This bias can creep into the second step, biasing the estimation of the parameter of interest. Furthermore, this bias is on many occasions not asymptotically negligible which in turn invalidates inference and computation of standard errors for the parameter of interest.

We give a general construction of alternative DMLU-estimators that are robust to these biases. As in [Chernozhukov et al. \(2018\)](#), [Chernozhukov et al. \(2022\)](#) and the literature therein, we call these alternative estimators debiased or locally robust estimators. The latter term comes from the fact that DMLU-estimators are in the limit locally insensitive to variation of the nuisance parameters around the truth. The key difference with previous literature is that we construct debiased estimators in a U-statistic setting.

As we show in the paper, our U-statistic setting leads to nontrivial methodological and conceptual problems relative to the existing results. The standard approach in the literature to handle U-statistics reduces the problem back to sample means (through the so-called Hajek projection), but this approach leads to additional nuisance parameters and complicated expressions for debiasing in our applications. We take an alternative route based directly on U-statistics, which provides simple estimators in our applications.

While the theoretical framework provides a robust foundation, its application to real-world economic problems highlights its practical relevance and potential for impact. For this reason, we provide an extensive empirical application to Inequality of Opportunity (IOp)

together with an R package that implements the debiased estimators and the inferential methods. Additionally, we also provide applications to dyadic regressions, gravity models, and to algorithmic fairness.

This work was firstly motivated by the example of IOp, and as such, we use this as our running and leading example. IOp refers to unfair inequality that stems from circumstances outside the control of the individual, e.g. parental income or race. Not only is IOp considered unfair, but it might also hinder economic growth. While inequalities based on different returns to different levels of effort can spur innovation and growth, inequalities based on circumstances might entail an important waste of talents and capabilities. If the determinants of success are circumstances, we lose the contribution of an important share of society.¹ Moreover, perceptions about IOp shape redistributive preferences and hence the design of social policy (e.g. [Alesina and La Ferrara \(2005\)](#)).

The well-grounded theoretical and normative framework developed in the seminal contributions by [Van De Gaer \(1993\)](#), [Fleurbaey \(1995\)](#) and [Roemer \(1998\)](#) has fueled a growing empirical literature quantifying IOp, see the reviews in [Roemer and Trannoy \(2016\)](#), [Ramos and Van de Gaer \(2016\)](#) or [Ferreira and Peragine \(2016\)](#). However, the statistical properties of current empirical approaches within this field remain by and large elusive. This problem becomes more acute with the recent use of black-box ML algorithms, which has become very popular.²

The leading measure of IOp is the Gini coefficient of fitted values or ML predictions. By predicting outcomes with circumstances and then measuring the inequality in the distribution of predictions we can quantify IOp. For example, suppose that the income of any individual is perfectly predicted by her circumstances. Then, all income inequality would be due to circumstances. Fortunately, this extreme case of unfair inequality is not empirically common and measures of variability of the predictions do not typically explain all the variability in income. From the econometrics perspective, the two-step nature of measures of IOp creates important challenges for the development of inferential methods, as we explain below. In this paper, we address these challenges, pointing out problems in the existing procedures and providing solutions to these problems.

First, we show that ML predictions lead to sizable regularization and model selection biases in the measurement of IOp. In simulations, the bias is so large that makes standard inference procedures invalid for commonly observed sample sizes. We then propose a novel and extremely simple debiased IOp estimator. The new estimator leads to valid and robust inference for a great variety of ML first steps. The debiased estimator is very simple to

¹See e.g. [Marrero and Rodríguez \(2013\)](#), [Ferreira et al. \(2018\)](#), [Aiyar and Ebeke \(2020\)](#) or [Carranza \(2020\)](#)

²See e.g. [Brunori et al. \(2021\)](#), [Brunori et al. \(2019a\)](#), [Brunori and Neidhöfer \(2021\)](#), [Salas-Rojo and Rodríguez \(2022\)](#), [Carranza \(2022\)](#) or [Hufe et al. \(2022c\)](#)

compute, with no additional nuisance parameters needed to obtain our estimator. Independently of our paper, [Heuchenne and Jacquemain \(2022\)](#) provide a bootstrap-based inference for IOp in the special case of a single index model with a strictly increasing link function and arrive to an estimator similar to ours. Our estimator can be seen as a nonparametric version of their Lorenz regression allowing for ML first steps.

A more popular approach in economics is to measure Intergenerational Mobility as opposed to Inequality of Opportunity. This literature has traditionally focused on simpler statistical measures such as the slope coefficient in an OLS regression of logged children’s income on logged parental income. This approach has the advantage of the availability of simple inference tools but it focuses on only one circumstance. In contrast, IOp has a stronger normative foundation, but it lacks inferential guarantees. This paper provides a step-change to address the statistical shortcomings in the IOp literature. Our results are firmly anchored in the semiparametric literature and provide the first valid and robust inferential methods for IOp.

We also provide an application to dyadic data motivated by gravity models in economics. Dyadic regression is a popular tool in econometrics with its origin in the influential report by [Tinbergen \(1962\)](#). For a thorough review and references of its application in international trade, war among nation-states, risk sharing across households, supply side linkages across firms, R&D, migration or network formation see [Graham \(2020b\)](#). CONTINUE cite [Chiang et al. \(2021a\)](#) and [Chiang et al. \(2021b\)](#).

We also discuss an application to fairness in algorithmic decision-making. ADD REFERENCES. Fairness (or lack thereof) is typically measured as deviations from (conditional) independence between a ML prediction and an attribute of interest. The extensive literature on fairness so far has overlooked the fact that fairness measures are two-step measures with a ML first step. Inspired by [Rychener et al. \(2022\)](#), we propose to measure unfairness by Integral Probability Metrics (IPMs) which take the form of MLU-estimators, and we provide, to the best of our knowledge, the first valid inferences on measures of algorithmic fairness.

Other examples of economic applications involving MLU-estimators include the MULTIWAY CLUSTERING, analysis of variance in high dimensions ([Lou et al. \(2023\)](#)), measures of economic polarization ([Duclos et al. \(2004\)](#)), optimal risk in the bipartite ranking problem ([Cléménçon et al. \(2008\)](#) or [Werner \(2021\)](#)), distance-based estimators in semiparametric conditional moment restrictions ([Domínguez and Lobato \(2004\)](#)), Causal U-statistics in ([Mao \(2018\)](#), [Yin et al. \(2024\)](#) and [Terschuur \(2023\)](#)), or semiparametric estimators of binary models with endogeneity ([Blundell and Powell \(2004\)](#)).

We are not the first to deal with U-estimators with nonparametric or high-dimensional first steps. Estimators of this sort with classical nonparametric first steps have been treated in [Powell \(1987\)](#), [Sherman \(1994\)](#) and [Honoré and Powell \(2005\)](#) and asymptotic theory has

been derived using uniform versions of the Hoeffding decomposition under Donsker conditions (see e.g. [Nolan and Pollard \(1988\)](#) or [Arcones and Giné \(1993\)](#)). Unfortunately, Donsker conditions do not hold or are not known to hold in high dimensional settings. REFERENCE HERE. To avoid Donsker conditions we use cross-fitting, as in [Chernozhukov et al. \(2018\)](#), but in a U-statistic setting. To our knowledge, we are the first to propose DMLU-estimators and valid inferences with high-dimensional first steps in U-statistic estimating equations.

The empirical application of this paper is to use the first debiased IOP estimations of income IOP in 29 European countries using the 2019 cross-sectional European Union Statistics on Income and Living Conditions (EU-SILC) survey. This survey is one of the main references for the analysis of the income distribution and poverty in Europe. In the years 2005, 2011 and 2019 it includes a module on intergenerational transmission of disadvantages with information on circumstances. We restrict our attention to the year 2019 since it is the most recent one and contains the richest set of circumstances. The 2019 wave has been used in [Carranza \(2022\)](#) and in [Terschuur \(2022\)](#) to measure income IOP in Europe with Mean Logarithmic Deviation IOP measures. [Hufe et al. \(2022b\)](#) also uses the 2019 wave to asses multidimensional IOP using gender, parental education and parental occupation as circumstances. [Brunori et al. \(2019b\)](#) and [Brunori et al. \(2021\)](#) use cross-validation for model selection in the first step and Conditional Inference Forest (CIF), a ML method, to measure IOP for EU-SILC 2011. [Marrero and Rodríguez \(2012\)](#) use EU-SILC 2005 to estimate IOP in Europe. Our paper is the first one to estimate IOP in EU-SILC 2019 using a variety of machine learners and debiased Gini IOP estimators with theoretically valid inference guarantees.

The countries with the highest IOP are Romania and Bulgaria, where around 60% of income inequality can be explained by circumstances, while Denmark is the country with the lowest IOP. Nordic countries, Germany, the Netherlands and some Eastern countries have low IOP. Southern countries have high IOP with Greece and Italy having 39% and Spain having 43% of total inequality explained by circumstances. We find that the plug-in estimator tends to overestimate IOP in general and particularly so for Denmark. The difference between our debiased estimates and the biased plug-in estimates of relative IOP (i.e. fraction of inequality explained by circumstances), can be as large as 25 percentage points in the case of Denmark. In the rest of the countries the difference is smaller, but we still observe differences of 5-10 percentage points. Hence, plug-in estimators give a biased view of IOP in Europe.

Our debiased IOP estimates are also much less sensitive to the choice of the machine learner than commonly used plug-in measures. We estimate relative IOP with six different machine learners with both the debiased and the plug-in estimator and see that plug-in estimates using different machine learners are much more dispersed than the debiased estimates.

Plug-in estimates for the same country but using different machine learners can be up to 60 percentage points apart. Even differences of 30-40 percentage points are not uncommon. In contrast, debiased estimates are much more concentrated and do not exhibit such sizeable variations across machine learners. While hard to compare with previous empirical results due to the use of plug-in methods and older data, the traditional divide of North and South into high and low IOp remains in our results.

We also explore which circumstances contribute the most towards explaining inequality. We find that parental education and occupation are the most important circumstances explaining unfair inequality in European countries. Our results complement the empirical findings in [Hufe et al. \(2022a\)](#), who also find these circumstances to be the most important drivers of the increasing unfair inequality in the US during the period 1980-2014. However, we are able to observe differences between the impact of maternal and paternal characteristics. While in the case of the mothers it is the education that matters most, in the case of the fathers the occupation is the most important variable.

A crucial improvement of our methodology is the ability to perform inference. This allows to perform statistical tests of IOp between independent populations, across time, or between different sets of circumstances. The ability to do inference is paramount to evaluate changes in IOp before and after a policy has been implemented or across time. Hence, we believe the inferential tools provided here can be of great use for policy design aiming at reducing IOp.

The rest of the paper is organized as follows. [Section 2](#) introduces the concept of orthogonal moments and DMLU-estimators using IOp as a running example. [Section 3](#) explains how to construct orthogonal moments generally. [Section 4](#) derives the asymptotic theory in the general setting and for the IOp example. [Section 5](#) shows the performance of the debiased IOp estimator in Monte Carlo simulations. [Section 6](#) studies IOp in Europe. [Sections 7](#) and [I.4](#) consider applications to dyadic data and algorithmic fairness. Mathematical proofs and complementary results are reported in the Appendix.

2 Methodology

We have independent and identically distributed (i.i.d.) data $W_i = (Y_i, X_i)$, for $i = 1, \dots, n$, distributed with unknown distribution F_0 , an unknown first step function γ_0 and a finite dimensional parameter of interest θ in some set $\Theta \subseteq \mathbb{R}^k$. We can form pairs (W_i, W_j) , where W_j is an independent copy of W_i , with realizations (w_i, w_j) . $\mathbb{E}[\cdot]$ is the expectation under F_0 , $\gamma(F)$ is the probabilistic limit (plim) of a first step estimator $\hat{\gamma}$ under F , like in [Newey \(1994\)](#), and $\gamma_0 = \gamma(F_0)$. We assume there is a vector $g(w_i, w_j, \gamma, \theta)$ of k known identifying

moment functions such that

$$\mathbb{E}[g(W_i, W_j, \gamma_0, \theta)] = 0 \iff \theta = \theta_0 \in \Theta.$$

Without loss of generality, we can assume that g is a symmetric function in w_i and w_j .³ We use the terminology of *quadratic moment* for $\mathbb{E}[g(W_i, W_j, \gamma_0, \theta)]$ since it is a quadratic function of the true distribution F_0 , as by independence between W_i and W_j

$$\mathbb{E}[g(W_i, W_j, \gamma_0, \theta)] = \int \int g(w_i, w_j, \gamma_0, \theta) F_0(dw_i) F_0(dw_j).$$

This feature is what differentiates our analysis from the standard debiased literature (see, e.g., Chernozhukov et al. (2022)), and it leads naturally to U-statistics (see LEE 2019). If we replace F_0 by the empirical distribution of $\{W_i\}_{i=1}^n$ in $\mathbb{E}[g(W_i, W_j, \gamma_0, \theta)]$ we obtain the V-statistic

$$V_n g(\cdot, \gamma_0, \theta) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n g(W_i, W_j, \gamma_0, \theta),$$

and its unbiased U-statistic version

$$U_n g(\cdot, \gamma_0, \theta) = \frac{2}{n(n-1)} \sum_{i < j} g(W_i, W_j, \gamma_0, \theta),$$

where, henceforth, we use the short notation $\sum_{i < j} = \sum_{i=1}^{n-1} \sum_{j=i+1}^n$ and $U_n f = \binom{n}{2}^{-1} \sum_{i < j} f(W_i, W_j)$, and the population analogue $Uf = \mathbb{E}[f(W_i, W_j)]$.⁴ Plugging in a first step estimator $\hat{\gamma}$ for γ_0 leads to the plug-in estimator $\hat{\theta}^P$ solving $U_n g(\cdot, \hat{\gamma}, \hat{\theta}^P) = 0$. We illustrate with our running example.

EXAMPLE (Inequality of Opportunity, IOp): The leading measure of IOp is given by the Gini coefficient of fitted values, which can be expressed as

$$\theta_0 = \frac{\mathbb{E}[|\gamma_0(X_i) - \gamma_0(X_j)|]}{\mathbb{E}[\gamma_0(X_i) + \gamma_0(X_j)]}. \quad (2.1)$$

Here X_i is a vector of circumstances individual i did not choose, such as parental wealth/income, parental education, gender, color of the skin or social origin, but many other variables including high dimensional genetic information might be available. Rearranging (2.1), we obtain

³Otherwise, replace $g(w_i, w_j, \gamma_0, \theta)$ with its symmetrization $g^*(w_i, w_j, \gamma_0, \theta) = (1/2)[g(w_i, w_j, \gamma_0, \theta) + g(w_j, w_i, \gamma_0, \theta)]$.

⁴Note that $\mathbb{E}[V_n g] = \mathbb{E}[g(W_i, W_j, \gamma_0, \theta)] + b_n$, where $b_n = (\mathbb{E}[g(W_i, W_i, \gamma_0, \theta)] - \mathbb{E}[g(W_i, W_j, \gamma_0, \theta)]) / n$ is a bias term. In contrast, by symmetry it follows that $\mathbb{E}[U_n g] = \mathbb{E}[g(W_i, W_j, \gamma_0, \theta)]$. Nevertheless, our debiased estimators address a typically larger bias term than b_n , so the difference between V and U statistics is not central to our results.

an identifying quadratic moment function

$$g(w_i, w_j, \gamma, \theta) = (\gamma(x_i) + \gamma(x_j))\theta - |\gamma(x_i) - \gamma(x_j)|. \quad (2.2)$$

A natural and popular estimator of θ_0 is the Plug-in sample Gini coefficient of fitted values

$$\hat{\theta}^P = \frac{\sum_{i < j} |\hat{\gamma}(X_i) - \hat{\gamma}(X_j)|}{\sum_{i < j} (\hat{\gamma}(X_i) + \hat{\gamma}(X_j))},$$

for a first step $\hat{\gamma}$ such as Conditional Inference Forests (CIF), which is a popular ML method in the IOp literature, see [Brunori et al. \(2021\)](#), [Brunori et al. \(2019a\)](#) or [Brunori and Neidhöfer \(2021\)](#). We show theoretically and empirically that $\hat{\theta}^P$ is highly biased when ML first steps are used. We develop inference methods for θ_0 and related parameters based on debiased estimators, which improve upon plug-in estimators.

As an illustration, see Figure 1 where we simulate the plug-in and the debiased estimators of IOp which we develop below. The Data Generating Process (DGP) is the same as in Section 5. The fitted values are estimated with CIF. The histograms approximate the distributions of the centered estimators and the curves are normal p.d.f.s with the same variance as the estimators but with no bias (i.e. centered at zero). Even for as large sample sizes as $n = 3000$ the plug-in estimator has a large bias compared to the DMLU estimator proposed in this paper. ■

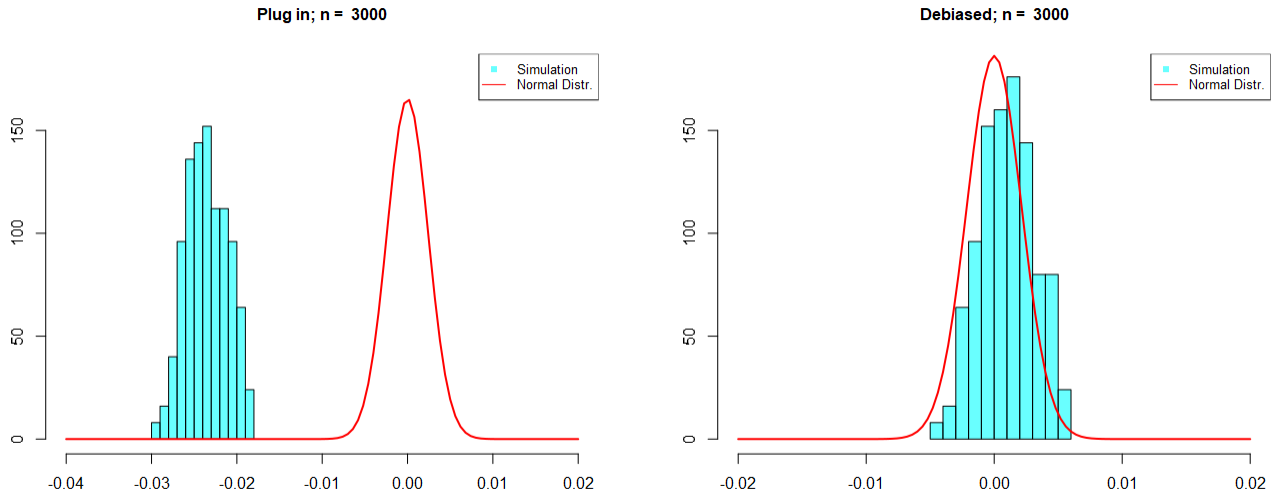


Figure 1: Comparison of plug-in and debiased estimator.

More broadly, plug-in estimators are generally biased by model selection and/or regularization in the first step. Intuitively, ML algorithms trade-off bias and variance. Hence, some

bias might be allowed in the first step if it aids prediction. This first step bias can creep into the second step biasing the parameter estimation. To explain the bias in the context of our running example, define the sign function $sgn(x) = 1(x > 0) - 1(x < 0)$ (equals $-1, 0$ and 1 for negative, zero and positive numbers, respectively) and consider an expansion of the numerator of the Gini of fitted values (the numerator is the most difficult term in an expansion of the bias)

$$\begin{aligned}
\sum_{i < j} |\hat{\gamma}(X_i) - \hat{\gamma}(X_j)| &= \sum_{i < j} (\hat{\gamma}(X_i) - \hat{\gamma}(X_j)) sgn(\hat{\gamma}(X_i) - \hat{\gamma}(X_j)) \\
&= \sum_{i < j} (Y_i - Y_j) sgn(\hat{\gamma}(X_i) - \hat{\gamma}(X_j)) \\
&+ \sum_{i < j} (\hat{\gamma}(X_i) - \gamma_0(X_i) - \hat{\gamma}(X_j) + \gamma_0(X_j)) sgn(\hat{\gamma}(X_i) - \hat{\gamma}(X_j)) \\
&+ \sum_{i < j} (Y_i - \gamma_0(X_i) - Y_j + \gamma_0(X_j)) sgn(\hat{\gamma}(X_i) - \hat{\gamma}(X_j)).
\end{aligned}$$

The first term is the numerator of our DMLU-estimator and leads to a (nonlinear) bias that is of second order under some conditions. The second term is the leading first order bias term. This term is the source of the bias problem. With ML estimators the bias $\mathbb{E}[\hat{\gamma}(X_i) - \gamma_0(X_i)]$ is so large that makes $\sqrt{n}(\hat{\theta}^P - \theta_0)$ to diverge. The third term is a stochastic term that leads to another second order bias.

More generally, we can explain the source of the bias problem through a standard expansion of $U_n g(\cdot, \hat{\gamma}, \hat{\theta}^P)$ around θ_0 ,

$$0 = U_n g(\cdot, \hat{\gamma}, \theta_0) + \frac{\partial U_n g(\cdot, \hat{\gamma}, \bar{\theta}^P)}{\partial \theta'} (\hat{\theta}^P - \theta_0), \quad (2.3)$$

where $\bar{\theta}^P$ is an intermediate point between θ_0 and $\hat{\theta}^P$. From (2.3) and expanding $\hat{\gamma} \rightarrow U_n g(\cdot, \hat{\gamma}, \theta_0)$ around γ_0 , we obtain the expansion

$$\begin{aligned}
\sqrt{n}(\hat{\theta}^P - \theta_0) &\approx - \left[\frac{\partial U_n g(\cdot, \gamma_0, \theta_0)}{\partial \theta'} \right]^{-1} \sqrt{n} U_n g(\cdot, \gamma_0, \theta_0) \\
&- \left[\frac{\partial U_n g(\cdot, \gamma_0, \theta_0)}{\partial \theta'} \right]^{-1} \sqrt{n} \frac{\partial}{\partial \gamma} U g(\cdot, \gamma_0, \theta_0) [\hat{\gamma} - \gamma_0].
\end{aligned}$$

The second term in right hand side quantifies the effect of estimating γ_0 in the MLU-estimator, and leads to a large bias when the $\sqrt{n}\mathbb{E}[\hat{\gamma} - \gamma_0]$ is large and the derivative $\partial U g(\cdot, \gamma_0, \theta_0)/\partial \gamma$ is not zero. We formalize these heuristics and the functional derivative with respect to γ in the next section.

2.1 The MLU-IF

As a first methodological contribution of this paper, we introduce the concept of *MLU-Influence Function* (MLU-IF in short), ϕ , which takes into account the impact that the first step $\hat{\gamma}$ has on the estimation of the parameter of interest θ_0 . We claim the MLU-IF explains the observed bias in Figure 1. To quantify such impact, let F be a cumulative distribution function (cdf) for W_i which is unrestricted except for regularity conditions such as the existence of $\gamma(F)$. For example, for $\gamma(F) = \mathbb{E}_F[Y|X]$ we require $\mathbb{E}_F[|Y|] < \infty$, where $\mathbb{E}_F$ denotes expectation under F . Henceforth, $d/d\tau$ denotes the derivative from the right with respect to τ evaluated at $\tau = 0$ and set $F_\tau = F_0 + \tau(H - F_0)$ for some alternative distribution H and $\tau \in [0, 1]$. The alternative distribution H is chosen such that $\gamma_\tau = \gamma(F_\tau)$ exists for τ small enough. The MLU-IF ϕ is such that for all θ and all H in $F_\tau = F_0 + \tau(H - F_0)$,

$$\mathbb{E}_{F_\tau}[\phi(w_i, w_j, \gamma(F_\tau), \alpha(F_\tau), \theta)] = 0 \text{ for all } \tau \in [0, \bar{\tau}), \bar{\tau} > 0, \quad (2.4)$$

and

$$\frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma(F_\tau), \theta)] = \int \int \phi(w_i, w_j, \gamma_0, \alpha_0, \theta) K_H(dw_i, dw_j), \quad (2.5)$$

with $K_H(dw_i, dw_j) = F_0(dw_i)H(dw_j) + H(dw_i)F_0(dw_j)$. Here, α_0 is an additional nuisance parameter which satisfies (2.5). We may assume without loss of generality that ϕ is symmetric in w_i, w_j .⁵ Equation (2.4) is a zero mean condition on ϕ . The pathwise derivative in satisfies (2.5) formalizes the heuristic derivative $\partial U g(\cdot, \gamma_0, \theta_0)/\partial \gamma$ of the previous section. The MLU-IF ϕ can be found by solving the functional equation (2.5) and it characterizes the local effect of the first step $\gamma(F)$ on the average moment functional $\mu(F) = \mathbb{E}[g(W_i, W_j, \gamma(F), \theta)]$ as F varies away from F_0 in any direction H . In Section 3 we give new results characterizing ϕ for first steps satisfying orthogonality conditions.

EXAMPLE (Inequality of Opportunity, cont.): We focus here on the case in which $\hat{\gamma}$ is a nonparametric estimator of the conditional mean of income Y given circumstances X , i.e. $\gamma(F) = \mathbb{E}_F[Y|X]$. A first technical challenge we face for computing the MLU-IF in this example is the lack of differentiability of the absolute value. Despite this lack of differentiability, we are able to compute the MLU-IF under certain assumptions from an application of our general results in the next section. Define $\Delta_0 = \gamma_0(X_i) - \gamma_0(X_j)$.

Assumption 1 (i) $\partial \gamma(F_\tau)/\partial \tau$ is bounded; either (ii) $\mathbb{P}[\Delta_0 = 0] = 0$ or (iii) $x_i \neq x_j \implies \gamma_0(x_i) - \gamma_0(x_j) \neq 0$.

⁵As with g , setting $\phi^*(w_i, w_j, \gamma_0, \alpha_0, \theta) = (1/2)[\phi(w_i, w_j, \gamma_0, \alpha_0, \theta) + \phi(w_j, w_i, \gamma_0, \alpha_0, \theta)]$, we have $\int \int \phi(w_i, w_j, \gamma_0, \alpha_0, \theta) K_H(dw_i, dw_j) = \int \int \phi^*(w_i, w_j, \gamma_0, \alpha_0, \theta) K_H(dw_i, dw_j)$ and ϕ^* is symmetric.

Assumption 1(i) is a mild regularity assumption. Assumption 1(ii) is satisfied if $\gamma_0(X_i)$ is absolutely continuous, for example, if γ_0 is strictly monotone on a circumstance which is absolutely continuous given all the other circumstances. Assumption 1(iii) says that two observations with different circumstances must have different fitted values. Recall $\text{sgn}(x) = 1(x > 0) - 1(x < 0)$.

Proposition 1 *Under Assumption 1, a MLU-IF in the Gini of fitted values is given by*

$$\phi(w_i, w_j, \gamma_0, \alpha_0, \theta) = \theta(y_i + y_j - \gamma_0(x_i) - \gamma_0(x_j)) - \alpha_0(X_i, X_j)(y_i - y_j - \gamma_0(x_i) + \gamma_0(x_j)), \quad (2.6)$$

where $\alpha_0(x_i, x_j) = \text{sgn}(\gamma_0(x_i) - \gamma_0(x_j))$.

The function ϕ satisfies equations (2.4) and (2.5) for the g of this example, and the fact that $\phi \neq 0$, or more precisely that (2.5) is not zero, explains the biased observed in Figure 1. ■

To motivate our definition of the MLU-IF, note that equation (2.4) is a zero mean condition and it implies by the chain rule

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}[\phi(W_i, W_j, \gamma(F_\tau), \alpha(F_\tau), \theta)] &= -\frac{d}{d\tau} \mathbb{E}_{F_\tau}[\phi(W_i, W_j, \gamma(F_0), \alpha(F_0), \theta)] \\ &= -\int \int \phi(w_i, w_j, \gamma_0, \alpha_0, \theta) K_H(dw_i, dw_j), \end{aligned} \quad (2.7)$$

for all θ and H . Equation (2.7) shows that the effect the first steps have on the MLU-IF “cancels out” with the effect they have on the original identifying moment in (2.5).

Thus, letting $\psi(w_i, w_j, \gamma, \alpha, \theta) = g(w_i, w_j, \gamma, \theta) + \phi(w_i, w_j, \gamma, \alpha, \theta)$, we have an identification condition (by taking $\tau = 0$ in (2.4))

$$\mathbb{E}[\psi(W_i, W_j, \gamma_0, \alpha_0, \theta)] = 0 \text{ iff } \theta = \theta_0,$$

and local robustness with respect to first steps (by (2.5) and (2.7))

$$\frac{d}{d\tau} \mathbb{E}[\psi(W_i, W_j, \gamma(F_\tau), \alpha(F_\tau), \theta)] = 0.$$

This local robustness motivates estimators based on the orthogonal moments ψ . This construction led to the expression of the new IOp estimator, which would not have been possible from the results in the debiased literature. In the general case, α_0 is an additional parameter on which the adjustment term ψ depends. In our leading IOp example this additional

parameter is only a function of γ_0 , and as such, it substantially simplifies the construction of orthogonal quadratic moments and debiased estimators.

EXAMPLE (Inequality of Opportunity, cont.): Adding ϕ of Proposition 1 to the identifying moment g from (2.2) leads after rearranging to the orthogonal quadratic moment function

$$\begin{aligned}\psi(W_i, W_j, \gamma_0, \alpha_0, \theta) &= (\gamma(X_i) + \gamma(X_j))\theta - |\gamma_0(x_i) - \gamma_0(x_j)| + \theta(y_i + y_j - \gamma_0(x_i) - \gamma_0(x_j)) \\ &\quad - \alpha_0(X_i, X_j)(y_i - y_j) + \alpha_0(X_i, X_j)(\gamma_0(x_i) - \gamma_0(x_j)) \\ &= \theta(y_i + y_j) - \alpha_0(X_i, X_j)(y_i - y_j)\end{aligned}$$

where we use that $\alpha_0(X_i, X_j) = \text{sgn}(\gamma_0(X_i) - \gamma_0(X_j))$ and that for $a, b \in \mathbb{R}$, $\text{sgn}(a-b)(a-b) = |a - b|$. In this example, we get a surprisingly simple orthogonal moment function. ■

2.2 Debiased MLU-Estimators

Estimation of nuisance parameters and moment conditions with the same observations can induce an “overfitting” bias. Also, machine learning first steps usually do not satisfy Donsker conditions. We use cross-fitting to overcome these issues (see Bickel (1982), Schick (1986), Klaassen (1987), Chernozhukov et al. (2018) and Chernozhukov et al. (2022)), but adapted to the U-statistics setting. First, we partition the observation indices $\mathcal{N} = \{1, \dots, n\}$ into K sets, $\mathcal{C} = \{C_1, \dots, C_K\}$. Then, $\mathcal{C}^2$ is a partition of all pairs of observations. Since we can focus on U-statistics with symmetric kernels, we use the partition of the set $\{(i, j) \in \mathcal{C}^2 : i < j\}$ into $L = K(K + 1)/2$ groups, $\mathcal{I} = \{I_1, \dots, I_L\}$. An illustration of this partition is given in Appendix K. Given a partition $\mathcal{I}$, let $\hat{\gamma}_l$ and $\hat{\alpha}_l$ be constructed using observations not present in the pairs in I_l . The debiased sample moment is

$$\hat{\psi}(\theta) = \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \left(g(W_i, W_j, \hat{\gamma}_l, \theta) + \phi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta) \right),$$

where $\hat{\alpha}_l$ is some estimator of α_0 . We call an estimator solving $\hat{\psi}(\hat{\theta}) = 0$ a DMLU estimator. We give sufficient conditions in Section 4 so that the asymptotic variance of $\sqrt{n}(\hat{\theta} - \theta_0)$ is $V = B^{-1}\Sigma B'^{-1}$, where

$$B = \frac{\partial \mathbb{E}[\psi(W_i, W_j, \gamma_0, \alpha_0, \theta)]}{\partial \theta}, \quad \Sigma = 4\text{Var}\left(\mathbb{E}[\psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) | W_i]\right).$$

The above asymptotic variance can be estimated as $\hat{V} = \hat{B}^{-1}\hat{\Sigma}\hat{B}'^{-1}$, where

$$\hat{B} = \binom{n}{2}^{-1} \sum_{i < j} \frac{\partial}{\partial \theta} \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \hat{\theta}),$$

$$\hat{\Sigma} = \frac{4}{n(n-1)^2} \sum_{i=1}^n \left[\sum_{j \neq i} \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \hat{\theta}) \right] \left[\sum_{j \neq i} \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \hat{\theta}) \right]'$$

Note that we do not use cross-fitting to estimate the variance, which simplifies the implementation and makes available standard U-statistics formulas for variances valid.

EXAMPLE (Inequality of Opportunity, cont.): Let $\hat{\alpha}_l(X_i, X_j) = \text{sgn}(\hat{\gamma}_l(X_i) - \hat{\gamma}_l(X_j))$, where $\hat{\gamma}_l(x)$ is a cross-fitted ML prediction of income given circumstances x . Solving the debiased orthogonal sample moment we get the DMLU estimator

$$\hat{\theta} = \frac{\sum_{l=1}^L \sum_{(i,j) \in I_l} \hat{\alpha}_l(X_i, X_j)(Y_i - Y_j)}{\sum_{i < j} (Y_i + Y_j)}.$$

The debiased IOP estimator resembles the standard Gini coefficient for income, but rather than weighting by the sign of the differences in income, the debiased IOP estimator weights by the sign of the difference in fitted values. This means that whenever two individuals have the same fitted values their difference in incomes cannot be attributed to inequality of opportunity.

To give the expression for the asymptotic variance, define $d(u, t) = \mathbb{E}[\text{sgn}(u - \gamma_0(X_j))(t - Y_j)]$. Then, in this example

$$\mathbb{E}[\psi(W_i, W_j, \gamma_0, \alpha_0, \theta) | W_i] = \theta(Y_i + \mu) - \mathbb{E}[d(\gamma_0(X_i), Y_i)].$$

We have that the asymptotic variance of the DMLU estimator is given by $V = B^{-2}\Sigma$, where

$$\Sigma = 4\text{Var}\left(\theta_0 Y_i - d(\gamma_0(X_i), Y_i)\right) \text{ and } B = 2\mathbb{E}[Y_i].$$

The asymptotic variance of $\hat{\theta}$ can be estimated by

$$\hat{V} = \frac{1}{n\bar{Y}^2} \sum_{i=1}^n \left(\hat{\theta} Y_i - \hat{d}_i - \hat{\theta} \bar{Y} - \bar{d} \right)^2, \quad (2.8)$$

where $\hat{d}_i = (n-1)^{-1} \sum_{j \neq i} [\text{sgn}(\hat{\gamma}(X_i) - \hat{\gamma}(X_j))(Y_i - Y_j)]$, $\bar{d} = n^{-1} \sum_{i=1}^n \hat{d}_i$, and $\bar{Y} = n^{-1} \sum_{i=1}^n Y_i$. ■

3 General Construction of the MLU-IF

The construction of the orthogonal moment requires to derive a MLU-IF for the given identifying quadratic moment. We have derived in Proposition 1 the MLU-IF corresponding to the identifying moment g in IOp, but we have not explained how this MLU-IF and others could be obtained. This section covers the theory of constructing MLU-IFs for a general class of two-step U-statistics. In Appendix B we sum up the whole process of building a MLU-IF for a simpler example than Inequality of Opportunity and provide two schematic figures which guide the process.

The expression of the MLU-IF ϕ depends on the original identifying moment g and the first step limit $\gamma(F)$. We consider first step functions in a linear set Γ which satisfy a linear orthogonality condition, as this setting is quite general and fits well the applications we are interested in. Let $W_i = (Y_i, X_i) \in \mathcal{Y} \times \mathcal{X}$ and $\gamma(F) \in \Gamma$, be such that

$$\mathbb{E}_F[\nu(X_i)(Y_i - \gamma(F))] = 0 \text{ for all } \nu \in \Gamma. \quad (3.1)$$

This setting covers high dimensional and nonparametric regression, additive regression and single index models, among others (see Ichimura and Newey (2022)). As an illustration, if $\Gamma = L_2(X_i)$ then $\gamma(F)(X_i) = \mathbb{E}_F(Y_i|X_i)$ and if $\Gamma = \{\beta'X_i : \beta \in \mathbb{R}^p\}$ with p the dimension of X_i , then $\gamma(F)(X_i) = \beta_F'X_i$ where $\beta_F = \mathbb{E}_F(X_iX_i'^{-1}\mathbb{E}_F(X_iY_i))$. We assume that $\Gamma \subseteq L_2(X_i)$, where $L_2(X_i)$ is the set of squared integrable functions of X_i (similarly, we define $L_2(X_i, X_j)$). Sometimes we drop the reference to random variables and use simply L_2 (the meaning will be clear from the context). For simplicity of exposition, we take $\gamma(F)$ to be real-valued, though the extension of our results to multiple first steps follows straightforwardly from the chain rule.

The starting point of our analysis is the linearization of the original identifying moment function g with respect to γ . We allow for the possibility that γ enters in g through $\gamma(X_i)$ and $\gamma(X_j)$, as in the IOp example. We assume there exist functions $\delta_m \in L_2$ and constants (c_{1m}, c_{2m}) , $m = 1, \dots, M$ and $M \geq 1$ such that

$$\frac{d}{d\tau}\mathbb{E}[g(W_i, W_j, \gamma(F_\tau), \theta)] = \sum_{m=1}^M \frac{d}{d\tau}\mathbb{E}[\delta_m(X_i, X_j, \gamma_0, \theta)(c_{1m}\gamma_\tau(X_i) + c_{2m}\gamma_\tau(X_j))], \quad (3.2)$$

where $\gamma_\tau(x) = \gamma(F_\tau)(x)$ denotes the orthogonal projection, identified from (3.1), pertaining to $F_\tau = F_0 + \tau(H - F_0)$. For simplicity of notation, we drop the possible dependence of δ_m , c_{1m} and c_{2m} on θ and also the possible dependence of c_{1m} and c_{2m} on γ_0 . Henceforth, we also use the short notation $\delta_{ij,m}(\gamma) \equiv \delta_m(X_i, X_j, \gamma)$ and $\delta_{ij} = (\delta_{ij,1}, \dots, \delta_{ij,M})'$.

EXAMPLE (Inequality of Opportunity, cont.): In the proof of Proposition 1 we show

that

$$\frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma(F_\tau), \theta)] = \frac{d}{d\tau} \mathbb{E}[\theta(\gamma_\tau(X_i) + \gamma_\tau(X_j)) + \text{sgn}(\gamma_0(X_i) - \gamma_0(X_j))(\gamma_\tau(X_i) - \gamma_\tau(X_j))],$$

so that $\delta_{ij,1}(\gamma_0) = \theta$, $c_{11} = c_{21} = 1$, $\delta_{ij,2}(\gamma_0) = \text{sgn}(\gamma_0(X_i) - \gamma_0(X_j))$, and $(c_{12}, c_{22}) = (1, -1)$. $\blacksquare$

After the linearization, we exploit that $c_{1m}\gamma_0(X_i) + c_{2m}\gamma_0(X_j)$ can be viewed as a projection of $c_{1m}Y_i + c_{2m}Y_j$ onto a closed set $\mathcal{S}$, $\mathcal{S} \subseteq L_2(X_i, X_j)$, satisfying some regularity conditions.

Assumption 2 $\mathcal{S}$ is a closed linear set of functions of (X_i, X_j) containing $\Gamma + \Gamma := \{\nu(X_i) + w(X_j) : \nu, w \in \Gamma\}$. Also, Γ is a closed linear set of functions that contains constant functions.

Define $\alpha_{0m}(X_i, X_j) \equiv \Pi_{\mathcal{S}}(\delta_m(X_i, X_j, \gamma_0))$, where δ_m is as in (3.2), and where henceforth $\Pi_V(\cdot)$ is the orthogonal projection operator onto V , for a closed linear set V .

Lemma 1 Suppose (3.1), (3.2) and Assumption 2 hold. If α_{0m} is such that $\alpha_{0m}(\cdot, x) \in \Gamma$ and $\alpha_{0m}(x, \cdot) \in \Gamma$ for all x and $m = 1, \dots, M$, then (2.5) holds with $\alpha_0 = (\alpha_{01}, \dots, \alpha_{0M})'$ and

$$\phi(W_i, W_j, \gamma_0, \alpha_0, \theta) = \sum_{m=1}^M \alpha_{0m}(X_i, X_j) (c_{1m}Y_i + c_{2m}Y_j - c_{1m}\gamma_0(X_i) - c_{2m}\gamma_0(X_j)). \quad (3.3)$$

Note that (2.4) is also satisfied.

This lemma gives some flexibility in the choice of $\mathcal{S}$, as long as it satisfies Assumption 2. Thus, we could apply Lemma 1 to different choices of $\mathcal{S}$ to obtain different weights $\alpha_{0m}(X_i, X_j) \equiv \Pi_{\mathcal{S}}(\delta_m(X_i, X_j, \gamma_0))$ and the corresponding MLU-IFs. However, depending on the choice of $\mathcal{S}$ and δ_m the projection $\alpha_{0m} \equiv \Pi_{\mathcal{S}}\delta_m$ may be hard to compute. The following result provides two instances where the conditions of Lemma 1 are satisfied for the given Γ and $\mathcal{S}$ and simplified expressions of α_{0m} are provided.

Lemma 2 Suppose (3.1) and (3.2) hold. For all $m = 1, \dots, M$ we have that:

- (i) For $\Gamma = L_2(X_i)$, if $\mathcal{S} = L_2(X_i, X_j)$, then $\alpha_{0m} = \delta_m(\cdot, \gamma_0)$.
- (ii) For any Γ , if $\mathcal{S} = \Gamma + \Gamma$, then

$$\alpha_{0m}(X_i, X_j) = \Pi_{\Gamma} \mathbb{E}[\delta_{ij,m}(\gamma_0)|X_i] + \Pi_{\Gamma} \mathbb{E}[\delta_{ij,m}(\gamma_0)|X_j] - \mathbb{E}[\delta_{ij,m}(\gamma_0)]. \quad (3.4)$$

We refer henceforth to the situation of Lemma 2(i) as the fully nonparametric case. The Γ and $\mathcal{S}$ in (i) lead to the simple expression for the debiased estimator in our application to

IOP. Lemma 2(ii) is most useful in settings where considering fully nonparametric estimation is not feasible (e.g. in high dimensional settings), and thus the researcher chooses as Γ a strict subset of $L_2(X_i)$. As discussed in Appendix A, a difference with respect to the debiased literature is that the adjustment term ϕ in our case is not unique. However, any MLU-IF leads to the same first order asymptotics, as we explain in the next section.

4 Asymptotic theory

The aim of using a debiased moment function and cross-fitting is to be able to perform valid inference. First, we will show the key result that

$$\sqrt{n} \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta_0) = \sqrt{n} \binom{n}{2}^{-1} \sum_{i < j} \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) + o_p(1). \quad (4.1)$$

This means that the first order asymptotic properties of the debiased $\hat{\theta}$ are not affected by the estimation of first steps. The estimators $\hat{\alpha}_l$ for $\alpha_0(X_i, X_j) \equiv \Pi_{\mathcal{S}}(\delta(X_i, X_j, \gamma_0))$ depend on the choice of $\mathcal{S}$ and the expression for $\delta(X_i, X_j, \gamma_0)$. In all our examples δ is known up to the first step γ_0 , so in the joint nonparametric case of Lemma 2(i) $\alpha_0 = \delta(\cdot, \gamma_0)$ is also known up to γ_0 and we can set $\hat{\alpha}_l = \delta(\cdot, \hat{\gamma}_l)$. For other cases we can obtain $\hat{\alpha}_l$ from Lemmas 1 and 2 by any ML method that estimates the projection of $\delta(\cdot, \hat{\gamma}_l)$ onto $\mathcal{S}$ or Γ (such as Lasso, neural nets, sieves, etc.). For example, using Lemma 2 (ii) and letting $n_l = \sum_{j:(i,j) \notin I_l} 1$, we can first compute

$$\tilde{\alpha}_{1l}(X_i) = \frac{1}{n_l} \sum_{j \neq i} \delta_{ij}(\hat{\gamma}_l) \text{ and } \tilde{\alpha}_{2l}(X_j) = \frac{1}{n_l} \sum_{i \neq j} \delta_{ij}(\hat{\gamma}_l),$$

and then we can estimate the orthogonal projection of $\tilde{\alpha}_{rl}(X_i)$ onto Γ with any ML estimator estimating projections and with observations not in I_l .

Let now $|\cdot|$ and $\|\cdot\|$ be the Euclidean and L_2 norms, respectively. The terms $\rightarrow_p$ and $\rightarrow_d$ denote convergence in probability and distribution, respectively.

Assumption 3 $\mathbb{E}[|\psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)|^2] < \infty$ and

- (i) $\int \int |g(w_i, w_j, \hat{\gamma}_l, \theta_0) - g(w_i, w_j, \gamma_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0$;
- (ii) $\int \int |\phi(w_i, w_j, \hat{\gamma}_l, \alpha_0, \theta_0) - \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0$;
- (iii) $\int \int |\phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta_0) - \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0$.

These are mild mean-square consistency conditions for $\hat{\gamma}_l$ and $\hat{\alpha}_l$ separately. A linearization argument like Equation (3.2) often implies that the left hand side of Assumption 3(i)-(ii) are bounded above by a constant times $\|\hat{\gamma}_l - \gamma_0\|^2$, so L_2 consistency suffices. Assumption 3(iii) typically follows from L_2 consistency of $\hat{\alpha}_l$, as for ϕ in Equation (3.3).

There is a large literature checking L_2 -convergence rates for different machine learners under low level sparsity or smoothness conditions on the nuisance parameters. The traditional non-parametric literature gives rates for kernel regression (e.g. TSYBAKOV'S BOOK) and sieves/series (e.g. Chen (2007)). For L_1 -penalty estimators such as Lasso see, e.g., Belloni and Chernozhukov (2011) and Belloni and Chernozhukov (2013). Also for low level conditions for shrinkage and kernel estimators see Appendix B in Sasaki and Ura (2021). Rates for L_2 -boosting in low dimensions are found in Zhang and Yu (2005), and more recently in Kueck et al. (2023) with high dimensional data. For results on versions of random forests see Wager and Walther (2015) and Athey et al. (2019). Finally, for single-layer, sigmoid-based neural networks see Chen and White (1999) and for a modern setting of deep neural networks with rectified linear (ReLU) activation function see Farrell et al. (2021) MORE REFERENCES HERE FROM ANNALS. For instance, Theorem 1 in Farrell et al. (2021) shows that for $X \in [-1, 1]^d$ continuously distributed and bounded Y and $\gamma_0(X)$, we have that with high probability for a deep ReLU network estimator $\hat{\gamma}$ for large n

$$\|\hat{\gamma} - \gamma\|^2 \leq C \left(n^{-\frac{\beta}{\beta+d}} \log^8 n + \frac{\log \log n}{n} \right),$$

where β is a smoothness parameter and C is some positive constant. Also, under some conditions, Kueck et al. (2023) give a L_2 -rate of $s \log(d \vee n) \log n / n$ for L_2 -boosting, which almost attains the usual Lasso rate $s \log(d \vee n) / n$, where d is the dimension of X and s is a sparsity parameter.

Define now the following interaction term

$$\hat{\xi}_l(w_i, w_j) = \phi(w_i, w_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta_0) - \phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta_0) - \phi(w_i, w_j, \hat{\gamma}_l, \alpha_0, \theta_0) + \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0).$$

Assumption 4 For each $l = 1, \dots, L$, either i)

$$\sqrt{n} \int \int \hat{\xi}_l(w_i, w_j) F_0(dw_i) F_0(dw_j) \rightarrow_p 0, \int \int |\hat{\xi}_l(w_i, w_j)|^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0,$$

or ii) $\sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} |\hat{\xi}_l(W_i, W_j)| \rightarrow_p 0$, or iii) $\sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} \hat{\xi}_l(W_i, W_j) \rightarrow_p 0$.

These are rate conditions on the remainder term $\hat{\xi}_l(w_i, w_j)$. For ϕ in Equation (3.4) in

the main text, the interaction term has the form

$$\hat{\xi}_l(w_i, w_j) = (\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j)) (c_1 \hat{\gamma}_l(X_i) + c_2 \hat{\gamma}(X_j) - c_1 \gamma_0(X_i) - c_2 \gamma_0(X_j)).$$

Therefore, Assumption 4 follows for first steps satisfying orthogonality restrictions if $\sqrt{n} \|\hat{\alpha}_l - \alpha_0\| \|\hat{\gamma}_l - \gamma_0\| = o_p(1)$. This is a product rate condition which allows for nuisance estimators to converge at slower rates as long as the product converges at $\sqrt{n}$ -rate. Define now $\bar{\psi}(\gamma, \alpha, \theta) \equiv \mathbb{E}[\psi(W_i, W_j, \gamma, \alpha, \theta)]$. Henceforth, C is a generic positive constant that may change from expression to expression.

Assumption 5 *For each $l = 1, \dots, L$ and θ , i) $\int \int \phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta) F_0(dw_i) F_0(dw_j) = 0$ with probability approaching one; and either ii) $\|\hat{\gamma}_l - \gamma_0\| = o_p(n^{-1/4})$ and $|\bar{\psi}(\gamma, \alpha_0, \theta_0)| \leq C \|\gamma - \gamma_0\|^2$ for all γ with $\|\gamma - \gamma_0\|$ small enough; or iii) $\sqrt{n} \bar{\psi}(\hat{\gamma}_l, \alpha_0, \theta_0) \rightarrow_p 0$.*

Assumption 5 (i) incorporates the global robustness property of α and is in most cases easy to check by inspection of ϕ . For example, it holds for ϕ in Equation (2.6). Assumption 5 (ii) and (iii) are small bias conditions. For a parameter of the form $\theta_0 = \mathbb{E}[\delta(X_i, X_j, \gamma_0)(c_1 \gamma_0(X_i) + c_2 \gamma_0(X_j))]$, Assumption 5 (ii) and (iii) hold if $\|\hat{\gamma}_l - \gamma_0\| = o_p(n^{-1/4})$ and $\|\delta(\cdot, \gamma) - \delta(\cdot, \gamma_0)\| \leq C \|\gamma - \gamma_0\|$.

Lemma 3 *If Assumptions 3-5 are satisfied then equation (4.1) holds.*

Lemma 3 is a key asymptotic result which implies asymptotic normality for the debiased estimator by standard U-statistics arguments when the identifying moment condition is non-degenerate.

For valid inference we also need convergence of the asymptotic variance estimators. To simplify the computation we implemented the variance estimator without cross-fitting, so standard U-statistics formulas are valid. Consistency of the variance follows under standard L_2 consistency of first steps. Define $\hat{g}_{ij} = g(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \hat{\theta})$ and $g_{ij} = g(W_i, W_j, \gamma_0, \alpha_0, \theta_0)$. Then define the following leave-one out average $\hat{g}_{-i} = (n-1)^{-1} \sum_{j \neq i} \hat{g}_{ij}$ and $g_{-i} = (n-1)^{-1} \sum_{j \neq i} g_{ij}$, let $\hat{\phi}_{-i}$ and ϕ_{-i} be defined in the same way.

Lemma 4 *If $\mathbb{E}[|\psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)|^2] < \infty$, $n^{-1} \sum_{i=1}^n |\hat{g}_{-i} - g_{-i}|^2 \rightarrow_p 0$ and $n^{-1} \sum_{i=1}^n |\hat{\phi}_{-i} - \phi_{-i}|^2 \rightarrow_p 0$, then $\hat{\Sigma} \rightarrow_p \Sigma$.*

We also need convergence of the Jacobian of the moment condition $\hat{B} \rightarrow_p B$. Define $\tilde{\psi}_{ij} = \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \theta_0)$.

Assumption 6 *B exists and there is a neighborhood $\mathcal{N}$ of θ_0 such that i) $\|\hat{\gamma} - \gamma_0\| \rightarrow_p 0$; ii) for all $\|\gamma - \gamma_0\|$ and $\|\alpha - \alpha_0\|$ small enough $\psi(W_i, W_j, \gamma, \alpha, \theta)$ is differentiable in θ*

on $\mathcal{N}$ with probability approaching one and there is $C > 0$ and $d(W_i, W_j, \gamma, \alpha)$ such that $\mathbb{E}[d(W_i, W_j, \gamma, \alpha)] \leq C$ and such that for $\theta \in \mathcal{N}$ and $\|\gamma - \gamma_0\|$ and $\|\alpha - \alpha_0\|$ small enough

$$\left| \frac{\partial \psi(W_i, W_j, \gamma, \alpha, \theta)}{\partial \theta} - \frac{\partial \psi(W_i, W_j, \gamma, \alpha, \theta_0)}{\partial \theta} \right| \leq d(W_i, W_j, \gamma, \alpha) |\theta - \theta_0|^{1/C}$$

iii) For each k , $\mathbb{E}[\partial \tilde{\psi}_{ij} / \partial \theta_k - \partial \psi_{ij} / \partial \theta_k] \rightarrow_p 0$.

Lemma 5 If Assumption 6 is satisfied and if $\bar{\theta} \rightarrow_p \theta_0$ then $\partial \hat{\psi}(\bar{\theta}) / \partial \theta \rightarrow_p B$.

Now we are ready to give the main asymptotic result of the paper,

Theorem 1 If Assumptions 3-6 and conditions in Lemma 4 are satisfied, $\hat{\theta} \rightarrow_p \theta_0$ and $V = B^{-1} \Sigma B'^{-1}$ is nonsingular, then

$$\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow_d \mathcal{N}(0, V).$$

Also, $\hat{V} \rightarrow_p V$.

Theorem 1 requires consistency of $\hat{\theta}$. In Appendix E we give conditions similar to the ones in this section under which $\hat{\theta}$ is consistent.

REMARK (On degeneracy of orthogonal moments). When the orthogonal (sample) moment is a degenerate U-statistic, $\Sigma = 0$ and hence $V = 0$. Therefore, our asymptotic distribution theory is useful only for non-degenerate orthogonal moments. We focus on this case because it is the most common one in applications, as for example in IOp. The analysis of the degenerate case is beyond the scope of this paper and deferred to future research. ■

4.1 Asymptotic properties of the debiased IOp

We give low-level conditions for the IOp example based on the general asymptotic properties given in the previous section. Define the difference of fitted values $\Delta_\gamma(X_i, X_j) = \gamma(X_i) - \gamma(X_j)$ and recall $\Delta_0 \equiv \Delta_{\gamma_0}(X_i, X_j)$ and $\alpha_0(X_i, X_j) = \text{sgn}(\gamma_0(X_i) - \gamma_0(X_j))$. To obtain asymptotic normality assume the following

Assumption 7 (i) $\mathbb{E}[(Y_i - \gamma_0(X_i))^2] \leq C$;

(ii) $\int |\hat{\gamma}_l(x) - \gamma_0(x)|^2 F_0(w) \rightarrow_p 0$; (iii) $\int \int |\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j)|^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0$;

(iv) $\sqrt{n} \int \int (\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j)) (\hat{\gamma}_l(x_i) - \gamma_0(x_i) - \hat{\gamma}_l(x_j) + \gamma_0(x_j)) F_0(dw_i) F_0(dw_j) \rightarrow_p 0$.

Assumption 7 (i) imposes a mild boundedness condition. Assumptions 7 (ii) and (iii) require $\hat{\gamma}_l$ and $\hat{\alpha}_l$ to be mean square consistent, which are also mild conditions and allow for generic estimators. In the next proposition we show that for Assumptions 7 (iii) and (iv) to hold it is enough for $\|\hat{\gamma}_l - \gamma_0\|$ to converge in probability sufficiently fast.

Proposition 2 *If $\|\hat{\gamma}_l - \gamma_0\| = o_p(n^{-1/4})$ then Assumptions 7 (iii) and (iv) hold.*

Now we move to the conditions for consistent estimation of the variance. Since cross-fitting is not needed to show the consistency of the variance, we repeat some of the assumptions in Assumption 7 for nuisance estimators which use all observations.

Assumption 8 (i) $\mathbb{E}[(\hat{\gamma}(X_i) - \gamma_0(X_i))^2] \rightarrow 0$; (ii) $\mathbb{E}[(\hat{\alpha}(X_i, X_j) - \alpha_0(X_i, X_j))^2] \rightarrow 0$.

Proposition 3 *Under Assumptions 7 and 8 and $\hat{\theta} \rightarrow_p \theta_0$, we have that $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow_d \mathcal{N}(0, V)$ and $\hat{V} \rightarrow_p V$, where $\hat{V}$ is given in (2.8).*

Consistency of $\hat{\theta}$ follows readily from the mild conditions given in Appendix E.

5 Simulations

We evaluate our debiased estimator for the Gini of the fitted values with Monte Carlo simulations. We consider three independent categorical covariates with eight equally likely levels, $X_{k,i} \in \{0, 1, \dots, 7\}$ for $k = 1, 2, 3$. Define dummy variables $D_{kj,i} = 1(X_{k,i} = j)$ for $j = 0, 1, \dots, 7$. We use the following DGP:

$$\ln(Y_i) = \beta_0 + \sum_{k=1}^3 \sum_{j=1}^7 \beta_{k,j} D_{kj,i} + \sum_{k < k'}^3 \sum_{r=1}^7 \sum_{s=1}^7 \delta_{kk',rs} D_{kr,i} D_{k's,i} + \sum_{r=1}^7 \sum_{s=1}^7 \sum_{t=1}^7 \gamma_{rst} D_{1r,i} D_{2s,i} D_{3t,i} + \varepsilon_i.$$

The above constitutes a saturated model with a reference group represented by the constant, $3 \times 7 = 21$ coefficients of main effects, $\binom{3}{2} \times 7^2 = 147$ coefficients for pairwise interactions among the levels and $7^3 = 343$ coefficients for threewise interactions among the levels. Hence, in total we have 512 parameters which means that estimating this model for low sample sizes is a high dimensional problem. The noise variable ε_i is distributed as $\mathcal{N}(0, \sigma^2)$. We set $\beta_0 = 5$ and for the vector $\beta = (\beta_{11}, \dots, \beta_{1,7}, \beta_{2,1}, \dots, \beta_{3,7})'$, we set $\beta_m = 0.2(-1)^{m+1}$ for $m = 1, \dots, 21$. For the vector of interaction coefficients $\xi = (\delta', \gamma')'$ we set $\xi_m = (2m^2)^{-1}$ for $m = 1, \dots, 490$.

In the simulations we set the variance of the unobservable term σ^2 to 0.1. This implies a Signal to Noise ratio of $StN = 10.5$. For the results for $\sigma \in \{0.2, 0.3\}$ (and correspondingly $StN \in \{2.6, 1.2\}$) see Appendix J. To estimate the fitted values we run Lasso and Ridge regressions of $\ln(Y_i)$ using 10-fold Cross-Validation for the regularization parameters. For the

IOP we apply the exponential function to the fitted values of the Lasso and Ridge regressions and then use the plug-in and debiased estimators since we are interested in the IOP of Y_i and not of $\ln(Y_i)$.⁶ For the debiased estimator we partition the observation indices into 5 sets for cross-fitting, i.e. $K = 5$. We also estimate the first step with Random Forests (RF), Conditional Inference Forests (CIF), Extreme Gradient Boosting (XGB) and Catboost (CB). The results are in Table 1.

Gini of the FVs								
plug-in Lasso		Debiased Lasso		plug-in Ridge		Debiased Ridge		
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.002	0.863	-0.026	0.567	-0.086	0.000	-0.022	0.629
n = 500	-0.005	0.746	-0.002	0.909	-0.081	0.000	-0.005	0.798
n = 1000	-0.005	0.710	-0.001	0.950	-0.002	0.877	-0.003	0.879
n = 3000	-0.002	0.752	0.000	0.940	0.000	0.927	-0.001	0.931
plug-in RF		Debiased RF		plug-in CIF		Debiased CIF		
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.032	0.157	-0.015	0.774	-0.072	0.002	-0.017	0.730
n = 500	-0.022	0.008	-0.002	0.917	-0.034	0.000	-0.003	0.923
n = 1000	-0.022	0.000	-0.001	0.937	-0.027	0.000	-0.002	0.923
n = 3000	-0.023	0.000	-0.001	0.933	-0.023	0.000	-0.001	0.942
plug-in XGBoost		Debiased XGBoost		plug-in Catboost		Debiased Catboost		
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	0.010	0.851	-0.039	0.310	-0.012	0.619	-0.032	0.429
n = 500	0.006	0.774	-0.005	0.863	0.002	0.815	-0.003	0.907
n = 1000	0.004	0.772	-0.002	0.931	0.002	0.837	-0.001	0.929
n = 3000	0.003	0.702	0.000	0.956	0.001	0.829	0.000	0.956

Table 1: Simulation based on 500 Monte Carlo iterations, true value for the Gini of the FVs is 0.18.

The debiased estimators are unstable for $n = 100$. This is expected since the first steps are using very few observations due to the cross-fitting procedure. This issue disappears fast as n increases. In the case of Lasso we see that for sample sizes larger than 100 the bias of the debiased estimators is considerably smaller in absolute size than the plug-in biases. Also, the coverage rate of the debiased estimator is much closer to the nominal 95% compared to the coverage of the plug-in estimator. In the case of Ridge there is not much difference

⁶Under a log-normal model $\ln(Y_i) = \beta'X_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, we have that $\mathbb{E}[Y_i|X_i] = e^{\beta'X_i + \sigma^2/2}$, hence the fitted values $\gamma(x) = e^{\beta'x}$ solve $\mathbb{E}[\nu(X_i)(Y_i/e^{\sigma^2/2} - \gamma(X_i))] = 0$ for all $\nu(\cdot) \in L_2$. Hence, we can use the estimator of the joint nonparametric case and the constant $e^{\sigma^2/2}$ cancels.

between the plug-in estimator and the debiased estimator, possibly due to the weak sparsity of the model, and the coverage of both confidence intervals slowly increases to the correct nominal size as the sample size increases.

The performance of RF and CIF is similar. The biases of the debiased estimators are much lower in absolute value than that of the plug-in estimators. Also, the coverage rates of the plug-in estimators are far from the nominal values while the coverage of the debiased estimators are close to the nominal value. Finally, the boosting methods XGB and CB show little bias for the debiased and plug-in estimators. However, only the debiased estimators achieve coverage rates close to the nominal values.

Hence, in all cases the debiased estimators achieve little bias and are the only estimators which allow for correct inference.

6 Inequality of Opportunity in Europe

We measure IOp in 29 European countries using our debiased IOp estimator and the 2019 wave of EUSILC. Our measure of income is equivalized household income and the level of observation is the individual.⁷ We restrict the sample to those aged between 25 and 59 years old to focus on the working age population.

The circumstances included in the intergenerational module include questions on characteristics of the parents and questions related to the individual’s life/household when he/she was around 14 years old. We use the following circumstances: sex, country of birth, whether he/she was living with the mother/father, the number of adults/working adults/kids in the household, population of the municipality, tenancy of the house, country of birth of the parents, nationality of the parents, education of the parents, occupational status of the parents, father’s managerial position, father’s occupation, basic school needs (whether he/she had access to books, materials, etc.), financial situation, food needs (whether he/she could eat meat/chicken/fish/vegetarian equivalent once a week and holidays outside of home once a year). Remember that it refers to when the individual was around 14 years old, so, for instance, financial situation refers to the financial situation of the household where the individual resided when he/she was around 14 years old.

All circumstances are discrete and there are many different combinations of the categories. This makes the problem high dimensional and hard to deal with without machine learning procedures. In this application we use Lasso, Ridge, RF, CIF, XGB and CB. For Lasso and Ridge we use dummy encoding and employ a dictionary including up to 8-wise interactions (1,658 regressors). A nice feature of the cross-fitting procedure is that one can retrieve a

⁷This variable is the total household disposable income in 2018 (variable HY020) divided by the equivalence scale given in the database (HX240) which is the modified OECD equivalence scale.

cross-validated estimate of the Root Mean Square Error (RMSE) in the first stage. We will show our results for each country only for the machine learner which attains the lowest RMSE when predicting income in the first stage; we call this machine learner the best performing machine learner.

For cross-fitting we split $\{1, \dots, n\}$ into $C_1, \dots, C_K$ with $K = 5$ which leads to $L = K(K + 1)/2$ blocks $I_1, \dots, I_L$ (see Appendix K for details). The debiased and plug-in relative IOp estimates (IOp as a fraction of the Gini of income) based on the best performing machine learner with 95 % confidence intervals for the debiased estimates can be seen in Figure 2.

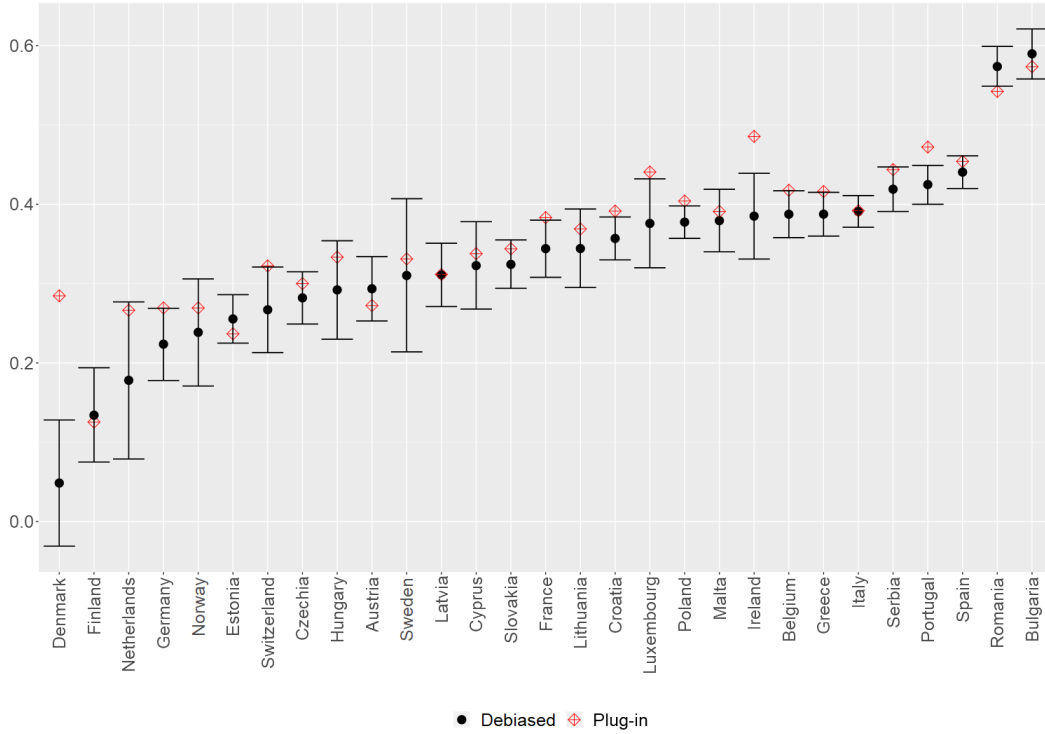


Figure 2: Gini IOp in Europe

We see a lot of heterogeneity in IOp across different European countries. Relative IOp, i.e. Gini of the fitted values over the Gini of income, takes values from 5% to almost 60%. As usual, Nordic countries such as Denmark, Finland or Norway are among the countries with the lowest relative IOp with 5%, 13% and 24% of inequality explained by circumstances respectively. In Denmark, predicting income from circumstances seems to be a formidable task, suggesting that income is close to mean independent from the observed circumstances. From the Nordic countries, the one with the highest relative IOp is Sweden with 31% of relative IOp. Netherlands and Germany are also in the lower range with relative IOp of 18% and 22% respectively. Eastern countries are more heterogeneous, in the lower range we have countries such as Czechia or Hungary with relative IOp of 28% and 29% respectively. In the higher range, we have Romania and Bulgaria with relative IOp being almost 60%. Southern

countries generally have high levels of IOp; Greece, Italy and Spain have 39%, 39% and 44% of total inequality explained by circumstances.

Comparing our empirical results with other studies we see similar patterns qualitatively. As in [Hufe et al. \(2022b\)](#), [Brunori et al. \(2021\)](#), [Marrero and Rodríguez \(2012\)](#) or [Carranza \(2022\)](#) we see that Nordic countries and countries such as Germany or the Netherlands have low IOp, Southern countries have high IOp and there is heterogeneity in Eastern countries. [Hufe et al. \(2022b\)](#) takes a multidimensional approach and shows that this divide between north and south is reduced when considering also IOp in education and health.

We report more detailed results in Table 2 where for each country we can see the income inequality, the plug-in and debiased estimates based on the best performing machine learner, relative IOp, the best performing machine learner, the cross-validated root mean squared error of the first step for the best performing machine learner and the sample size.

Country	Mean	Gini	Plug-in	Debiased	Debiased/Gini (%)	Best Performer	RMSE	n
Austria	29824	0.268	0.073	0.079 (0.066,0.091)	29 (25,33)	Ridge	16364	5119
Belgium	28432	0.237	0.099	0.092 (0.083,0.101)	39 (36,42)	CIF	12542	6359
Bulgaria	6603	0.415	0.238	0.244 (0.222,0.267)	59 (56,62)	CIF	8073	5906
Switzerland	52753	0.273	0.088	0.073 (0.056,0.09)	27 (21,32)	CIF	30580	4705
Cyprus	20281	0.299	0.101	0.096 (0.077,0.115)	32 (27,38)	Ridge	16083	4647
Czechia	12215	0.23	0.069	0.065 (0.056,0.073)	28 (25,32)	XGB	5944	6330
Germany	28813	0.271	0.073	0.061 (0.047,0.074)	22 (18,27)	XGB	17261	5967
Denmark	36768	0.26	0.074	0.013 (-0.009,0.034)	5 (-3,13)	CB	20807	2068
Estonia	14310	0.283	0.067	0.072 (0.063,0.081)	26 (22,29)	Lasso	7026	5649
Greece	9915	0.31	0.129	0.12 (0.109,0.131)	39 (36,42)	CB	6985	14144
Spain	17732	0.326	0.148	0.144 (0.136,0.152)	44 (42,46)	CIF	10587	16864
Finland	29305	0.263	0.033	0.035 (0.019,0.052)	13 (8,19)	Lasso	17033	4398
France	27014	0.274	0.105	0.094 (0.082,0.106)	34 (31,38)	CIF	15801	7821
Croatia	8916	0.281	0.11	0.1 (0.091,0.109)	36 (33,38)	CIF	4625	7120
Hungary	6929	0.282	0.094	0.082 (0.061,0.104)	29 (23,35)	CIF	4673	4557
Ireland	32223	0.278	0.135	0.107 (0.087,0.127)	39 (33,44)	CB	19362	3653
Italy	20004	0.319	0.125	0.125 (0.117,0.132)	39 (37,41)	CIF	12041	16359
Lithuania	10383	0.347	0.128	0.12 (0.1,0.139)	34 (30,39)	XGB	7150	3643
Luxembourg	45490	0.329	0.145	0.124 (0.1,0.147)	38 (32,43)	CIF	35653	3516
Latvia	10789	0.337	0.105	0.105 (0.09,0.12)	31 (27,35)	Lasso	7129	3337
Malta	19271	0.266	0.104	0.101 (0.088,0.114)	38 (34,42)	CIF	9280	3623
Netherlands	30636	0.259	0.069	0.046 (0.017,0.075)	18 (8,28)	CIF	18445	4441
Norway	41855	0.245	0.066	0.058 (0.04,0.077)	24 (17,31)	XGB	21998	2364
Poland	8621	0.292	0.118	0.11 (0.103,0.118)	38 (36,40)	CIF	5014	14281
Portugal	12032	0.305	0.144	0.13 (0.121,0.138)	42 (40,45)	XGB	7217	13776
Romania	4826	0.343	0.186	0.197 (0.185,0.209)	57 (55,60)	CIF	2667	5932
Serbia	3984	0.329	0.146	0.138 (0.126,0.149)	42 (39,45)	CIF	2448	5648
Sweden	29189	0.305	0.101	0.095 (0.057,0.132)	31 (21,41)	XGB	28923	1894
Slovakia	9197	0.221	0.076	0.072 (0.064,0.079)	32 (29,36)	CIF	3483	5727

Table 2: EUSILC Gini IOP: debiased vs. plug-in estimators, bestmachine learner and sample size

The bias of the plug-in estimator varies from country to country. Denmark, Croatia, Luxembourg, France, Poland, Ireland, Portugal and Romania have plug-in estimates outside the 95 % confidence interval of the debiased estimates. In Figure 3 we see the difference between the (relative) debiased and plug-in estimators. For most countries the plug-in estimator overestimates IOp but this is not always the case. The differences can be quite stark, in Denmark we have almost 25 percentage point difference. In the rest of the countries the differences are much smaller but we still observe differences of 5-10 percentage points in some countries.

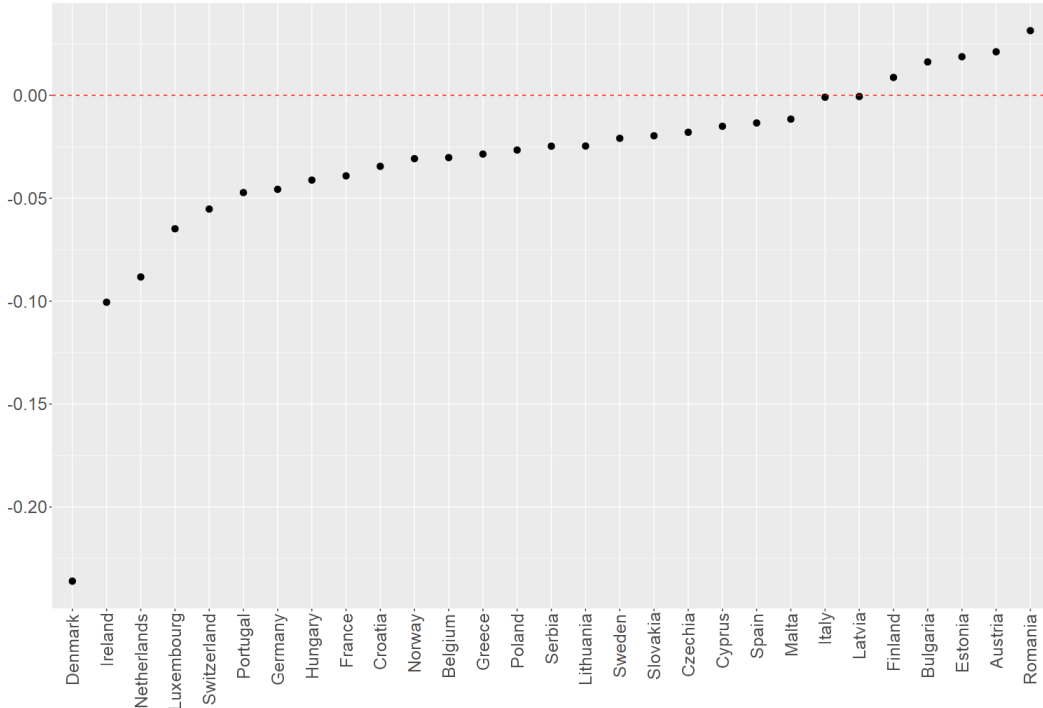


Figure 3: Difference between relative debiased IOp and plug-in IOp.

As already pointed out, the debiased estimates are much less sensitive to the choice of the machine learner. In Figure 4 we report the debiased and plug-in estimates for the 6 different machine learners. We can see that the debiased estimates are much more concentrated and near to each other than the plug-in estimates. The plug-in estimates are much more dispersed and in some cases there are differences of 60 percentage points from one machine learner to the other. Even differences of 30-40 percentage points between plug-in estimates using different machine learners are not uncommon. This result alone shows the importance of using locally robust debiased estimators when estimating IOp. We also see that Random Forest substantially overestimates in all countries, however, taking away Random Forests from the Figure we still see that the plug-in estimates are much more dispersed than the debiased estimates.

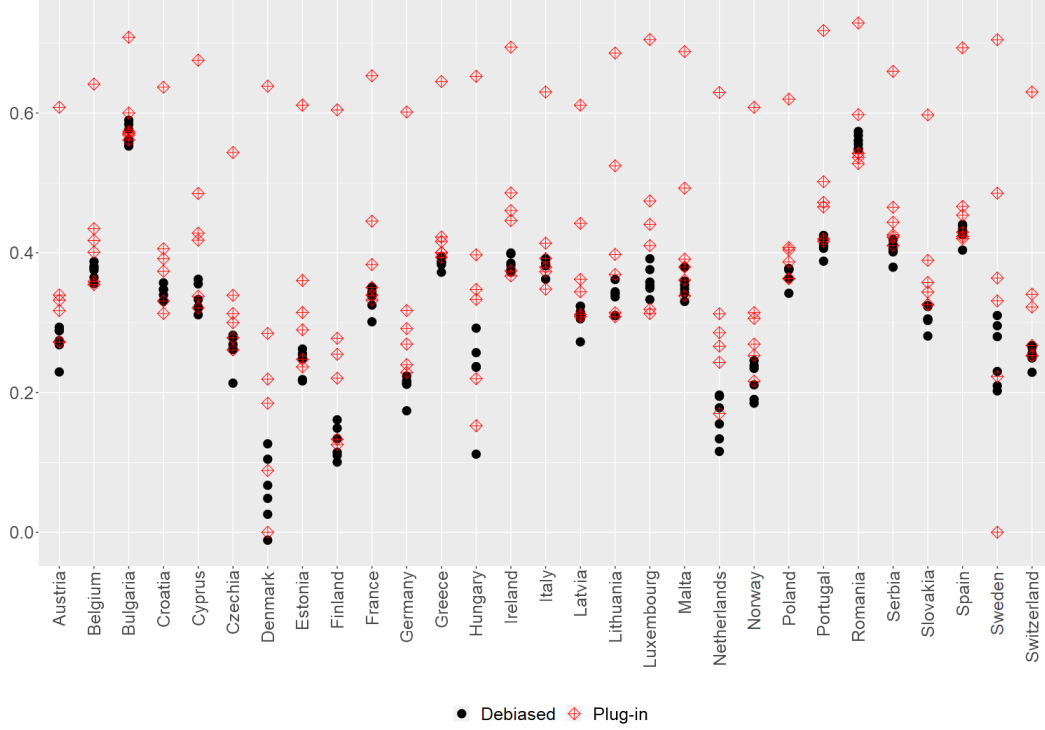


Figure 4: Debiased vs plug-in estimates for the 6 different machine learners

Finally, we want to have an idea of which circumstances are the most relevant to explain inequality. To do this, we compute the relative change in IOp when we exclude a given circumstance. In Table 3 we report the circumstance in each country for which this relative change is the greatest. For instance, in Austria the most important circumstance is living with the father. This means that if we exclude this circumstance, the relative IOp would decrease 2.6%. We see that the most important circumstance varies from country to country. However, parental education and occupation stand out as recurrent circumstances which help explain inequality the most. For instance in countries such as the Netherlands or Germany, excluding parental occupation leads to a 12-13% decrease in IOp. Meanwhile, excluding maternal education leads to a 4% decrease of IOp in Czechia or almost a 13% decrease in IOp in Hungary. This is in line with recent findings in Hufe et al. (2022b) who attribute the increase in IOp in the US mainly to the increased importance of parental occupation and education. Here we go a bit deeper, as we are able to separate paternal and maternal influences and provide uncertainty quantification. We see that while for the mother it is the education which matters the most to explain inequality, for the father it is the occupation which matters the most.

It might seem surprising that sex does not appear as an important contributor to explaining inequality in any country. However, as pointed out in Hufe et al. (2022a), this is expected since we are looking at equalised household income. Hence, any intra-household

differences are not accounted for. Any explanatory power sex might have in our analysis would come only from single-headed households.

7 Dyadic models

Suppose now that we have random vector $(O_{ij}, Z_i, Z_j) \sim F_0$, with $O_{ij} = (Y_{ij}, D_{ij}, U_{ij})$ and $Z_i = (X_i, A_i, B_i)$. We observe $\{W_{ij}\}_{i=1, j=1}^{i=n, j=n} = \{Y_{ij}, D_{ij}, X_i, X_j\}_{i=1, j=1}^{i=n, j=n}$ and we do not observe $\{A_i, B_i\}_{i=1}^n$ and $\{U_{ij}\}_{i=1, j=1}^{i=n, j=n}$. (O_{ij}, O_{kp}) are allowed to be dependent whenever they share at least one index in common but are otherwise independent, $X_1, \dots, X_n$ are i.i.d.

EXAMPLE (Gravity Model): A popular application, originally proposed in [Tinbergen \(1962\)](#), is gravity models. Y_{ij} is trade flow or migration between country i and country j . Let $X_i = (G_i, C_i)$ be observable country characteristics, such as GDP and coordinates from which you can compute distance. Let

$$Y_{ij} = \frac{G_i^{\beta_0} G_j^{\beta_1}}{r(D_i, D_j)} A_i B_j U_{ij}, \quad (7.1)$$

where $r(\cdot, \cdot)$ is known, for instance, the inverse of haversine function used to compute distance from coordinates. A_i and B_j are unobserved country fixed effects. These fixed effects are usually called multilateral resistance in the trade literature and allow for exporter and importer fixed effects. $\{U_{ij}\}_{(i,j) \in \{1, \dots, n\}^2}$ are iid unobservables and $\mathbb{E}[A_i B_j U_{ij} | X_i, X_j] = 1$. This assumption rules out mean dependence between unobservable fixed effects and observables. For example, if X_i includes membership to some trade organization, we rule out that this is correlated with some unobserved "openness to trade". We keep this strong assumption in the main text and relax it in [Appendix I.2](#). ■

Now suppose that we are interested in a finite-dimensional parameter θ_0 identified by the moment condition

$$\mathbb{E}[g(W_{ij}, \gamma_0, \theta)] = 0 \text{ iff } \theta = \theta_0.$$

As before, $\gamma_0 = \gamma(F_0)$ is a possibly infinite-dimensional first step. Let Y_{ij} be a component of O_{ij} . We consider now first steps γ such that

$$\mathbb{E}_F[\nu(X_i, X_j) (Y_{ij} - \gamma(F)(X_i, X_j))] = 0 \text{ for all } \nu \in \Gamma, \quad (7.2)$$

where now Γ is a set of functions of (X_i, X_j) . For instance, if $\Gamma = L_2$, then $\gamma(F)(X_i, X_j) = \mathbb{E}_F[Y_{ij} | X_i, X_j]$. [Graham et al. \(2021\)](#) study the nonparametric estimation of such nuisance parameters, which they call nonparametric dyadic regression⁸. They provide asymptotic

⁸Whenever $Y_{ij} = h(Y_i, Y_j)$ for known h and observed (Y_i, Y_j) , we are back in the world of U-statistics.

and supremum norm rate results and propose to use a Nadaraya-Watson estimator. Further research on machine learners and their rates for such objects is needed.

EXAMPLE (Semiparametric Gravity Model): Let now D_{ij} be some dyad-level variable, for example, some pair-level treatment such as a trade treaty or a bilateral migration agreement, and X_i be observable country characteristics. Let the model be

$$Y_{ij} = e^{\theta_0 D_{ij}} m(X_i, X_j) A_i B_j U_{ij},$$

We let the functional form of m unspecified. In practical applications, m is usually assumed, e.g. in (7.1) where X_i includes GDP and coordinates. However, we believe that there exist unknown complicated nonlinear complementarities (e.g. between the GDPs of two countries). Up to our knowledge, we are the first to leave m unspecified in such a multiplicative model. Since [Silva and Tenreiro \(2006\)](#) it has become common to base the estimation of gravity models by assuming that the conditional mean of the outcome is proportional to its conditional variance and employ Poisson Pseudo-maximum-likelihood (PPML)⁹. Then, under the assumption that $\mathbb{E}[A_i B_j U_{ij} | X_i, X_j, D_{ij}] = 1$ (relaxed in [Appendix I.2](#)), an identifying moment for θ_0 is

$$\mathbb{E}[(Y_{ij} - e^{\theta_0 D_{ij}} m(X_i, X_j)) D_{ij}] = 0.$$

Under $\mathbb{E}[A_i B_j U_{ij} | X_i, X_j, D_{ij}] = 1$ it also follows that $m(X_i, X_j) = \mathbb{E}[Y_{ij} | X_i, X_j] / \mathbb{E}[e^{\theta_0 D_{ij}} | X_i, X_j]$. Hence, we have two nuisance parameters

$$\gamma_{01}(X_i, X_j) = \mathbb{E}[e^{\theta_0 D_{ij}} | X_i, X_j], \quad \gamma_{02}(X_i, X_j) = \mathbb{E}[Y_{ij} | X_i, X_j].$$

■

While the setting and dependence structure of the data discussed in this section are strikingly similar to that of U-statistics, the double-indexed random variables take us away from the U-statistics setting. However, as we show below, by slightly redefining the MLU-IF and using projection arguments, we will still be able to use our U-statistic theory. While before the moment condition depended on the product $F_0(w_i)F_0(w_j)$, i.e. was quadratic on F_0 , now the moment condition depends on $F_0(o_{ij}, x_i, x_j) = F_0(o_{ij} | x_i, x_j)F_0(x_i)F_0(x_j)$, i.e. is quadratic on $F_0(x)$ and linear on $F_0(o_{ij} | x_i, x_j)$ where we abuse the notation to refer to the conditional distribution of O_{ij} given (X_i, X_j) and the marginal distribution of X_i and X_j .

Such an object has been called Conditional U-statistic and its nonparametric estimation has been studied in [Stute \(1991\)](#).

⁹The first order condition for non-linear least squares (NLS) is $\mathbb{E}[(Y_{ij} - e^{\theta_0 D_{ij}} m(X_i, X_j)) e^{\theta_0 D_{ij}} D_{ij}]$, however, under the assumption of the conditional variance being proportional to the conditional mean, we can employ a weighted NLS to be more efficient. Doing this provides the moment in the text. For more on the advantages of this method instead of NLS see [Silva and Tenreiro \(2006\)](#).

The characterization of the MLU-IF is the same as before, but now

$$\begin{aligned} F_\tau(w_{ij}) &= (F_0(o_{ij}|x_i, x_j) + \tau(H(o_{ij}|x_i, x_j) - F_0(o_{ij}|x_i, x_j))) \\ &\quad \times (F_0(x_i) + \tau(H(x_i) - F_0(x_i))) \\ &\quad \times (F_0(x_j) + \tau(H(x_j) - F_0(x_j))), \end{aligned}$$

where H is an alternative distribution for W_{ij} for which, as before, (W_{ij}, W_{kp}) are allowed to be dependent whenever they share at least one index in common but are otherwise independent and $X_1, \dots, X_n$ are i.i.d. Then, the MLU-IF is characterized by

$$\int \phi(w_{ij}, \gamma(F_\tau), \alpha(F_\tau), \theta) F_\tau(dw_{ij}) = 0 \text{ for all } \tau \in [0, \bar{\tau}), \bar{\tau} > 0, \quad (7.3)$$

and

$$\frac{d}{d\tau} \mathbb{E}[g(W_{ij}, \gamma(F_\tau), \theta)] = \int \phi(w_{ij}, \gamma_0, \alpha_0, \theta) K_H(dw_{ij}), \quad (7.4)$$

where now

$$K_H(w_{ij}) = H(o_{ij}|x_i, x_j)F_0(x_i)F_0(x_j) + F_0(o_{ij}|x_i, x_j)H(x_i)F_0(x_j) + F_0(o_{ij}|x_i, x_j)F_0(x_i)H(x_j).$$

Now, by assuming that the following linearization is possible

$$\frac{d}{d\tau} \mathbb{E}[g(W_{ij}, \gamma(F_\tau), \theta)] = \frac{d}{d\tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0, \theta) \gamma_\tau(X_i, X_j)], \quad (7.5)$$

we can use the same projection arguments as in Lemma 1 to show that the MLU-IF takes the following form

Lemma 6 Suppose (7.2) and (7.5) hold. Then, the MLU-IF is given by

$$\phi(W_{ij}, \gamma_0, \alpha_0, \theta) = \alpha_0(X_i, X_j)(Y_{ij} - \gamma_0(X_i, X_j)),$$

where $\alpha_0(X_i, X_j) = \Pi_\Gamma(\delta(X_i, X_j, \gamma_0, \theta_0))$.

EXAMPLE (Semiparametric Gravity Model (Cont.)): We can start computing the MLU-IF using the chain rule

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}[(Y_{ij} - e^{\theta_0 D_{ij}} m_\tau(X_i, X_j)) D_{ij}] &= -\frac{d}{d\tau} \mathbb{E} \left[\frac{e^{\theta_0 D_{ij}} D_{ij}}{\mathbb{E}[e^{\theta_0 D_{ij}} | X_i, X_j]} \mathbb{E}_\tau[Y_{ij} | X_i, X_j] \right] \\ &\quad + \frac{d}{d\tau} \mathbb{E} \left[\frac{e^{\theta_0 D_{ij}} D_{ij} \mathbb{E}[Y_{ij} | X_i, X_j]}{\mathbb{E}[e^{\theta_0 D_{ij}} | X_i, X_j]^2} \mathbb{E}_\tau[e^{\theta_0 D_{ij}} | X_i, X_j] \right]. \end{aligned}$$

By a simple generalization of Lemma 6 to multiple nuisance parameters, we have that the MLU-IF is

$$\phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) = \mathbb{E}[\alpha_{01}(X_i, X_j)(e^{\theta_0 D_{ij}} - \gamma_{01}(X_i, X_j))] - \mathbb{E}[\alpha_{02}(X_i, X_j)(Y_{ij} - \gamma_{02}(X_i, X_j))],$$

where

$$\alpha_{01}(X_i, X_j) = \frac{\mathbb{E}[e^{\theta_0 D_{ij}} D_{ij} Y_{ij} | X_i, X_j]}{\gamma_{01}(X_i, X_j)}, \quad \alpha_{02}(X_i, X_j) = \frac{\mathbb{E}[e^{\theta_0 D_{ij}} D_{ij} | X_i, X_j]}{\gamma_{01}(X_i, X_j)}.$$

■

Now let

$$\psi(W_{ij}, \gamma_0, \alpha_0, \theta) = \frac{g(W_{ij}, \gamma_0, \theta) + g(W_{ji}, \gamma_0, \theta) + \phi(W_{ij}, \gamma_0, \alpha_0, \theta) + \phi(W_{ji}, \gamma_0, \alpha_0, \theta)}{2}.$$

While before we assumed without loss of generality that ψ was symmetric, here we explicitly impose it since it is common to have non-symmetric models in dyadic data. For example, in the trade gravity model, we have importer and exporter fixed effects (A_i, B_j) . This does not affect our theory or computations. We can define the debiased estimator as the $\hat{\theta}$ that solves

$$\binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_{ij}, \hat{\gamma}_l, \hat{\alpha}_l, \hat{\theta}) = 0.$$

Now we need to prove an analogous result to Lemma 3.

Lemma 7 *If Assumptions 3 -5 are satisfied with (W_i, W_j) replaced by W_{ij} everywhere, then*

$$\binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_{ij}, \hat{\gamma}_l, \hat{\alpha}_l, \theta_0) = \binom{n}{2}^{-1} \sum_{i < j} \psi(W_{ij}, \gamma_0, \alpha_0, \theta_0) + o_p(1).$$

Finally, for the rest of our theory to apply we now show that our object of interest behaves up to first order as a U-statistic. Let $\psi_{ij} \equiv \psi(W_{ij}, \gamma_0, \alpha_0, \theta_0)$ and $Z_{ij} = (X_i, X_j, A_i, B_j, A_j, B_i)$, WHERE A_i, B_j, A_j, B_i ARE... Then we assume the following conditional independence.

Assumption 9 *For $j \neq k$*

$$\mathbb{E}[\psi_{ij}\psi_{ik}|Z_{ij}, Z_{ik}] = \mathbb{E}[\psi_{ij}|Z_{ij}]\mathbb{E}[\psi_{ik}|Z_{ik}].$$

This assumption relates to the concept of relative exchangeability of an array with respect to X in Crane and Towsner (2018) and to the Conditionally-independent dyadic model

(CID, e.g. [Dabbs et al. \(2020\)](#)). Exchangeability results in [Aldous \(1981\)](#), [Hoover \(1979\)](#), [Kallenberg \(1989\)](#) and [Crane and Towsner \(2018\)](#) are used in, for example, [Davezies et al. \(2021\)](#), [Graham \(2020b\)](#) or [Graham \(2020a\)](#).

EXAMPLE (Semiparametric Gravity Model (Cont.)): Assumption [9](#) holds, for example, if for some measurable function λ

$$D_{ij} = \lambda(X_i, X_j, A_i, B_j, U_{ij}),$$

and $U_{ij} \perp U_{ik} \mid Z_{ij}, Z_{ik}$. That is, unobservable dyadic components determining Y_{ij} and D_{ij} are allowed to be dependent across dyads sharing an index in common if this dependence vanishes after controlling for individual observed and unobserved variables. ■

Let now define

$$s_{ij} = \mathbb{E}[\psi(W_{ij}, \gamma_0, \alpha_0, \theta_0) \mid Z_{ij}].$$

The next Lemma shows how we can use our U-statistic asymptotic theory by showing that the average of ψ behaves up to first order in the same way as a U-statistic with kernel characterized by s_{ij} .

Lemma 8 *Under Assumption [9](#),*

$$\sqrt{n} \binom{n}{2}^{-1} \sum_{i < j} \psi(W_{ij}, \gamma_0, \alpha_0, \theta_0) = \sqrt{n} \binom{n}{2}^{-1} \sum_{i < j} s_{ij} + o_p(1),$$

and $\sqrt{n} \binom{n}{2}^{-1} \sum_{i < j} s_{ij} \rightarrow_d \mathcal{N}(0, \Sigma)$ with $\Sigma = 4\mathbb{E}[\psi(W_{ij}, \gamma_0, \alpha_0, \theta_0)\psi(W_{ik}, \gamma_0, \alpha_0, \theta_0)]$.

An estimator of the asymptotic variance which can be shown to be consistent under Assumptions analogous to those in [Section 4](#) is

$$\hat{\Sigma} = \frac{4}{n} \sum_{i=1}^n \left(\frac{1}{n-1} \sum_{j \neq i} \psi(W_{ij}, \hat{\gamma}, \hat{\alpha}, \hat{\theta}) \right)^2.$$

In [Appendix I.1](#) and [Appendix I.3](#) we explore further dyadic examples such as partially linear dyadic regressions, dyadic selection models or causal gravity models.

8 Fairness

Now we have random variables $W = (Y, X, A) \in \{0, 1\} \times \mathcal{X} \times \{0, 1\}$ distributed as F_0 , where A might be an element in X and is called a sensitive feature. We are interested in solving

$$\gamma_0 = \arg \min_{\gamma \in \Gamma} \mathbb{E}[L(\gamma(X), Y)],$$

where L is some loss function. For example, a bank is deciding whether to give a loan to a customer. Let $Y = 1$ if the customer repays the loan and $Y = 0$ otherwise. The bank does not observe Y for this individual but has access to features X of the individual and an i.i.d. training sample $(Y_i, X_i, A_i)_{i=1}^n$ from F_0 . In a classification problem, $\gamma_0(X)$ is a binary rule in the space of measurable functions $\Gamma : \mathcal{X} \rightarrow \{0, 1\}$ that minimizes some classification loss. More generally, the loss function could allow for any supervised learning problem, such as classification, regression, or ranking, where γ_0 is not necessarily binary. For instance, we might be interested in learning the regression function $\mathbb{E}[Y|X]$.

A is called the sensitive attribute. There is widespread concern that the decision rule γ may be unfair to individuals with certain values of A . In Table 4 we summarize different conceptions of fairness in the literature.

Statistical Parity, also known as Demographic Parity, requires $\gamma_0(X)$ to be independent of A . Equal Opportunity has been proposed independently from the Equality of Opportunity literature in economics. It requires that the true positive rate is the same for both groups. For instance, those who repay the loan should be treated the same regardless of their sensitive attribute. However, we might treat those who will not repay the loan differently based on their sensitive attribute. Probabilistic Predictive Equality requires the same as equal opportunity but for $Y = 0$ instead of $Y = 1$. Equalized Odds adds Probabilistic Predictive Equality to Equal Opportunity to equalize true positive and false positive rates across sensitive groups. Risk parity imposes that we are equally accurate (in terms of the loss) across all sensitive groups. Finally, Conditional Statistical Parity only imposes statistical parity at given values $x \in \mathcal{X}$.

Notice that only Risk Parity depends on the loss function directly. The other definitions are based on the equality of conditional distributions of $\gamma_0(X)$. In the next section, we focus on $\gamma_0(X) = \mathbb{E}(Y|X)$ and derive debiased fairness measures for deviations from the generic fairness definition

$$\Pr(\gamma_0(X) \leq t | A = 1, C) - \Pr(\gamma_0(X) \leq t | A = 0, C) = 0,$$

where different choices of events C lead to different fairness definitions in Table 4. For instance, with $C = \{Y = 1\}$, the expression above becomes the Equal Opportunity definition.

For conciseness, we leave out the analysis of Risk Parity as all the results would depend on the specific loss function being used. Also notice that any classification setting in which the optimal classifier takes the form of thresholding $\gamma_0(X_i) = \mathbb{E}(Y|X)$ is included in our analysis since if the fairness definitions we treat hold for $\gamma_0(X)$ they also hold for $1(\gamma_0(X) \geq t)$ for any $t \in \mathbb{R}$.

The previous fairness definitions are based on equalities between different (conditional) distributions. Rychener et al. (2022) propose to use Integral Probability Metrics (IPM). IPM are well-known measures in probability theory used to measure distance between distributions defined by some class of functions $\mathcal{F}$.

Definition 2 (Integral Probability Metrics (IPM)) *Let F and H be distributions over a set $\mathcal{Z}$. Let $Z \sim F$, $X \sim H$ and $\mathcal{F}$ be a class of real-valued functions on $\mathcal{Z}$. Then, the $\mathcal{F}$ -IPM is*

$$D_{\mathcal{F}}(F, H) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_F[f(Z)] - \mathbb{E}_H[f(X)] \right|, \quad (8.1)$$

where $\mathbb{E}_F$ is the expectation with respect to distribution F .

$D_{\mathcal{F}}$ is a pseudo-metric (non-negative, symmetric, zero if $F = H$ and satisfies triangle inequality). If $\mathcal{F}$ separates points in the space of probability distributions (i.e. for any $F \neq H$, there exists $f \in \mathcal{F}$ such that $\mathbb{E}_F[f(X)] \neq \mathbb{E}_H[f(Z)]$), then $D_{\mathcal{F}}$ is a metric and is equal to zero if and only if $F = H$.

The idea in Rychener et al. (2022) is that for appropriate choices of $\mathcal{F}$, one can measure deviations from the fairness definitions above by estimating $D_{\mathcal{F}}(F_{\gamma(X)|A=1,C}, F_{\gamma(X)|A=0,C})$ for different choices of the event C , where $F_{\gamma(X)|A,C}$ is the distribution of $\gamma(X)$ conditional on (A, C) . As long as $\mathcal{F}$ separates points in the space of distributions, these measures will be zero if and only if the fairness definitions are satisfied.

As noted in Rychener et al. (2022), (8.1) has an interpretation as a worst-case difference between expected utilities from two different distributions where the utility functions belong in $\mathcal{F}$.¹⁰ While there are many possible choices for $\mathcal{F}$, here we focus on the Maximum Mean Discrepancy (MMD) IPM. First, we define the concept of a positive definite kernel.

Definition 3 (Positive Definite Kernel) *A kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is positive definite if*

¹⁰IPM are similar to the concept of second-order stochastic dominance which is a well-known concept in decision theory. A distribution F is said to second-order dominate H if for any non-decreasing and concave utility function u : $\mathbb{E}_F[u(X)] \geq \mathbb{E}_H[u(X)]$. As noted in Atkinson et al. (1970), this is equivalent to the Lorenz curves not intersecting (if means are the same), so that we can claim one distribution is more unequal than the other only imposing non-decreasing and concave utility.

for any $n \geq 1$, $x_1, \dots, x_n \in \mathcal{X}$ and $c_1, \dots, c_n \in \mathbb{R}$, it holds that

$$\sum_{i,j=1}^n c_i c_j K(x_i, x_j) \geq 0.$$

For a positive definite symmetric kernel $K : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, the MMD is defined as

$$D_{MMD}(F, H) = \left(\int \int K(u, u') F(du) F(du') + \int \int K(u, u') H(du) H(du') - 2 \int \int K(u, u') F(du) H(du') \right)^{1/2}.$$

It is known that this distance is the IPM induced by $\mathcal{F} = \{f \in \mathbb{H}_K : \|f\|_{\mathbb{H}_K} \leq 1\}$ in the reproducing kernel Hilbert space $\mathbb{H}_K$ corresponding to K . For a given K , our parameter of interest is $\theta_{0C} \equiv D_{MMD}^2(F_{\gamma_0(X)|A=1,C}, F_{\gamma_0(X)|A=0,C})$, i.e. deviations from a fairness definition measured as

$$\begin{aligned} \theta_{0C} &= \int \int K(u, u') F_{\gamma_0(X)|A=1,C}(du) F_{\gamma_0(X)|A=1,C}(du') \\ &\quad + \int \int K(u, u') F_{\gamma_0(X)|A=0,C}(du) F_{\gamma_0(X)|A=0,C}(du') \\ &\quad - 2 \int \int K(u, u') F_{\gamma_0(X)|A=1,C}(du) F_{\gamma_0(X)|A=0,C}(du'). \end{aligned}$$

Rewriting we get

$$\begin{aligned} \theta_{0C} &= \sum_{(a,b) \in \{0,1\}^2} \zeta_{ab} \mathbb{E}[K(\gamma_0(X_i), \gamma_0(X_j)) | A_i = a, A_j = b, C_i, C_j] \\ &= \sum_{(a,b) \in \{0,1\}^2} \zeta_{ab} \frac{\mathbb{E}[K(\gamma_0(X_i), \gamma_0(X_j)) 1(A_i = a, C_i) 1(A_j = b, C_j)]}{P(A_i = a, C_i) P(A_j = b, C_j)}, \end{aligned}$$

where again, C_i refers to the conditioning event in the fairness definition. For instance, in Equal Opportunity $C_i = \{Y_i = 1\}$ and in Conditional Statistical Parity $C_i = \{X_i = x\}$. $\zeta_{ab} = 1$ if $a = b$ and $\zeta_{ab} = -1$ otherwise. To shorten notation let $K_{ij}(\gamma_0) \equiv K(\gamma_0(X_i), \gamma_0(X_j))$. If we knew γ_0 , we could simply estimate this quantity by the following ratio of U-statistics:

$$\tilde{\theta}_C = \sum_{(a,b) \in \{0,1\}^2} \zeta_{ab} \frac{\sum_{i < j} K_{ij}(\gamma_0) 1(A_i = a, C_i) 1(A_j = b, C_j)}{\sum_{i < j} 1(A_i = a, C_i) 1(A_j = b, C_j)}.$$

If γ_0 is unknown and we have some estimator $\hat{\gamma}$, we can just create a plug-in estimator by plugging $\hat{\gamma}$ in the above expression. However, there can be an effect of estimating γ_0 . Here

we provide the orthogonal moment and debiased estimator for the case of the popular kernel $K(u, u') = e^{-0.5(u-u')^2}$. At the end of this section, we show the construction of the orthogonal moments for a general kernel and for this particular kernel. Let $p_{a,C}(X_i) = P(A_i = a, C_i|X_i)$ and let also $\Delta_{ij}(\gamma_0) = \gamma_0(X_i) - \gamma_0(X_j)$. The identifying function for a fixed $(a, b) \in \{0, 1\}^2$ is

$$\mathbb{E}\left(K_{ij}(\gamma_0)1(A_i = a)1(A_j = b) - \theta_{0C}^{(ab)}1(A_i = a)1(A_j = b)\right) = 0,$$

The U-FSIF for a fixed $(a, b) \in \{0, 1\}^2$ is given by the following Proposition.

Proposition 4 *Let $K(u, u') = e^{-0.5(u-u')^2}$. Then, the U-FSIF for a fixed $(a, b) \in \{0, 1\}^2$ is*

$$\phi^{(ab)}(W_i, W_j, \gamma_0, \alpha_0) = \alpha_0^{(ab)}(X_i, X_j)(Y_i - Y_j - \gamma_0(X_i) + \gamma_0(X_j)),$$

where $\alpha_0^{(ab)}(X_i, X_j) = -K_{ij}(\gamma_0)(\gamma_0(X_i) - \gamma_0(X_j))P(A_i = a, C_i|X_i)P(A_j = b, C_j|X_j)$.

In table 5 we summarize the different $P(A_i = a, C_i|X_i)$ for different fairness definitions. Whenever A_i is included in X_i , which is the typical case, $P(A_i = a, C_i|X_i)$ is known or known up to γ_0 in the case of Equal Opportunity and Probabilistic Predictive Equality. Hence, in this case, we do not need to estimate additional nuisance parameters. However, if A_i is not included in X_i , we need to estimate $P(A_i = a, C_i|X_i)$.

The debiased estimator is

$$\hat{\theta}_C = \sum_{(a,b) \in \{0,1\}^2} \zeta_{ab} \hat{\theta}_C^{(ab)},$$

where

$$\hat{\theta}_C^{(ab)} = \frac{\sum_{l=1}^L \sum_{(i,j) \in I_l} K_{ij}(\hat{\gamma}_l)1(A_i = a, C_i)1(A_j = b, C_j) + \hat{\alpha}_l^{(ab)}(X_i, X_j)(Y_i - Y_j - \hat{\gamma}_l(X_i)) + \hat{\gamma}_l(X_j)}{\sum_{i < j} 1(A_i = a, C_i)1(A_j = b, C_j)}.$$

The asymptotic variance follows from the asymptotic properties of $\{\hat{\theta}_C^{(ab)}\}_{(a,b) \in \{0,1\}^2}$, which follow under the conditions given in this paper, and the Delta Method. For further detail see below. One problem with MMD measures is that they are hard to interpret whenever they are different from zero. To alleviate this concern we propose to measure MMD relative to the MMD in the most unfair case. We take the most unfair case to be such that $\gamma(X) = A$. For example, a bank gives loans to anyone with $A = 1$ and no one with $A = 0$ regardless of any other characteristics¹¹. For the kernel we are using, this maximum unfairness (regardless of the fairness definition) is

$$D_M \equiv D^2(F_{A|A=1}, F_{A|A=0}) = 2 - 2e^{-0.5} \approx 0.7869.$$

¹¹It does not matter if loans are only given to all those with $A = 1$ or all those with $A = 0$.

Hence, we define the relative MMD as

$$\theta_C^R = \frac{\theta_C}{D_M}.$$

9 Appendices

A On Uniqueness of the U-FSIF and the FSIF

The U-FSIF is unique up to the addition of functions with mean zero with respect to K_H for any H . This follows since, for any function $\xi(w_i, w_j)$ such that $\int \int \xi(w_i, w_j) K_H(dw_i, dw_j) = 0$,

$$\int \int [\phi(w_i, w_j, \gamma, \alpha) + \xi(w_i, w_j)] K_H(dw_i, dw_j) = \int \int \phi(w_i, w_j, \gamma, \alpha) K_H(dw_i, dw_j), \quad (9.1)$$

i.e. if ϕ is a U-FSIF so is $\phi + \xi$. If no restrictions are placed on the alternative distribution H , such functions $\xi(w_i, w_j)$ are called degenerate kernels (see, e.g., [Lee \(2019\)](#)).

For any symmetric U-FSIF ϕ , we define $\phi_1(W_i, \gamma, \alpha) = \int \phi(W_i, w, \gamma, \alpha) F_0(dw)$. Then, we can decompose any such $\phi(W_i, W_j, \gamma, \alpha)$ with mean zero as

$$\phi(W_i, W_j, \gamma, \alpha) = \phi_1(W_i, \gamma, \alpha) + \phi_1(W_j, \gamma, \alpha) + \xi(W_i, W_j, \gamma, \alpha),$$

where $\xi(w_i, w_j, \gamma, \alpha) = \phi(w_i, w_j, \gamma, \alpha) - \phi_1(w_i, \gamma, \alpha) - \phi_1(w_j, \gamma, \alpha)$ is a degenerate kernel. Hence,

$$\int \int \phi(w_i, w_j, \gamma, \alpha) K_H(dw_i, dw_j) = \int 2\phi_1(w, \gamma, \alpha) H(dw), \quad (9.2)$$

where the right hand side (RHS) coincides with the characterization of $2\phi_1$ as the FSIF in [Chernozhukov et al. \(2022\)](#). Thus, (9.1) shows that U-FSIFs are unique only up to the addition of degenerate kernels, while (9.2) shows all U-FSIFs give rise to the same FSIF (hence same first-order asymptotics).

An alternative approach based only on linear moments and FSIFs is to let $\bar{g}(w_i, \gamma_0, F_0, \theta) \equiv \int g(w_i, w_j, \gamma_0, \theta) F_0(dw_j)$ be the identifying moment function with unknown nuisance parameters (γ_0, F_0) . The derivative with respect to the first steps in this case is by the chain rule

$$\frac{d}{d\tau} \mathbb{E}[\bar{g}(W_i, \gamma(F_\tau), F_\tau, \theta)] = \frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma(F_\tau), \theta)] + \frac{d}{d\tau} \mathbb{E}[\bar{g}(W_i, \gamma(F_0), F_\tau, \theta)].$$

The first term of the RHS can be characterized by Equation (2.2) in the main text and the second term implies additional nuisance parameters. Hence, this alternative approach does not lead to a simpler methodology or simpler estimators than ours and it relies on our derivations for its completion. Thus, we recommend following instead our methods based on U-FSIFs.

The discussion about uniqueness also helps in constructing a U-FSIF from the FSIF:

symmetrizing $2\phi_1$, i.e. by computing $\phi_1^*(w_i, w_j, \gamma, \alpha) = \phi_1(w_i, \gamma, \alpha) + \phi_1(w_j, \gamma, \alpha)$. However, we have found that the construction of U-FSIF that we provide leads to simpler estimators such as our debiased IOp estimator.

REMARK (No first step estimation effect): A condition for the first step not having an effect on the first order asymptotics for the parameter of interest is $\phi_1(\cdot, \gamma_0, \alpha_0) \equiv 0$, i.e. that the U-FSIF is degenerate. We have shown that this condition does not hold in a number of applications, thereby invalidating inferences that do not account for first steps in standard errors. ■

Another alternative approach would be to consider the random vector $(W_i, W_j) \sim \tilde{F}_0$ and characterize the FSIF as in [Chernozhukov et al. \(2022\)](#)

$$\frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma_\tau, \theta)] = \int \int \phi(w_i, w_j, \gamma_0, \alpha_0) \tilde{H}(dw_i, dw_j),$$

where $\tilde{F}_\tau = \tilde{F}_0 + \tau(\tilde{H} - \tilde{F}_0)$. Firstly, this seems to be a bit artificial. The difference with our approach is that we are deviating the joint distribution on (W_i, W_j) towards arbitrary joint distributions in which W_i and W_j might not be either independent or identically distributed. Hence, it considers deviations towards distributions that do not satisfy our data-generating process sampling design. On top of this, complications arise when computing the influence function under this approach. For simplicity of exposition assume that Assumption 3.2 holds with $M = 1$ and $(c_{11}, c_{21}) = (1, -1)$. A key step in the construction of the U-FSIF (see proof of Lemma 1) is to use the fact that the first step is a projection to get that $\mathbb{E}_\tau[\delta(X_i, X_j)(\gamma_\tau(X_i) - \gamma_\tau(X_j))]$ is equal to $\mathbb{E}_\tau[\delta(X_i, X_j)(Y_i - Y_j)]$. In this approach

$$\mathbb{E}_{\tilde{F}_\tau}(\delta(X_i, X_j)(Y_i - Y_j)) = \mathbb{E}_{\tilde{F}_\tau}(\delta(X_i, X_j)(\mathbb{E}_{\tilde{F}_\tau}(Y_i|X_i, X_j) - \mathbb{E}_{\tilde{F}_\tau}(Y_j|X_i, X_j)))$$

which is not equal in general to $\mathbb{E}_{\tilde{F}_\tau}(\delta(X_i, X_j)(\mathbb{E}_{\tilde{F}_\tau}(Y_i|X_i) - \mathbb{E}_{\tilde{F}_\tau}(Y_j|X_j)))$ since under $\tilde{F}_\tau$ (Y_i, X_j) might be dependent. A final alternative would be to consider deviations where (W_i, W_j) are independent but the distribution of W_i is perturbed towards H while the distribution of W_j is perturbed towards another distribution G . In this case, we get an alternative characterization of the U-FSIF as

$$\frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma_\tau, \theta)] = \int \int \phi(w_i, w_j, \gamma_0, \alpha_0)(H(dw_i)F_0(dw_j) + F_0(dw_i)G(dw_j)).$$

Also, the right-hand side above would be equal to $\int \phi_1(w, \gamma_0, \alpha_0)H(dw) + \int \phi_1(w, \gamma_0, \alpha_0)G(dw)$. This approach is also artificial since W_j is a copy of W_i , furthermore, the set of solutions to the functional equation characterizing the U-FSIF under this approach is the same as in ours so nothing changes in practice.

B Simple example and summary of construction of U-FSIFs

In this section, we go through the process of constructing a U-FSIF for the variance of $\gamma_0(X_i)$ which can be written as

$$\theta_0 = \mathbb{E} \left[\frac{(\gamma_0(X_i) - \gamma_0(X_j))^2}{2} \right].$$

While interesting in its own right, this example is rather arbitrary since we could simply apply the results in [Chernozhukov et al. \(2022\)](#) to $\theta_0 = \mathbb{E}[\gamma_0(X_i)^2] - \mathbb{E}[Y_i]^2$. Still, it is a very useful example to illustrate the construction of U-FSIFs. By an application of the chain rule

$$\frac{d}{d\tau} \mathbb{E} \left[\frac{(\gamma_\tau(X_i) - \gamma_\tau(X_j))^2}{2} - \theta \right] = \frac{d}{d\tau} \mathbb{E}[(\gamma_0(X_i) - \gamma_0(X_j))(\gamma_\tau(X_i) - \gamma_\tau(X_j))] \quad (9.3)$$

Hence, in this case we have $M = 1$, $\delta_{ij,1}(\gamma_0) = \gamma_0(X_i) - \gamma_0(X_j)$ and $(c_{11}, c_{21}) = (1, -1)$. Since $\delta_{ij}(\gamma_0)$ is linear in γ_0 and $\mathcal{S}$ contains $\Gamma + \Gamma$, we also have that $\alpha_0(X_i, X_j) = \gamma_0(X_i) - \gamma_0(X_j)$ regardless of the choice of $\mathcal{S}$. Now, we have that

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}[\alpha_0(X_i, X_j)(\gamma_\tau(X_i) - \gamma_\tau(X_j))] &= \frac{d}{d\tau} \mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(\gamma_\tau(X_i) - \gamma_\tau(X_j))] \\ &\quad - \frac{d}{d\tau} \mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(\gamma_0(X_i) - \gamma_0(X_j))] \\ &= \frac{d}{d\tau} \mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(Y_i - Y_j)] \\ &\quad - \frac{d}{d\tau} \mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(\gamma_0(X_i) - \gamma_0(X_j))] \\ &= - \int \int \alpha_0(x_i, x_j)(y_i - y_j - \gamma_0(x_i) + \gamma_0(x_j)) K_H(dw_i, dw_j), \end{aligned}$$

where in the first equality we use the chain rule and in the second equality we use the fact that $\gamma_\tau(X_i)$ is a projection of Y_i on Γ (and hence $\gamma_\tau(X_i) - \gamma_\tau(X_j)$ is a projection of $Y_i - Y_j$ on any $\mathcal{S}$ containing $\Gamma + \Gamma$). K_H comes from differentiating F_τ with respect to τ . Thus, the U-FSIF is

$$\phi(W_i, W_j, \gamma_0, \alpha_0, \theta) = \alpha_0(X_i, X_j) (Y_i - Y_j - \gamma_0(X_i) + \gamma_0(X_j)). \quad (9.4)$$

We can also exploit the additivity of expectations and write

$$\mathbb{E} \left[\frac{(\gamma_\tau(X_i) - \gamma_\tau(X_j))^2}{2} - \theta \right] = \frac{d}{d\tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0) \gamma_\tau(X_i)] - \frac{d}{d\tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0) \gamma_\tau(X_j)]$$

to obtain the alternative U-FSIF with $M = 2$, $c_{11} = 1$, $c_{21} = 0$, $c_{12} = 0$, $c_{22} = -1$, $\alpha_{01}(X_i, X_j) \equiv \alpha_{01}(X_i) = \gamma_0(X_i) - \mathbb{E}[Y_i]$ and $\alpha_{02}(X_i, X_j) \equiv \alpha_{02}(X_j) = \mathbb{E}[Y_i] - \gamma_0(X_j)$.

$$\phi(W_i, W_j, \gamma_0, \alpha_0, \theta) = \alpha_{01}(X_i) (Y_i - \gamma_0(X_i)) - \alpha_{02}(X_j) (Y_j - \gamma_0(X_j)). \quad (9.5)$$

For a practitioner with a given semiparametric quadratic moment in mind, we ease the process of finding the U-FSIF by summing up the whole process for the general case and for our running example in Figures 5 and 6.

To complete the discussion in the previous section about uniqueness note that in this example ϕ_1 is

$$\phi_1(W_i, \gamma, \alpha) = [\gamma_0(X_i) - \mathbb{E}[Y_i]] \times (Y_i - \gamma_0(X_i)). \quad (9.6)$$

and

$$\xi(w_i, w_j) = (\mathbb{E}[Y_j] - \gamma_0(x_j)) (y_i - \gamma_0(x_i)) - (\mathbb{E}[Y_i] - \gamma_0(x_i)) (y_j - \gamma_0(x_j)), \quad (9.7)$$

which is a degenerate kernel. $\phi_1 = 0$ only when γ_0 is constant. Thus, only when $\gamma_0(\cdot)$ is a constant there is no estimation effect from the first steps. When the fitted value is not constant, e.g. under IOp, there is an estimation effect from the first step, inference not accounting for the first step is invalid and valid inference on the variance of fitted values can be based on orthogonal quadratic moments proposed here.

Country	Most important circumstance	Relative importance	CI
Austria	Living with the father	0.026	(0.022, 0.03)
Belgium	Father's education	0.038	(0.036, 0.04)
Bulgaria	Father's occupation	0.031	(0.027, 0.035)
Switzerland	Financial situation	0.051	(0.049, 0.053)
Cyprus	Population	0.017	(0.011, 0.023)
Czechia	Mother's education	0.044	(0.042, 0.046)
Germany	Father's occupation	0.131	(0.127, 0.135)
Denmark	Number of adults	0.689	(0.681, 0.697)
Estonia	Mother's education	0.064	(0.06, 0.068)
Greece	Country of birth	0.029	(0.027, 0.031)
Spain	Father's occupation	0.046	(0.044, 0.048)
Finland	Country of birth	0.057	(0.049, 0.065)
France	Father's occupation	0.089	(0.085, 0.093)
Croatia	Mother's education	0.033	(0.031, 0.035)
Hungary	Mother's education	0.126	(0.124, 0.128)
Ireland	Holidays	0.058	(0.054, 0.062)
Italy	Father's occupation	0.022	(0.02, 0.024)
Lithuania	Population	0.098	(0.092, 0.104)
Luxembourg	Country of birth	0.079	(0.077, 0.081)
Latvia	Number of adults	0.026	(0.022, 0.03)
Malta	Father's managerial status	0.075	(0.073, 0.077)
Netherlands	Father's occupation	0.12	(0.116, 0.124)
Norway	Father's managerial status	0.21	(0.202, 0.218)
Poland	Father's occupation	0.038	(0.036, 0.04)
Portugal	Father's occupation	0.052	(0.048, 0.056)
Romania	Population	0.037	(0.033, 0.041)
Serbia	Holidays	0.072	(0.068, 0.076)
Sweden	Financial situation	0.255	(0.249, 0.261)
Slovakia	Mother's education	0.053	(0.051, 0.055)

Table 3: Relative importance of most important circumstance in each country. The relative importance of a circumstance is computed by estimating the relative effect of dropping that circumstance.

Fairness Definition	For all $t \in \mathbb{R}$ and some $x \in \mathcal{X}$ it holds that
Statistical Parity	$\Pr(\gamma_0(X) \leq t A = 1) - \Pr(\gamma_0(X) \leq t A = 0) = 0$
Equal Opportunity	$\Pr(\gamma_0(X) \leq t A = 1, Y = 1) - \Pr(\gamma_0(X) \leq t A = 0, Y = 1) = 0$
Probabilistic Predictive Equality	$\Pr(\gamma_0(X) \leq t A = 1, Y = 0) - \Pr(\gamma_0(X) \leq t A = 0, Y = 0) = 0$
Risk Parity	$\Pr(L(\gamma_0(X), Y) \leq t A = 1) - \Pr(L(\gamma_0(X), Y) \leq t A = 0) = 0$
Conditional Statistical Parity	$\Pr(\gamma_0(X) \leq t A = 1, X = x) - \Pr(\gamma_0(X) \leq t A = 0, X = x) = 0$
Equalized Odds	Equal Opportunity and Probabilistic Predictive Equality

Table 4: Different fairness definitions.

	$P(A_i = a, C_i X_i)$	
	A_i in X_i	A_i not in X_i
Statistical Parity	$1(A_i = a)$	$\mathbb{P}(A_i = a X_i)$
Equal Opportunity	$1(A_i = a)\gamma_0(X_i)$	$\mathbb{P}(A_i = a, Y_i = 1 X_i)$
Probabilistic Predictive Equality	$1(A_i = a)(1 - \gamma_0(X_i))$	$\mathbb{P}(A_i = a, Y_i = 0 X_i)$
Conditional Statistical Parity	$1(A_i = a)1(X_i = x)$	$\mathbb{P}(A_i = a X_i)1(X_i = x)$

Table 5: $p_{a,C}(X_i)$ for different fairness definitions.

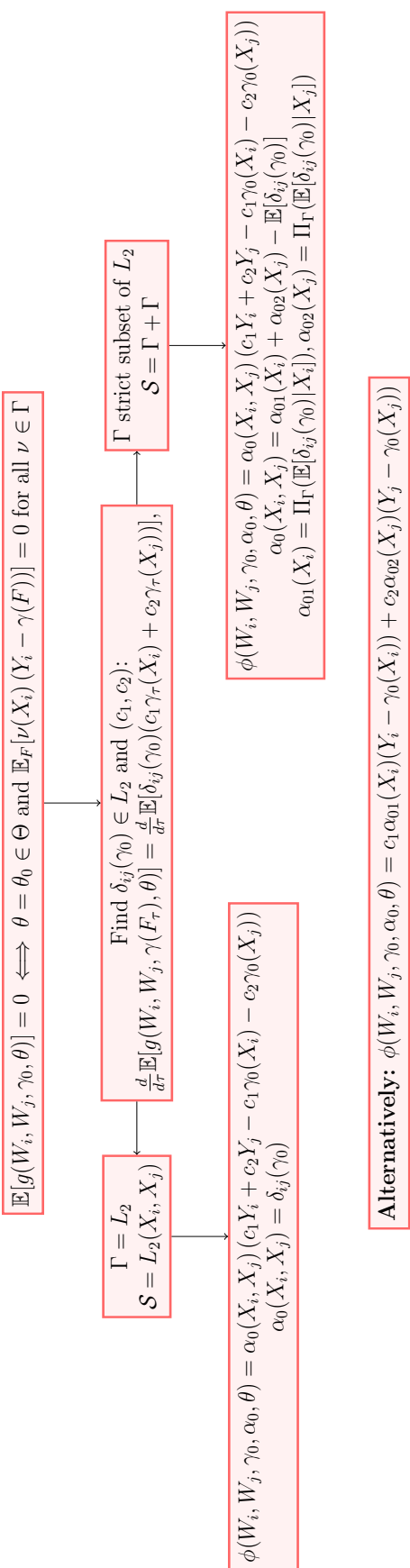


Figure 5: How to find U-FSIFs for general quadratic moment condition.

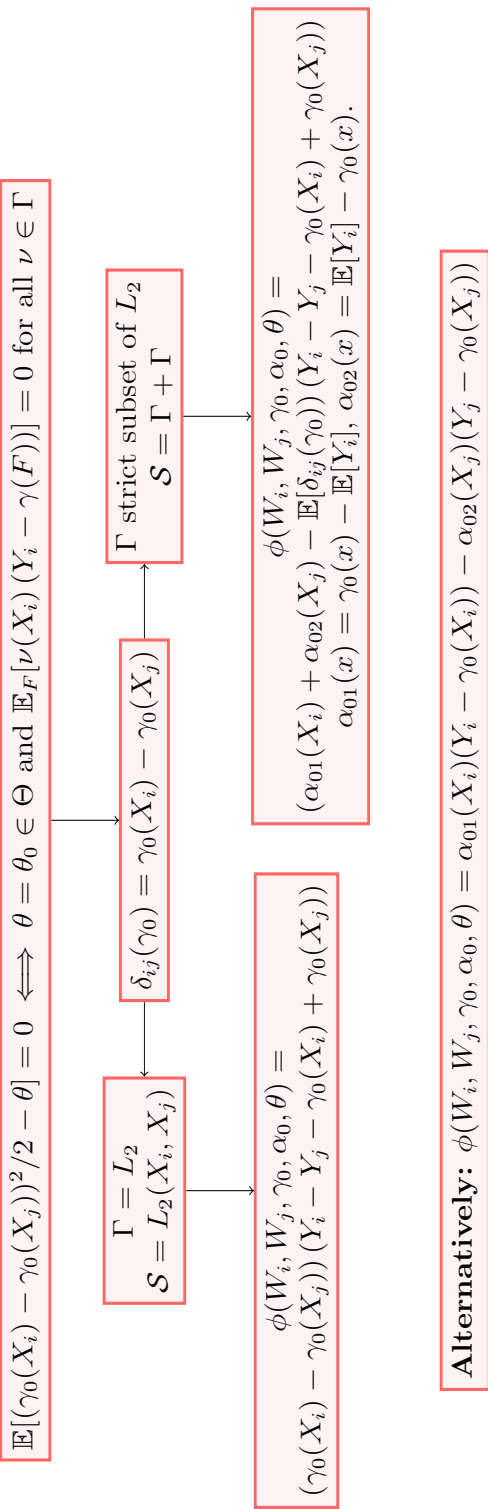


Figure 6: How to find U-FSIFs for the variance of the fitted values.

C Proofs of General Results

C.1 Main results

We prove all results for the case in which $M = 1$ in (3.2) to not carry the subscript m and the summation $\sum_{m=1}^M$ in the notation. The general case follows by applying the same arguments for each m in $\{1, \dots, M\}$.

Proof of Lemma 1: Note that

$$\mathbb{E}[\delta(W_i, W_j, \gamma_0)(c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j))] = \mathbb{E}[\alpha_0(W_i, W_j, \gamma_0)(c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j))],$$

because

$$\mathbb{E}[(\delta(X_i, X_j, \gamma_0) - \alpha_0(X_i, X_j))(c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j))] = 0,$$

$\alpha_0(X_i, X_j) = \Pi_{\mathcal{S}}(\delta(X_i, X_j, \gamma_0))$ and $c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j) \in \Gamma + \Gamma \subseteq \mathcal{S}$. Since $\alpha_0(\cdot, x) \in \Gamma$ and $\alpha_0(x, \cdot) \in \Gamma$ for all $x \in \mathcal{X}$,

$$\mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(c_1(Y_i - \gamma_\tau(X_i)) + c_2(Y_j - \gamma_\tau(X_j)))] = 0.$$

Thus, by the chain rule and because $\alpha_0 \in \mathcal{S}$,

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}[g(W_i, W_j, \gamma(F_\tau), \theta)] &= \frac{d}{d\tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0)(c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j))] \\ &= \frac{d}{d\tau} \mathbb{E}[\alpha_0(X_i, X_j)(c_1\gamma_\tau(X_i) + c_2\gamma_\tau(X_j))] \\ &= \frac{d}{d\tau} \mathbb{E}_{F_\tau}[\alpha_0(X_i, X_j)(c_1Y_i + c_2Y_j - c_1\gamma_0(X_i) - c_2\gamma_0(X_j))] \\ &= \int \int \phi(w_i, w_j, \gamma_0, \alpha_0, \theta) K_H(dw_i, dw_j) \end{aligned}$$

with $\phi(w_i, w_j, \gamma_0, \alpha_0, \theta) = \alpha_0(x_i, x_j)(c_1y_i + c_2y_j - c_1\gamma_0(x_i) - c_2\gamma_0(x_j))$, which gives the desired result. ■

Proof of Lemma 2: The proof of (i) follows from $\alpha = \delta$. For (ii), use the short notation $\delta_{ij}(\gamma) \equiv \delta(X_i, X_j, \gamma)$, and note that by iterated expectations and independence, we can write

$$\begin{aligned} \delta_{ij}(\gamma) &= \mathbb{E}[\delta_{ij}(\gamma)|X_i] + \mathbb{E}[\delta_{ij}(\gamma)|X_j] - \mathbb{E}[\delta_{ij}(\gamma)] + U_{ij}, \\ &\equiv \tilde{\delta}_{ij}(\gamma) + U_{ij}, \end{aligned}$$

where $\mathbb{E}[U_{ij}|X_i] = \mathbb{E}[U_{ij}|X_j] = 0$. This expansion implies that only $\tilde{\delta}_{ij}$ matters for the derivative in (3.2), so δ_{ij} could be replaced everywhere by $\tilde{\delta}_{ij}$. Thus, substituting δ_{ij} from

the last display in the expression for α_0 we obtain

$$\begin{aligned}\alpha_0(X_i, X_j) &= \Pi_{\mathcal{S}}(\delta(X_i, X_j, \gamma_0)) \\ &= \Pi_{\mathcal{S}}\mathbb{E}[\delta_{ij}(\gamma_0)|X_i] + \Pi_{\mathcal{S}}\mathbb{E}[\delta_{ij}(\gamma_0)|X_j] - \mathbb{E}[\delta_{ij}(\gamma)] \\ &= \Pi_{\Gamma}\mathbb{E}[\delta_{ij}(\gamma_0)|X_i] + \Pi_{\Gamma}\mathbb{E}[\delta_{ij}(\gamma_0)|X_j] - \mathbb{E}[\delta_{ij}(\gamma)],\end{aligned}$$

where last expression uses that $\mathcal{S} = \Gamma + \Gamma$. ■

D Asymptotic theory proofs

Proof of Lemma 3: Define

$$\begin{aligned}\hat{R}_{1,ij,l} &= g(W_i, W_j, \hat{\gamma}_l, \theta_0) - g(W_i, W_j, \gamma_0, \theta_0), \quad \hat{R}_{2,ij,l} = \phi(W_i, W_j, \hat{\gamma}_l, \alpha_0, \theta_0) - \phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0), \\ \hat{R}_{3,ij,l} &= \phi(W_i, W_j, \gamma_0, \hat{\alpha}_l, \theta_0) - \phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0), \quad (i, j) \in I_l.\end{aligned}$$

Then

$$g(W_i, W_j, \hat{\gamma}_l, \theta_0) + \phi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta_0) - \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) = \hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} + \hat{R}_{3,ij,l} + \hat{\xi}_l(W_i, W_j). \quad (9.8)$$

Let N_l^c be the observations not in I_l and note that $\hat{\gamma}_l$ and $\hat{\alpha}_l$ depend only on N_l^c . Hence,

$$\mathbb{E}[\hat{R}_{1,ij,l}|N_l^c] = \int \int g(w_i, w_j, \hat{\gamma}_l, \theta_0) F_0(dw_i) F_0(dw_j), \quad (9.9)$$

$$\mathbb{E}[\hat{R}_{2,ij,l}|N_l^c] = \int \int \phi(w_i, w_j, \hat{\gamma}_l, \alpha_0, \theta_0) F_0(dw_i) F_0(dw_j), \quad (9.10)$$

$$\mathbb{E}[\hat{R}_{3,ij,l}|N_l^c] = \int \int \phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta_0) F_0(dw_i) F_0(dw_j) = 0, \quad (9.11)$$

where the two first equalities follow since $\mathbb{E}[g(W_i, W_j, \gamma_0, \theta_0)] = 0$ and $\mathbb{E}[\phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)] = 0$ and in the third equality we use Assumption 5(i). Since pairs in I_l are dependent only when one or two of the members of the pair coincide (also we omit the fact that we are dealing with vectors since the convergence of the vector is the convergence of its elements)

$$\begin{aligned}\mathbb{E}\left[\left(\sqrt{n}\binom{n}{2}^{-1} \sum_{(i,j) \in I_l} (\hat{R}_{1,ij,l} - \mathbb{E}(\hat{R}_{1,ij,l}|N_l^c))\right)^2 \middle| N_l^c\right] = \\ n\binom{n}{2}^{-2} \left[\kappa_{2,l} \text{Var}(\hat{R}_{1,ij,l}|N_l^c) + \kappa_{1,l} \text{Cov}(\hat{R}_{1,ij,l}, \hat{R}_{1,ik,l}|I_l^c) \right],\end{aligned}$$

where $\kappa_{r,l}$ is the number of ways of choosing a pair of pairs in I_l that have $r = 1, 2$, elements in common. These depend on the way you partition the data but generally $n\binom{n}{2}^{-2} \kappa_{2,l} \rightarrow 0$ as

$n \rightarrow \infty$ and $n \binom{n}{2}^{-2} \kappa_{1,l} \leq 4$. For instance, with our partition (see Supplementary Material), $\kappa_{1,l} = \binom{|C_k|}{2} 2(|C_k| - 2)$ and $\kappa_{2,l} = \binom{|C_k|}{2}$ for the elements of the partition $\{C_k \times C_k : i < j\}$ for $k = 1, \dots, K$. For the elements of the partition $\{C_k \times C_m : m > k\}$ we have $\kappa_{1,l} = |C_k||C_m|(|C_k| + |C_m|)$ and $\kappa_{2,l} = |C_k||C_m|$. Since $|C_k| \leq n$ for $k = 1, \dots, K$, it follows that $n \binom{n}{2}^{-2} \kappa_{2,l} \rightarrow 0$ as $n \rightarrow \infty$ and $n \binom{n}{2}^{-2} \kappa_{1,l} \leq 4$. Hence

$$\begin{aligned} & n \binom{n}{2}^{-2} \left[\kappa_{2,l} \mathbb{V}ar(\hat{R}_{1,ij,l} | N_l^c) + \kappa_{1,l} \mathbb{C}ov(\hat{R}_{1,ij,l}, \hat{R}_{1,ik,l} | I_l^c) \right] \\ & \leq 4 \mathbb{C}ov(\hat{R}_{1,ij,l}, \hat{R}_{1,ik,l} | N_l^c) + o_P(1) \\ & \leq 4 \sqrt{\mathbb{V}ar(\hat{R}_{1,ij,l} | N_l^c) \mathbb{V}ar(\hat{R}_{1,ik,l} | N_l^c)} + o_P(1) \\ & \leq 4 \sqrt{\mathbb{E}(\hat{R}_{1,ij,l}^2 | N_l^c) \mathbb{E}(\hat{R}_{1,ik,l}^2 | N_l^c)} + o_P(1) \rightarrow_p 0, \end{aligned}$$

where the convergence in probability follows from Assumption 3(i). The same can be shown for $\hat{R}_{2,ij,l}$ with Assumption 3(ii) and for $\hat{R}_{3,ij,l}$ with Assumption 3(iii). By Assumption 5(i) and the conditional Markov inequality

$$\sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} (\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} + \hat{R}_{3,ij,l} - \mathbb{E}[\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} | N_l^c]) \rightarrow_p 0.$$

By equations (9.9) and (9.10), $\mathbb{E}[\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} | N_l^c] = \mathbb{E}[\psi(W_i, W_j, \hat{\gamma}_l, \alpha_0, \theta_0)]$ (with $\hat{\gamma}_l$ fixed due to the conditioning on N_l^c). Therefore, by Assumption 5

$$\left| \sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} \mathbb{E}(\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} | N_l^c) \right| \leq 2\sqrt{n} |\bar{\psi}(\hat{\gamma}_l, \alpha_0, \theta_0)| \rightarrow_p 0.$$

By the triangle inequality

$$\sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} (\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} + \hat{R}_{3,ij,l}) \rightarrow_p 0.$$

Also, by Assumption 4

$$\sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} \hat{\xi}_l(W_i, W_j) \rightarrow_p 0.$$

Hence, by the triangle inequality and equation (9.8)

$$\begin{aligned} & \sqrt{n} \binom{n}{2}^{-1} \sum_{(i,j) \in I_l} \psi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta_0) - \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) \\ &= \sum_{l=1}^L \sqrt{n} (\hat{R}_{1,ij,l} + \hat{R}_{2,ij,l} + \hat{R}_{3,ij,l} + \hat{\xi}_l(W_i, W_j)) \rightarrow_p 0. \end{aligned}$$

■

Proof of Lemma 4: Let $\hat{\psi}_{ij} = \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \hat{\theta})$, $\psi_{ij} = \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)$ and define $\hat{\chi}_i = (n-1)^{-1} \sum_{j \neq i} \hat{\psi}_{ij}$, $\tilde{\chi}_i = (n-1)^{-1} \sum_{j \neq i} \psi_{ij}$ and $\chi_i = \mathbb{E}[\psi_{ij} | W_i]$. Without loss of generality we focus on the scalar case. Note that $\hat{\Sigma} = 4n^{-1} \sum_{i=1}^n \hat{\chi}_i^2$ and we want to show that $\hat{\Sigma} \rightarrow_p 4\mathbb{E}[\chi_i^2] = \Sigma$. To do this note that

$$\frac{1}{n} \sum_{i=1}^n (\hat{\chi}_i - \tilde{\chi}_i + \tilde{\chi}_i)^2 = \frac{1}{n} \sum_{i=1}^n \tilde{\chi}_i^2 + \frac{1}{n} \sum_{i=1}^n (\hat{\chi}_i - \tilde{\chi}_i)^2 + \frac{2}{n} \sum_{i=1}^n (\hat{\chi}_i - \tilde{\chi}_i) \tilde{\chi}_i.$$

The first term in the RHS goes in probability to $\mathbb{E}[\chi_i^2]$ by standard U-statistic theory. By Cauchy-Schwartz the third term can be bounded as

$$\frac{2}{n} \sum_{i=1}^n (\hat{\chi}_i - \tilde{\chi}_i) \tilde{\chi}_i \leq 2 \left(\frac{1}{n} \sum_{i=1}^n (\hat{\chi}_i - \tilde{\chi}_i)^2 \right)^{\frac{1}{2}} \left(\frac{1}{n} \sum_{i=1}^n \tilde{\chi}_i^2 \right)^{\frac{1}{2}}.$$

Hence, if we show that $n^{-1} \sum_{i=1}^n |\hat{\chi}_i - \tilde{\chi}_i|^2 \rightarrow_p 0$ we have that $n^{-1} \sum_{i=1}^n \hat{\chi}_i^2 = \mathbb{E}[\chi_i^2] + o_p(1)$. It follows by the C_r inequality and the assumptions in the lemma that

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n |\hat{\chi}_i - \tilde{\chi}_i|^2 &= \frac{1}{n} \sum_{i=1}^n \left| \frac{1}{n-1} \sum_{j \neq i} [\hat{g}_{ij} - g_{ij} + \hat{\phi}_{ij} - \phi_{ij}] \right|^2 \\ &\leq \frac{2}{n} \sum_{i=1}^n |\hat{g}_{-i} - g_{-i}|^2 + \frac{2}{n} \sum_{i=1}^n |\hat{\phi}_{-i} - \phi_{-i}|^2 \rightarrow_p 0. \end{aligned}$$

■

Proof of Lemma 5: Define

$$\hat{B} = \binom{n}{2}^{-1} \sum_{i < j} \partial \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \bar{\theta}) / \partial \theta, \quad \tilde{B} = \binom{n}{2}^{-1} \sum_{i < j} \partial \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \theta_0) / \partial \theta.$$

By ii), with probability approaching 1

$$\mathbb{E} \left[\binom{n}{2}^{-1} \sum_{i < j} d(W_i, W_j, \hat{\gamma}, \hat{\alpha}) \right] = \mathbb{E}[d(W_i, W_j, \hat{\gamma}, \hat{\alpha})] \leq C,$$

By Markov inequality, $\binom{n}{2}^{-1} \sum_{i < j} d(W_i, W_j, \hat{\gamma}, \hat{\alpha}) = O_p(1)$. So by ii) and $\bar{\theta} \rightarrow_p \theta_0$

$$\begin{aligned} \left| \hat{B} - \tilde{B} \right| &\leq \binom{n}{2}^{-1} \sum_{i < j} \left| \frac{\partial \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \bar{\theta})}{\partial \theta} - \frac{\partial \psi(W_i, W_j, \hat{\gamma}, \hat{\alpha}, \theta_0)}{\partial \theta} \right| \\ &\leq \binom{n}{2}^{-1} \sum_{i < j} d(W_i, W_j, \hat{\gamma}, \hat{\alpha}) |\bar{\theta} - \theta_0|^{1/C} \\ &= O_p(1) o_p(1) \rightarrow_p 0. \end{aligned}$$

It follows from Assumption 6 (iii) and Markov Inequality that

$$\left| \tilde{B} - \binom{n}{2}^{-1} \sum_{i < j} \partial \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) / \partial \theta \right| \rightarrow_p 0$$

and $\binom{n}{2}^{-1} \sum_{i < j} \partial \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) / \partial \theta \rightarrow_p B$ by usual U-statistics theory. Hence, the result follows from the triangle inequality. ■

Proof of Theorem 1: Let $\hat{\theta}(\gamma_0, \alpha_0)$ be the solution to the cross-fitted orthogonal sample moment if (γ_0, α_0) are known. It follows from Lemma 3 that $\hat{\theta}$ and $\hat{\theta}(\gamma_0, \alpha_0)$ are asymptotically equivalent. By the mean value theorem

$$\begin{aligned} 0 &= \sqrt{n} \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_i, W_j, \gamma_0, \alpha_0, \hat{\theta}(\gamma_0, \alpha_0)) \\ &= \sqrt{n} \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) \\ &\quad + \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi_\theta(W_i, W_j, \gamma_0, \alpha_0, \bar{\theta}) \sqrt{n} (\hat{\theta}(\gamma_0, \alpha_0) - \theta_0), \end{aligned}$$

where ψ_θ is the derivative with respect to θ and $\bar{\theta}$ is some intermediate value. Let $Q_n(\theta) \equiv \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_i, W_j, \gamma_0, \alpha_0, \theta)$ and let $\binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi_\theta(W_i, W_j, \gamma_0, \alpha_0, \bar{\theta}) = M_n$. Then,

$$\begin{aligned} \sqrt{n} (\hat{\theta}(\gamma_0, \alpha_0) - \theta_0) &= M_n^{-1} \sqrt{n} Q_n(\theta_0) = - \left(M_n^{-1} - B^{-1} + B^{-1} \right) \sqrt{n} Q_n(\theta_0) \\ &= -B^{-1} \sqrt{n} Q_n(\theta_0) - \left(M_n^{-1} - B^{-1} \right) \sqrt{n} Q_n(\theta_0) \\ &= -B^{-1} \sqrt{n} Q_n(\theta_0) + o_p(1) O_p(1) \rightarrow_d \mathcal{N}(0, V), \end{aligned}$$

where

$$V = B^{-1} \mathbb{V}ar \left(\mathbb{E}[\psi(W_i, W_j, \gamma_0, \alpha_0, \theta_0) | W_i] \right) B'^{-1}.$$

The normality result follows from standard U-statistic theory (see Theorem 12.3 in [Van der Vaart \(2000\)](#)) and Slutsky's lemma. Hence $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow_d \mathcal{N}(0, V)$. Consistency of the variance estimator follows from Lemmas 4 and 5 and standard arguments.

■

E Consistency

Re-define the interaction term in Section 4 as

$$\hat{\xi}_l(w_i, w_j, \theta) = \phi(w_i, w_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta) - \phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta) - \phi(w_i, w_j, \hat{\gamma}_l, \alpha_0, \theta_0) + \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0).$$

Theorem 4 *If (i) $\mathbb{E}[g(W_i, W_j, \gamma_0, \theta)] = 0$ iff $\theta = \theta_0$, (ii) Θ is compact, (iii) $\int \int |g(w_i, w_j, \hat{\gamma}_l, \theta) - g(w_i, w_j, \gamma_0, \theta)| F_0(dw_i) F_0(dw_j) \rightarrow_p 0$ and $\mathbb{E}[|g(W_i, W_j, \gamma_0, \theta)|] < \infty$ for all $\theta \in \Theta$, (iv) there is $C > 0$ and $d(W_i, W_j, \gamma)$ such that for $\|\gamma - \gamma_0\|$ small enough and all $\tilde{\theta}, \theta \in \Theta$*

$$|g(W_i, W_j, \gamma, \tilde{\theta}) - g(W_i, W_j, \gamma, \theta)| \leq d(W_i, W_j, \gamma) |\tilde{\theta} - \theta|^{1/C},$$

and $\mathbb{E}[d(W_i, W_j, \gamma)] < C$ and (v) Assumption 3 (ii), (iii), $\int \int |\hat{\xi}_l(w_i, w_j, \theta)| F_0(dw_i) F_0(dw_j) \rightarrow_p 0$ and $\mathbb{E}[|\phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)|] < \infty$. Then $\hat{\theta} \rightarrow_p \theta_0$.

Proof: Define $\hat{g}(\theta) \equiv \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} g(W_i, W_j, \hat{\gamma}_l, \theta)$ and $\bar{g}(\theta) \equiv \mathbb{E}[g(W_i, W_j, \gamma_0, \theta)]$. It follows from the conditional Markov inequality and (iii) that $\hat{g}(\theta) \rightarrow_p \bar{g}(\theta)$ for all $\theta \in \Theta$. Let $\tilde{\phi}_{ij} \equiv \binom{n}{2}^{-1} \sum_{i < j} \phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)$ and $\hat{\phi}_{ij}(\theta) \equiv \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \phi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta)$. In the notation of Lemma 3, $\hat{\phi}_{ij}(\theta) - \tilde{\phi}_{ij} = \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \hat{R}_{2,ij,l} + \hat{R}_{3,ij,l} + \hat{\xi}_l(W_i, W_j, \theta)$, so $\hat{\phi}_{ij}(\theta) - \tilde{\phi}_{ij} \rightarrow_p 0$ for all $\theta \in \Theta$ by Assumption (v) and the conditional Markov inequality. By consistency of U-statistics we have that $\tilde{\phi}_{ij} \rightarrow_p \mathbb{E}[\phi(W_i, W_j, \gamma_0, \alpha_0, \theta_0)] = 0$ so by the triangle inequality we have that $\hat{\phi}_{ij}(\theta) \rightarrow_p 0$ for all $\theta \in \Theta$. Therefore, defining $\hat{\psi}_{ij}(\theta) \equiv \binom{n}{2}^{-1} \sum_{l=1}^L \sum_{(i,j) \in I_l} \psi(W_i, W_j, \hat{\gamma}_l, \hat{\alpha}_l, \theta)$, we have that $\hat{\psi}_{ij,l}(\theta) = \hat{g}(\theta) + o_p(1)$. By triangle inequality and (iv) we know that with probability approaching one

$$|\hat{g}(\hat{\theta}) - \hat{g}(\theta)| \leq \binom{n}{2}^{-1} \sum_{i < j} |g(W_i, W_j, \hat{\gamma}_l, \hat{\theta}) - g(W_i, W_j, \hat{\gamma}_l, \theta)| \leq \underbrace{\binom{n}{2}^{-1} \sum_{i < j} d(W_i, W_j, \hat{\gamma}_l) |\hat{\theta} - \theta|^{1/C}}_{\equiv \hat{M}_l},$$

and by (iv) and the conditional Markov inequality $\hat{M}_l = O_p(1)$. By Corollary 2.2 in [Newey \(1991\)](#) we have that $\sup_{\theta \in \Theta} |\hat{\psi}(\theta) - \bar{g}(\theta)| = o_p(1)$. We also know that $\bar{g}(\theta)$ is continuous

by (iv). So the conclusion follows from the proof of Theorem 2.6 in [Newey and McFadden \(1994\)](#) applied to the Hájek projection of $\hat{g}(\hat{\theta})$. ■

F Proofs of Inequality of Opportunity

Proof of Proposition 1: We want to compute

$$\frac{\partial}{\partial \tau} \mathbb{E}(g(W_i, W_j, \gamma(F_\tau), \theta)) = \frac{\partial}{\partial \tau} \mathbb{E}[\theta(\gamma_\tau(X_i) + \gamma_\tau(X_j))] - \frac{\partial}{\partial \tau} \mathbb{E}(|\gamma_\tau(X_i) - \gamma_\tau(X_j)|).$$

The first term is already in the form of (3.2) so we can directly apply Lemma 1 to find

$$\phi_1(w_i, w_j, \gamma_0, \theta) = \theta(y_i + y_j - \gamma_0(x_i) - \gamma_0(x_j)).$$

Define short notation $\Delta_0 \equiv \gamma_0(X_i) - \gamma_0(X_j)$. First we find the Gateaux derivative of the map $\Delta \mapsto \mathbb{E}(|\Delta|)$ at Δ_0 for directions $v \in V = \{v : v(x_{ij}) = \tilde{v}(x_i) - \tilde{v}(x_j), \tilde{v} \in \Gamma\}$. We use that $\mathbb{E}(|\Delta|) = \mathbb{E}([1(\Delta \geq 0) - 1(\Delta < 0)]\Delta)$. Consider a deviation in the direction of the function v , $\Delta_0 + tv$, where, under Assumption 1(i) v is such that $\|v\|_\infty = \sup_{x_i, x_j \in \mathcal{X}^2} |v(x_i, x_j)| \leq C$ and $t > 0$, then

$$\begin{aligned} & \left[\mathbb{E} \left([1(\Delta_0 \geq -tv) - 1(\Delta_0 < -tv)] [\Delta_0 + tv] \right) - \mathbb{E}(|\Delta_0|) \right] / t = \\ & \underbrace{\left[\mathbb{E} \left([1(\Delta_0 \geq -tv) - 1(\Delta_0 < -tv)] \Delta_0 \right) - \mathbb{E}(|\Delta_0|) \right] / t}_{(1)} + \underbrace{\mathbb{E} \left([1(\Delta_0 \geq -tv) - 1(\Delta_0 < -tv)] v \right)}_{(2)}. \end{aligned}$$

Note that

$$\left| (2) - \mathbb{E} \left([1(\Delta_0 > 0) - 1(\Delta_0 < 0)] v \right) \right| \leq 2C \mathbb{E}[1(-tC \leq \Delta_0 \leq tC)],$$

so (2) $\rightarrow \mathbb{E} \left([1(\Delta_0 > 0) - 1(\Delta_0 < 0)] v \right)$ by Assumption 1(ii) as $t \downarrow 0$. Alternatively, under 1(iii)

$$\left| (2) - \mathbb{E} \left([1(\Delta_0 > 0) - 1(\Delta_0 < 0)] v \right) \right| \leq 2C \mathbb{E}[1(-tC \leq \Delta_0 \leq tC) 1(X_i \neq X_j)] \rightarrow 0.$$

Now we show that (1) $\rightarrow 0$ as $t \downarrow 0$. We use that $\Delta_0 = \Delta_0^+ - \Delta_0^-$, where $\Delta_0^+ = 1(\Delta_0 \geq 0)\Delta_0$ and $\Delta_0^- = -1(\Delta_0 < 0)\Delta_0$.

$$\begin{aligned}
|(1)| &= \left| \left[\mathbb{E}([1 - 2 \cdot 1(\Delta_0 < -tv)]\Delta_0^+ - [-1 + 2 \cdot 1(\Delta_0 < -tv)]\Delta_0^-) - \mathbb{E}(|\Delta_0|) \right] / t \right| \\
&= \left| \left[-2\mathbb{E}[1(\Delta_0 < -tv)\Delta_0^+] - 2\mathbb{E}[1(\Delta_0 \geq -tv)\Delta_0^-] + \underbrace{\mathbb{E}[\Delta_0^+ + \Delta_0^- - |\Delta_0|]}_{=0} \right] / t \right| \\
&\leq 2 \left| \left[\mathbb{E}(1(\Delta_0 < -tv)1(\Delta_0 \geq 0)\Delta_0) \right] / t \right| + 2 \left| \left[\mathbb{E}(1(\Delta_0 \geq -tv)1(\Delta_0 < 0)\Delta_0) \right] / t \right| \quad (\star).
\end{aligned}$$

In the first equality we use that $\Delta_0 = \Delta_0^+ - \Delta_0^-$ and that $[1(\Delta_0 \geq -tv) - 1(\Delta_0 < -tv)] = 1 - 2 \cdot 1(\Delta_0 < -tv) = -1 + 2 \cdot 1(\Delta_0 \geq -tv)$. In the second equality we note that $\Delta_0^+ + \Delta_0^- = [1(\Delta_0 \geq 0) - 1(\Delta_0 < 0)]\Delta_0 = |\Delta_0|$. In the inequality we use the triangle inequality.

Now we note that $1(\Delta_0 < -tv) \leq 1(\Delta_0 \leq tC)$ and $1(\Delta_0 \geq -tv) \leq 1(\Delta_0 \geq -tC)$, then

$$\begin{aligned}
(\star) &\leq 2 \left| \mathbb{E}[1(\Delta_0 \leq tC)1(\Delta_0 \geq 0)\Delta_0] / t \right| + 2 \left| \mathbb{E}[1(\Delta_0 \geq -tC)1(\Delta_0 < 0)\Delta_0] / t \right| \\
&= 2 \left| \mathbb{E}[1(0 \leq \Delta_0 \leq tC)\Delta_0] / t \right| + 2 \left| \mathbb{E}[1(-tC \leq \Delta_0 < 0)\Delta_0] / t \right|.
\end{aligned}$$

Hence, by Assumption 1(ii) or (iii), $(\star) \rightarrow 0$ and the Gateaux derivative in direction v of $\Delta \mapsto \mathbb{E}(|\Delta|)$ is

$$\mathbb{E} \left([1(\Delta_0 > 0) - 1(\Delta_0 < -0)]v \right).$$

Then,

$$\frac{\partial \mathbb{E}(|\gamma_\tau(X_i) - \gamma_\tau(X_j)|)}{\partial \tau} = \frac{\partial}{\partial \tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0)(\gamma_\tau(X_i) - \gamma_\tau(X_j))],$$

where $\delta(X_i, X_j, \gamma) = \text{sgn}(\gamma(X_i) - \gamma(X_j))$. By Lemma 1 the U-FSIF of the second term is

$$\phi_2(w_i, w_j, \gamma_0) = \alpha_0(x_i, x_j, \gamma_0)(y_i - y_j - \gamma_0(x_i) + \gamma_0(x_j))$$

where $\alpha_0(x_i, x_j, \gamma) = \Pi_{\mathcal{S}}(\delta(x_i, x_j, \gamma))$. Thus, the U-FSIF is

$$\phi(w_i, w_j, \gamma_0, \alpha_0, \theta) = \phi_1(w_i, w_j, \gamma_0, \theta) - \phi_2(w_i, w_j, \gamma_0, \alpha_0).$$

■

Proof of Proposition 2: For Assumption 7 (iii) we have that

$$\begin{aligned}
\mathbb{E}(|\hat{\alpha}_l - \alpha_0|^2 | I_l^c) &= \mathbb{E}(|\text{sgn}(\Delta_{\hat{\gamma}_l}) - \text{sgn}(\Delta_0)|^2 | I_l^c) \\
&\leq 4\mathbb{E}(|\text{sgn}(\Delta_{\hat{\gamma}_l} - \Delta_0)|^2 | I_l^c) \\
&= 4\mathbb{E}(|\text{sgn}(\Delta_{\hat{\gamma}_l} - \Delta_0)| | I_l^c) \\
&= 4\mathbb{E}(|1(\Delta_{\hat{\gamma}_l} - \Delta_0 > 0) - 1(\Delta_{\hat{\gamma}_l} - \Delta_0 < 0)| | I_l^c) \\
&= 4\mathbb{E}(|1(|\Delta_{\hat{\gamma}_l} - \Delta_0| > 0)| | I_l^c) \\
&= 4\mathbb{E}(1 - |1(|\Delta_{\hat{\gamma}_l} - \Delta_0| = 0)| | I_l^c),
\end{aligned}$$

where in the inequality we have used that for all $(x, y) \in \mathbb{R}^2$ we have that $|\text{sgn}(x) - \text{sgn}(y)| \leq 2|\text{sgn}(x - y)|$. Now note that for $x \in \mathbb{R}$ we have that $|x| = 0$ is equivalent to: for all $\varepsilon > 0$ we have that $|x| < \varepsilon$. Therefore, we have that the above expression is equivalent to saying that for all $\varepsilon > 0$

$$\begin{aligned}
4\mathbb{E}(1 - |1(|\Delta_{\hat{\gamma}_l} - \Delta_0| < \varepsilon)| | I_l^c) &= 4\mathbb{E}(|1(|\Delta_{\hat{\gamma}_l} - \Delta_0| \geq \varepsilon)| | I_l^c) \\
&= 4\mathbb{P}(|\Delta_{\hat{\gamma}_l} - \Delta_0| \geq \varepsilon | I_l^c) \\
&\leq 4 \frac{\mathbb{E}[|\Delta_{\hat{\gamma}_l} - \Delta_0|^2]}{\varepsilon^2} \rightarrow_p 0,
\end{aligned}$$

where we have used Markov's inequality and mean square consistency of $\hat{\gamma}_l$. For Assumption 7 (iv) we have that

$$\begin{aligned}
&\sqrt{n} \int \int (\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j))(\hat{\gamma}_l(x_i) - \gamma_0(x_i) - \hat{\gamma}_l(x_j) + \gamma_0(x_j)) F_0(dw_i) F_0(dw_j) \\
&\leq 2\sqrt{n} \sqrt{\int \int |\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j)|^2 F_0(dw_i) F_0(dw_j)} \sqrt{\int |\hat{\gamma}_l(x_i) - \gamma_0|^2 F_0(dw_i)} \\
&= 2\sqrt{n} o_p(n^{-1/4}) o_p(n^{-1/4}) = o_p(1),
\end{aligned}$$

where we have used Cauchy-Schwartz inequality and the fact that we proved above that we can bound $|\hat{\alpha}_l - \alpha_0|$ by $|\hat{\gamma}_l - \gamma_0|$ when showing Assumption 7 (iii) holds. ■

Proof of Proposition 3: We need to show that the general assumptions of the asymptotic theory in Section 4 hold in this example. Assumption 1 (i) follows since using the C_r inequality multiple times and by the reverse triangle inequality

$$\int \int |g(w_i, w_j, \hat{\gamma}_l, \theta_0) - g(w_i, w_j, \gamma_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \leq C \int |\hat{\gamma}_l(x) - \gamma_0(x)|^2 F_0(dx),$$

which converges to zero by Assumption 7 (ii). Similarly,

$$\begin{aligned}
& \int \int |\phi(w_i, w_j, \hat{\gamma}_l, \alpha_0, \theta_0) - \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \\
&= \int \int |(\theta_0 - \alpha_0(x_i, x_j))(\gamma_0(x_i) - \hat{\gamma}_l(x_i) + \gamma_0(x_j) - \hat{\gamma}_l(x_j))|^2 F_0(dw_i) F_0(dw_j) \\
&\leq C \int |\hat{\gamma}_l(x) - \gamma_0(x)|^2 F_0(dw) \rightarrow_p 0,
\end{aligned}$$

where we used the C_r inequality and boundedness of $\theta_0 - \alpha_0(x_i, x_j)$. Hence Assumption 1 (ii) is also satisfied. For Assumption 1 (iii)

$$\begin{aligned}
& \int \int |\phi(w_i, w_j, \gamma_0, \hat{\alpha}_l, \theta_0) - \phi(w_i, w_j, \gamma_0, \alpha_0, \theta_0)|^2 F_0(dw_i) F_0(dw_j) \\
&= \int \int |(y_i - y_j - \gamma_0(x_i) + \gamma_0(x_j))(\alpha_0(x_i, x_j) - \hat{\alpha}_l(x_i, x_j))|^2 F_0(dw_i) F_0(dw_j) \\
&\leq C \int \int |\alpha_0(x_i, x_j) - \hat{\alpha}_l(x_i, x_j)|^2 F_0(dw) \rightarrow_p 0,
\end{aligned}$$

where in the inequality we used Assumption 7 (i) and the convergence to zero follows from Assumption 7 (iii). To check Assumption 2 (i) note that

$$\begin{aligned}
& \sqrt{n} \int \int \hat{\xi}(w_i, w_j) F_0(dw_i) F_0(dw_j) \\
&= \sqrt{n} \int \int (\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j))(\hat{\gamma}_l(x_i) - \gamma_0(x_i) - \hat{\gamma}_l(x_j) + \gamma_0(x_j)) F_0(dw_i) F_0(dw_j) \rightarrow_p 0,
\end{aligned}$$

where the convergence follows by Assumption 7 (iv). Also,

$$\begin{aligned}
& \int \int |\hat{\xi}(w_i, w_j)|^2 F_0(dw_i) F_0(dw_j) \\
&= \int \int (\hat{\alpha}_l(x_i, x_j) - \alpha_0(x_i, x_j))^2 (\hat{\gamma}_l(x_i) - \gamma_0(x_i) - \hat{\gamma}_l(x_j) + \gamma_0(x_j))^2 F_0(dw_i) F_0(dw_j) \rightarrow_p 0,
\end{aligned}$$

where the convergence follows from boundedness of α_0 and by Assumption 7 (ii). Assumption 3 (i) is global robustness and follows from visual inspection. Assumption 3 (ii) follows from Assumption 7 (ii) and by result 1.8.4 in Yamamuro (1974). To use the last result note that the second order Gateaux derivative of $\bar{\psi}(\gamma, \alpha_0, \theta_0)$ with respect to γ for $\|\gamma - \gamma_0\|$ small enough is 0 under Assumption 1 (ii) since then the sgn function has zero slope a.s. For the conditions for the consistency of the variance let $\mathbb{E}_{n,i} = (n-1)^{-1} \sum_{j \neq i}$ and note that

$$\begin{aligned}
\frac{1}{n} \sum_{i=1}^n |\hat{g}_{-i} - g_{-i}|^2 &\leq \frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[\hat{\gamma}(X_i) + \hat{\gamma}(X_j)]|^2 |\hat{\theta} - \theta_0|^2 \\
&\quad + \frac{C\theta_0}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[\hat{\gamma}(X_i) + \hat{\gamma}(X_j) - \gamma_0(X_i) - \gamma_0(X_j)]|^2 \\
&\quad + \frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[\gamma_0(X_i) - \gamma_0(X_j) - \hat{\gamma}(X_i) + \hat{\gamma}(X_j)]|^2.
\end{aligned}$$

Now let us show that each term vanishes. First we note that $(1/n) \sum_{i=1}^n |\mathbb{E}_{n,i}[\hat{\gamma}(X_i) + \hat{\gamma}(X_j)]|^2 = O_p(1)$, let $C_\varepsilon > 0$, then

$$\begin{aligned}
&\mathbb{P} \left[\frac{1}{n} \sum_{i=1}^n \frac{1}{(n-1)^2} \left(\sum_{j \neq i} \hat{\gamma}(X_i) + \hat{\gamma}(X_j) \right)^2 > C_\varepsilon \right] \\
&\leq \frac{\mathbb{E} \left[\frac{1}{(n-1)^2} \left((n-1)\hat{\gamma}(X_i) + \sum_{j \neq i} \hat{\gamma}(X_j) \right)^2 \right]}{C_\varepsilon} \\
&\leq \frac{C \left(\frac{1}{n-1} \mathbb{E}[(\hat{\gamma}(X_i) - \gamma_0(X_i) + \gamma_0(X_i))^2] + \frac{1}{n-1} \mathbb{E} \left[\left(\sum_{j \neq i} \hat{\gamma}(X_j) \right)^2 \right] \right)}{C_\varepsilon} \\
&\leq \frac{C \left(\frac{1}{n-1} C(\mathbb{E}[(\hat{\gamma}(X_i) - \gamma_0(X_i))^2] + \mathbb{E}[\gamma_0(X_i)^2]) + C(\mathbb{E}[(\hat{\gamma}(X_j) - \gamma_0(X_j))^2] + \mathbb{E}[\gamma_0(X_j)^2]) \right)}{C_\varepsilon} \\
&\rightarrow_p \frac{C\mathbb{E}[\gamma_0(X_j)^2]}{C_\varepsilon}.
\end{aligned}$$

The first inequality uses Markov's inequality, the second one uses C_r inequality, the third one uses C_r and Cauchy Schwartz inequalities and the convergence follows from L2 convergence of $\hat{\gamma}$. Hence, for all $\varepsilon > 0$ and sufficiently large n we can choose $C_\varepsilon > 0$ sufficiently large so as to make $\mathbb{P} \left[\frac{1}{n} \sum_{i=1}^n \frac{1}{(n-1)^2} \left(\sum_{j \neq i} \hat{\gamma}(X_i) + \hat{\gamma}(X_j) \right)^2 > C_\varepsilon \right] < \varepsilon$. Hence the first term is $O_p(1)|\hat{\theta} - \theta_0|^2 \rightarrow_p 0$ by consistency of $\hat{\theta}$. Following the same steps we can bound the probability that the second terms exceed some $\varepsilon > 0$ by $\mathbb{E}[(\hat{\gamma} - \gamma_0)^2] \rightarrow_p 0$ so the second

term vanishes. The exact same arguments cause the third term to vanish too. Now

$$\begin{aligned}
\frac{1}{n} \sum_{i=1}^n |\hat{\phi}_{-i} - \phi_{-i}|^2 &\leq \frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[Y_i + Y_j - \hat{\gamma}(X_i) - \hat{\gamma}(X_j)]|^2 |\hat{\theta} - \theta_0|^2 \\
&\quad + \frac{C\theta_0}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[\gamma_0(X_i) + \gamma_0(X_j) - \hat{\gamma}(X_i) - \hat{\gamma}(X_j)]|^2 \\
&\quad + \frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[(\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))(Y_i - Y_j - \hat{\gamma}(X_i) + \hat{\gamma}(X_j))]|^2 \\
&\quad - \frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[\alpha_0(X_i, X_j)(Y_i - Y_j - \hat{\gamma}(X_i) + \hat{\gamma}(X_j))]|^2.
\end{aligned}$$

For the first term note that by the same arguments as before

$$\begin{aligned}
&\mathbb{P}\left(\frac{C}{n} \sum_{i=1}^n |\mathbb{E}_{n,i}[Y_i + Y_j - \hat{\gamma}(X_i) - \hat{\gamma}(X_j)]|^2 > C_\varepsilon\right) \\
&\leq \frac{\mathbb{E}\left[\frac{1}{(n-1)^2} \left(\sum_{j \neq i} Y_i + Y_j - \gamma_0(X_i) - \gamma_0(X_j) + \gamma_0(X_i) - \gamma_0(X_j) - \hat{\gamma}(X_i) - \hat{\gamma}(X_j)\right)^2\right]}{\varepsilon} \\
&\rightarrow_p C\mathbb{E}[(Y_j - \gamma_0(X_j))^2]/C_\varepsilon.
\end{aligned}$$

So the first term is also $O_p(1)|\hat{\theta} - \theta_0|^2 \rightarrow_p 0$. The second term vanishes as we have already shown. For the third term note that

$$\begin{aligned}
&\mathbb{P}\left(\frac{C}{n(n-1)^2} \sum_{i=1}^n \left|\sum_{j \neq i} [(\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))(Y_i - Y_j - \Delta_0 - \Delta_{\hat{\gamma}})]\right|^2 > \varepsilon\right) \\
&\leq \frac{\mathbb{E}\left[\frac{C}{(n-1)^2} \left(\sum_{j \neq i} (\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))(Y_i - Y_j - \Delta_0 + \Delta_0 - \Delta_{\hat{\gamma}})\right)^2\right]}{\varepsilon} \\
&\leq C \frac{\mathbb{E}\left[\frac{C}{(n-1)^2} \left(\sum_{j \neq i} (\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))(Y_i - Y_j - \Delta_0)\right)^2\right]}{\varepsilon} \\
&\quad + C \frac{\mathbb{E}\left[\frac{C}{(n-1)^2} \left(\sum_{j \neq i} (\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))(\Delta_0 - \Delta_{\hat{\gamma}})\right)^2\right]}{\varepsilon}.
\end{aligned}$$

Where we have used Markov's inequality and C_r inequality. For the first term above use Cauchy Schwartz, law of iterated expectations on (X_i, X_j) and a.s. finiteness of $\mathbb{E}[(Y_i - Y_j - \Delta_0)^2 | X_i, X_j]$ to show that it converges in probability to zero by L2 convergence of $\hat{\alpha}$. For the second term above use the fact that $(\alpha_0(X_i, X_j) - \hat{\alpha}(X_i, X_j))$ is bounded and L2 convergence

of $\hat{\gamma}$. Finally, for the fourth term use Markov's inequality as usual, add and subtract Δ_0 and use that

$$\mathbb{E}[\alpha_0(X_i, X_j)(Y_i - Y_j - \Delta_0)] = 0,$$

by global robustness and that $\alpha_0(X_i, X_j) \leq 1$ and L2 convergence of $\hat{\gamma}$ as before. Convergence of the Jacobian follows trivially since $\frac{\partial}{\partial \theta} \psi(w_i, w_j, \gamma, \alpha, \theta) = y_i + y_j$. ■

G Proofs of Dyadic data

Proof of Lemma 6:

$$\begin{aligned} \frac{d}{d\tau} \mathbb{E}[\delta(X_i, X_j, \gamma_0, \theta) \gamma_\tau(X_i, X_j)] &= \frac{d}{d\tau} \mathbb{E}[\alpha_0(X_i, X_j) \gamma_\tau(X_i, X_j)] \\ &= \frac{d}{d\tau} \mathbb{E}_\tau[\alpha_0(X_i, X_j) \gamma_\tau(X_i, X_j)] - \frac{d}{d\tau} \mathbb{E}_\tau[\alpha_0(X_i, X_j) \gamma_0(X_i, X_j)] \\ &= \int \int \alpha_0(x_i, x_j) (y_{ij} - \gamma_0(x_i, x_j)) K_H(dw_{ij}). \end{aligned}$$

■

Proof of Lemma 7: Same proof as that of Lemma 3 but exchanging (W_i, W_j) by (W_{ij}) everywhere. ■

Proof of Lemma 8: The proof follows classic variance computation of U-statistics in [Hoeffding \(1948\)](#). For notational convenience let $\psi_{ij} \equiv \psi(W_{ij}, \gamma_0, \alpha_0, \theta_0)$. Let

$$T_n = \binom{n}{2}^{-1} \psi_{ij} \text{ and } \bar{T}_n = \binom{n}{2}^{-1} s_{ij}.$$

Showing Lemma 8 amounts to showing that

$$R_n \equiv \sqrt{n}(T_n - \bar{T}_n) = o_p(1).$$

First note that $\mathbb{E}[R_n] = 0$. Also, by the fact that $\bar{T}_n$ is a projection of T_n on L_2 , we have that

$$\text{Cov}(T_n, \bar{T}_n) = \text{Var}(\bar{T}_n).$$

Hence,

$$\text{Var}(R_n) = n(\text{Var}(T_n) - \text{Var}(\bar{T}_n)).$$

Now, by counting the number of times that a pair has one or two elements in common we

have that

$$\begin{aligned}\mathbb{V}ar(T_n) &= \binom{n}{2}^{-1} \left(\binom{n}{2} \binom{2}{2} \binom{n-2}{0} \mathbb{C}ov(\psi_{ij}\psi_{ij}) + \binom{n}{2} \binom{2}{1} \binom{n-2}{1} \mathbb{C}ov(\psi_{ij}\psi_{ik}) \right) \\ &= \frac{2}{n(n-1)} \mathbb{C}ov(\psi_{ij}\psi_{ij}) + \frac{4(n-2)}{n(n-1)} \mathbb{C}ov(\psi_{ij}\psi_{ik}),\end{aligned}$$

also, by classical U-statistics asymptotic theory, $\mathbb{V}(\bar{T}_n) = (4/n) \mathbb{C}ov(s_{ij}s_{ik})$. Let $Z_{ij} = (X_i, X_j, A_i, B_i, A_j, B_j)$, then note that

$$\begin{aligned}\mathbb{E}[s_{ij}s_{ik}] &= \mathbb{E}[\mathbb{E}(\psi_{ij}|Z_{ij})\mathbb{E}(\psi_{ik}|Z_{ik})] \\ &= \mathbb{E}[\mathbb{E}(\psi_{ij}\psi_{ik}|Z_{ij}, Z_{ik})] \\ &= \mathbb{E}[\psi_{ij}\psi_{ik}],\end{aligned}$$

where we have used that ψ_{ij} and ψ_{ik} are mean independent given Z_{ij} and Z_{ik} , which is granted by Assumption 9. Hence, finally

$$\mathbb{V}ar(R_n) = n \left(\frac{2}{n(n-1)} \mathbb{C}ov(\psi_{ij}\psi_{ij}) + \frac{4(n-2)}{n(n-1)} \mathbb{C}ov(\psi_{ij}\psi_{ik}) - \frac{4}{n} \mathbb{C}ov(\psi_{ij}\psi_{ik}) \right) \rightarrow 0.$$

Hence, $R_n \rightarrow_p 0$ by Markov's inequality. ■

H Extension to higher order kernels

The extension to semiparametric U-statistics with higher-order kernels is also straightforward. There is only a slight change in the characterization of the U-FSIF. Suppose that for some $k < \infty$ we have the identifying moment condition

$$\mathbb{E}[g(W_{i1}, \dots, W_{ik}, \gamma_0, \theta)] = 0 \text{ iff } \theta = \theta_0,$$

where $W_{i1}, \dots, W_{ik}$ are i.i.d. copies of W_i . The U-FSIF is then characterized by a mean zero condition and the functional equation

$$\frac{d}{d\tau} \mathbb{E}[g(W_{i1}, \dots, W_{ik}, \gamma_\tau, \theta)] = \int \dots \int \phi(w_{i1}, \dots, w_{ik}, \gamma_0, \alpha_0, \theta) dK_H^{(k)},$$

where $K_H^{(k)}(w_{i1}, \dots, w_{ik}) = \sum_{t=1}^k H(w_{it}) \prod_{r \neq t} F_0(w_{ir})$. The cross-fitting in with $k > 2$, although not more conceptually complicated than in the case of $k = 2$, is much more cumbersome. In Appendix I we provide an example of a setting with a higher-order kernel.

I Further Applications

I.1 Partially Linear Dyadic Regression and Sample Selection

Let $Y_{ij} = h(Y_i^*, Y_j^*)$ where Y_{ij} is observed but Y_i^* and Y_j^* might not be observed. For example, in the context of international trade, Y_{ij} could be the trade volume between countries i and j , in migration it could be the net migration flow between two countries. Not knowing h or Y_i^* and Y_j^* , is non-essential for us as long as we observe Y_{ij} .

$$Y_{ij} = \theta_0 D_{ij} + m(X_i, X_j) + \varepsilon_{ij},$$

where D_{ij} is some dyad-level random variable, for example, a binary treatment of whether two countries are in a trade agreement or not. Assume $\mathbb{E}[\varepsilon_{ij}|D_{ij}, X_i, X_j] = 0$. The parameter of interest is θ_0 and the function $m(\cdot)$ is unknown and left unrestricted. Let us compute the U-FSIF of the identifying moment in this model

$$\frac{d}{d\tau} \mathbb{E}[(Y_{ij} - \theta_0 D_{ij} - m_\tau(X_i, X_j)) D_{ij}] = \mathbb{E}[\alpha_0(X_i, X_j) m_\tau(X_i, X_j) D_{ij}],$$

where $\alpha_0(X_i, X_j) = \mathbb{E}[D_{ij}|X_i, X_j]$. Noting that $m(X_i, X_j) = \mathbb{E}[Y_{ij} - \theta_0 D_{ij}|X_i, X_j]$ and employing the usual arguments we find that the U-FSIF is

$$\phi(W_i, W_j, \gamma_0, \alpha_0) = \alpha_0(X_i, X_j)(Y_{ij} - \theta_0 D_{ij} - m(X_i, X_j)).$$

Adding it to the identifying moment function and rearranging gives us an analogous locally robust moment to the one proposed by [Robinson \(1988\)](#) for partially linear models.

$$\psi(W_i, W_j, m, \alpha_0, \theta_0) = (Y_{ij} - \mathbb{E}[Y_{ij}|X_i, X_j] - \theta_0(D_{ij} - \alpha_0(X_i, X_j)))(D_{ij} - \alpha_0(X_i, X_j)).$$

This framework can be applied to models with selection. For example, in a network, we might only observe Y_{ij} for members who form a link, e.g. two companies connected in a supply chain. We might be also interested in jointly modeling the decisions of two countries to trade and the trade volume. In such cases a dyadic regression model with sample selection is appropriate as studied for instance in [Sakamoto \(2024\)](#). Suppose we have now the following model

$$\begin{aligned} Y_{ij} &= (\theta_0 m(X_i, X_j) + \varepsilon_{ij}) S_{ij}, \\ S_{ij} &= 1(b(Z_i, Z_j) > V_{ij}), \end{aligned}$$

where Z_i contains X_i and other variables that determine selection and are not in X_i . If the unobservables $(\varepsilon_{ij}, V_{ij})$ are correlated, we have a selection issue. Since $\mathbb{E}[Y_{ij}|Z_i, Z_j, S_{ij}] =$

1] = $\theta_0 m(X_i, X_j) + f(W_i, W_j)$ for some unknown function f , we can apply the results above to find a debiased estimator for θ_0 .

I.2 Semiparametric Gravity Models with Fixed Effects

In this section, we expand the results in [Jochmans \(2017\)](#) as an example with a higher order kernel. The model now is

$$Y_{ij} = e^{\theta_0 D_{ij}} m(X_i, X_j) A_i B_j U_{ij}.$$

We assume now that $\mathbb{E}[U_{ij}|D_{ij}, X_i, X_j] = 1$. As in [Jochmans \(2017\)](#), we treat the fixed effects as given, i.e. we condition on them throughout. In this model

$$\mathbb{E}[U_{ij}U_{i'j'}|D_{ij}, D_{i'j'}, X_i, X_j, X_{i'}, X_{j'}] = \mathbb{E}[U_{ij'}U_{i'j}|D_{ij'}, D_{i'j}, X_i, X_j, X_{i'}, X_{j'}],$$

which provides us with infinite unconditional moment conditions. Let

$$\Phi_{ij i' j'} = \Phi(D_{ij}, D_{i'j'}, D_{ij'}, D_{i'j}, X_i, X_j, X_{i'}, X_{j'}),$$

for some chosen function Φ , then

$$\mathbb{E} \left[\Phi_{ij i' j'} \left(\frac{Y_{ij} Y_{i'j'}}{e^{\theta_0(D_{ij} + D_{i'j'})} m(X_i, X_j) m(X_{i'}, X_{j'})} - \frac{Y_{ij'} Y_{i'j}}{e^{\theta_0(D_{ij'} - D_{i'j})} m(X_i, X_{j'}) m(X_{i'}, X_j)} \right) \right] = 0.$$

Let $\tilde{m}(X_i, X_j, X_{i'}, X_{j'}) = m(X_i, X_j) m(X_{i'}, X_{j'})$, and note that

$$\tilde{m}(X_i, X_j, X_{i'}, X_{j'}) = \frac{\mathbb{E}[Y_{ij} Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}]}{\mathbb{E}[e^{\theta_0(D_{ij} + D_{i'j'})} | X_i, X_j, X_{i'}, X_{j'}]}.$$

Let us just compute the U-FSIF for the first term above and the second term follows analogously. By the chain rule, we have

$$\begin{aligned} & \frac{d}{d\tau} \mathbb{E} \left[\Phi_{ij i' j'} \frac{Y_{ij} Y_{i'j'} \mathbb{E}_\tau[e^{\theta_0(D_{ij} + D_{i'j'})} | X_i, X_j, X_{i'}, X_{j'}]}{e^{\theta_0(D_{ij} + D_{i'j'})} \mathbb{E}_\tau[Y_{ij} Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}]} \right] \\ &= \frac{d}{d\tau} \mathbb{E} \left[\Phi_{ij i' j'} \frac{Y_{ij} Y_{i'j'} \mathbb{E}_\tau[e^{\theta_0(D_{ij} + D_{i'j'})} | X_i, X_j, X_{i'}, X_{j'}]}{e^{\theta_0(D_{ij} + D_{i'j'})} \mathbb{E}[Y_{ij} Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}]} \right] \\ & - \frac{d}{d\tau} \mathbb{E} \left[\Phi_{ij i' j'} \frac{Y_{ij} Y_{i'j'} \mathbb{E}[e^{\theta_0(D_{ij} + D_{i'j'})} | X_i, X_j, X_{i'}, X_{j'}]}{e^{\theta_0(D_{ij} + D_{i'j'})} \mathbb{E}[Y_{ij} Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}]^2} \mathbb{E}_\tau[Y_{ij} Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}] \right], \end{aligned}$$

so the U-FSIF is (with abuse of notation)

$$\begin{aligned}\phi(W_i, W_j, W_{i'}, W_{j'}, \gamma_0, \alpha_0) &= \alpha_{01}(X_i, X_j, X_{i'}, X_{j'}) (e^{\theta_0(D_{ij}+D_{i'j'})} - \gamma_{01}(X_i, X_j, X_{i'}, X_{j'})) \\ &\quad - \alpha_{02}(X_i, X_j, X_{i'}, X_{j'}) (Y_{ij}Y_{i'j'} - \gamma_{02}(X_i, X_j, X_{i'}, X_{j'})),\end{aligned}$$

where

$$\begin{aligned}\gamma_{01}(X_i, X_j, X_{i'}, X_{j'}) &= \mathbb{E}[e^{\theta_0(D_{ij}+D_{i'j'})} | X_i, X_j, X_{i'}, X_{j'}], \\ \gamma_{02}(X_i, X_j, X_{i'}, X_{j'}) &= \mathbb{E}[Y_{ij}Y_{i'j'} | X_i, X_j, X_{i'}, X_{j'}], \\ \alpha_{01}(X_i, X_j, X_{i'}, X_{j'}) &= \mathbb{E}\left[\frac{\Phi_{i,j,i',j'}Y_{ij}Y_{i'j'}}{e^{\theta_0(D_{ij}+D_{i'j'})}} \middle| X_i, X_j, X_{i'}, X_{j'}\right] \gamma_{02}^{-1}(X_i, X_j, X_{i'}, X_{j'}), \\ \alpha_{02}(X_i, X_j, X_{i'}, X_{j'}) &= \mathbb{E}\left[\frac{\Phi_{i,j,i',j'}Y_{ij}Y_{i'j'}}{e^{\theta_0(D_{ij}+D_{i'j'})}} \middle| X_i, X_j, X_{i'}, X_{j'}\right] \frac{\gamma_{01}(X_i, X_j, X_{i'}, X_{j'})}{\gamma_{02}^2(X_i, X_j, X_{i'}, X_{j'})}.\end{aligned}$$

I.3 Causal Gravity Models

We can also extend the popular AIPW framework in [Robins et al. \(1995\)](#) to causal gravity models or gravity models with missing data. Let $(Y_{ij}(1), Y_{ij}(0))$ be the potential outcomes for dyad (i, j) and D_{ij} be the treatment. Assume that $(Y_{ij}(1), Y_{ij}(0)) \perp D_{ij} \mid X_i, X_j$, let $e(X_i, X_j) = P(D_{ij} = 1 | X_i, X_j)$ with $0 < e(X_i, X_j) < 1$ and let $Y_{ij} = Y_{ij}(1)D_{ij} + Y_{ij}(0)(1 - D_{ij})$. Finally, let $m(D_{ij}, X_i, X_j) = \mathbb{E}[Y_{ij} | D_{ij}, X_i, X_j]$. The parameter of interest is the Average Treatment effect: $\theta_0 = \mathbb{E}[Y_{ij}(1) - Y_{ij}(0)]$. We can compute the U-FSIF by noticing that the identifying moment is linear in $m(D_{ij}, X_i, X_j)$

$$\begin{aligned}\theta_0 &= \mathbb{E}[m(1, X_i, X_j) - m(0, X_i, X_j)] \\ &= \mathbb{E}\left[\left(\frac{D_{ij}}{e(X_i, X_j)} - \frac{1 - D_{ij}}{1 - e(X_i, X_j)}\right) m(D_{ij}, X_i, X_j)\right].\end{aligned}$$

Hence, the orthogonal function is

$$\psi(W_i, W_j, m, \alpha_0, \theta_0) = m(1, X_i, X_j) - m(0, X_i, X_j) + \alpha_0(D_{ij}, X_i, X_j)(Y_{ij} - m(D_{ij}, X_i, X_j)),$$

$$\text{where } \alpha_0(D_{ij}, X_i, X_j) = \left(\frac{D_{ij}}{e(X_i, X_j)} - \frac{1 - D_{ij}}{1 - e(X_i, X_j)}\right).$$

I.4 Fairness technical details

In this example, we have four identifying moment functions, one for each $(a, b) \in \{0, 1\}^2$. It is enough to characterize the U-FSIF for arbitrary $(a, b) \in \{0, 1\}^2$ and then add them up.

The identifying moment for a fixed $(a, b) \in \{0, 1\}^2$ is

$$\mathbb{E}[(K_{ij}(\gamma_0) - \theta_{0,C}^{(ab)})1(A_i = a, C_i)1(A_j = b, C_j)] = 0,$$

Now we need to first assume some smoothness as in (3.2)

Assumption 10 *There exist $\alpha_{ab,p,C,0}$, $P < \infty$, and (c_{1p}, c_{2p}) for $p = 1, \dots, P$, such that for all $(a, b) \in \{0, 1\}^2$ the following linearization holds*

$$\frac{d}{d\tau} \mathbb{E}[K_{ij}(\gamma_\tau)1(A_i = a, C_i)1(A_j = b, C_j)] = \frac{d}{d\tau} \mathbb{E} \left[\sum_{p=1}^P \alpha_{ab,p,C,0}(X_i, X_j)(c_{1p}\gamma_\tau(X_i) + c_{2p}\gamma_\tau(X_j)) \right].$$

Then, by Lemma 1 we have that an orthogonal moment for $\theta_{0,C}^{(ab)}$ is

$$\underbrace{\mathbb{E} \left[(K_{ij}(\gamma_0) - \theta_{0,C}^{(ab)})1(A_i = a, C_i)1(A_j = b, C_j) + \sum_{p=1}^P \alpha_{ab,p,C,0}(X_i, X_j)(c_{1p}(Y_i - \gamma_0(X_i)) + c_{2p}(Y_j - \gamma_0(X_j))) \right]}_{\equiv \psi^{(ab)}(W_i, W_j, \gamma_0, \alpha_{ab,C,0}, \theta_{0,C}^{(ab)})}.$$

A locally robust estimator is

$$\begin{aligned} \hat{\theta}_C^{(ab)} = & \left(\sum_{l=1}^L \sum_{(i,j) \in I_l} K_{ij}(\hat{\gamma}_l)1(A_i = a, C_i)1(A_j = b, C_j) \right. \\ & \left. + \sum_{p=1}^P \hat{\alpha}_{ab,p,C,l}(X_i, X_j)(c_{1p}(Y_i - \hat{\gamma}_l(X_i)) + c_{2p}(Y_j - \hat{\gamma}_l(X_j))) \right) / \sum_{i < j} 1(A_i = a, C_i)1(A_j = b, C_j), \end{aligned}$$

where under the conditions given in this paper

$$\begin{aligned} \sqrt{n}(\hat{\theta}_C^{(ab)} - \theta_{0,C}^{(ab)}) &\rightarrow_d N(0, B^{-2}\sigma^2), \\ B &= P(A_i = a, C_i)P(A_j = b, C_j), \\ \sigma^2 &= \mathbb{V}ar[\mathbb{E}(\psi^{(ab)}(W_i, W_j, \gamma_0, \alpha_{ab,C,0}, \theta_{0,C}^{(ab)}) \mid W_i)]. \end{aligned}$$

Finally, a locally robust estimator for $\theta_{0,C}$ is

$$\hat{\theta}_C = \sum_{(a,b) \in \{0,1\}^2} \zeta_{ab} \hat{\theta}_C^{(ab)},$$

for which the asymptotic properties follow from applying the Delta Method. For the popular kernel $K(u, u') = e^{-0.5(u-u')^2}$, Assumption 10 holds with $P = 1$, $(c_{11}, c_{21}) = (1, -1)$ and $\alpha_{ab,1,C,0} = -K(\gamma(X_i), \gamma(X_j))(\gamma(X_i) - \gamma(X_j))P(A_i = a, C_i|X_i)P(A_j = b, C_j|X_j)$, as shown in

Proposition 4 which is proven below.

Proof of Proposition 4: By repeated application of the chain rule, we have

$$\begin{aligned}
& \frac{d}{d\tau} \mathbb{E}[K(\gamma_\tau(X_i), \gamma_\tau(X_j))1(A_i = a, C_i)1(A_j = b, C_j)] \\
&= \frac{d}{d\tau} \mathbb{E}[K(\gamma_\tau(X_i), \gamma_0(X_j))1(A_i = a, C_i)1(A_j = b, C_j)] \\
&+ \frac{d}{d\tau} \mathbb{E}[K(\gamma_0(X_i), \gamma_\tau(X_j))1(A_i = a, C_i)1(A_j = b, C_j)] \\
&= -\frac{d}{d\tau} \mathbb{E}[K(\gamma_0(X_i), \gamma_0(X_j))(\gamma_0(X_i) - \gamma_0(X_j))1(A_i = a, C_i)1(A_j = b, C_j)(\gamma_\tau(X_i) - \gamma_\tau(X_j))] \\
&= -\frac{d}{d\tau} \mathbb{E}[K(\gamma_0(X_i), \gamma_0(X_j))(\gamma_0(X_i) - \gamma_0(X_j))P(A_i = a, C_i|X_i)P(A_j = b, C_j|X_j)(\gamma_\tau(X_i) - \gamma_\tau(X_j))]
\end{aligned}$$

Then, by Lemma 1, we have that the first step influence function is

$$\phi(W_i, W_j, \gamma_0, \alpha_0) = \alpha_0^{(ab)}(X_i, X_j)(Y_i - Y_j - \gamma_0(X_i) + \gamma_0(X_j)),$$

where

$$\alpha_0^{(a,b)}(X_i, X_j) = -K_{ij}(\gamma_0)(\gamma_0(X_i) - \gamma_0(X_j))P(A_i = a, C_i|X_i)P(A_j = b, C_j|X_j).$$

■

I.5 Causal U-statistics

Using the results in this paper, [Terschuur \(2023\)](#) generalizes the causal estimands proposed in [Mao \(2018\)](#) and [Yin et al. \(2024\)](#). Let $W_i = (Y_i, D_i, X_i)$ and $Z_i = (Y_i(1), Y_i(0))$ where $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. Consider the following family of causal estimands where g is a known function and $\gamma^{(1)}$ and $\gamma^{(0)}$ are potential nuisance parameters (e.g. $\mathbb{E}[Y_i(1)|X_i]$ and $\mathbb{E}[Y_i(0)|X_i]$)

$$\theta_0(\pi) = \mathbb{E} \left[\sum_{(a,b) \in \{0,1\}^2} g(Y_i(a), X_i, Y_j(b), X_j, \gamma^{(a)}, \gamma^{(b)}) \pi_{ab}(X_i, X_j) \right] \text{ with } \pi_{ab} : \mathcal{X} \times \mathcal{X} \rightarrow \{0, 1\}.$$

In [Terschuur \(2023\)](#), $\theta(\pi)$ is a welfare function and π_{ab} is a treatment allocation rule. [Terschuur \(2023\)](#) proposes methods to estimate an optimal allocation rule π_{ab} so as to maximize welfare. However, in the context of causal U-statistics, we can fix a given π_{ab} and interpret $\theta(\pi)$ as a causal estimand. For example, in [Mao \(2018\)](#) and [Yin et al. \(2024\)](#), $\pi_{ab}(X_i, X_j) = 1$

if $(a, b) = (1, 0)$ and $\pi_{ab}(X_i, X_j) = 0$ otherwise. Also, in [Mao \(2018\)](#) and [Yin et al. \(2024\)](#)

$$g(y_i(a), x_i, y_j(b), x_j, \gamma^{(a)}, \gamma^{(b)}) = h(y_i(a), y_j(b)),$$

for some known function h . For such a π_{ab} , we have that under an RCT

$$\theta_0(\pi) = \mathbb{E}[h(Y_i(1), Y_j(0))] = \mathbb{E}[h(Y_i, Y_j) | D_i = 1, D_j = 0],$$

so that we are comparing how a treated group fares compared to a control group in terms of h . For example, if $h(y_i(a), y_j(b)) = y_i(a) - y_j(b)$ we recover the typical average treatment effect, i.e. on average how much better off are the treated compared to the control group. The use of causal U-statistics is better motivated for nonlinear h where typical methods cannot be used, for example, the Wilcoxon statistic is based on $h(y_i(a), y_j(b)) = 1(y_i(a) \geq y_j(b))$. For such an h , we are interested in the probability that the treated are better off than the control. This can be important when most treated fare worse but the average treatment effect is positive due to a few treated that fare much better. In these situations, the average treatment effect might be large but still, the Wilcoxon statistic might be small.

The more general setting in [Terschuur \(2023\)](#) allows for interesting causal estimands whenever we allow π_{ab} to depend non-trivially on some covariates. For instance, let $\pi : \mathcal{X} \rightarrow \{0, 1\}$ and $\pi_{ab}(X_i, X_j) = 1(\pi(X_i) = a, \pi(X_j) = b)$. Suppose, for instance, that $\pi(X_i) = 1$ only for female high school dropouts and zero otherwise, then we can estimate the counterfactual value of θ_0 whenever only female high school dropouts are treated. For example, if we are interested in the Gini Mean Difference (the Gini coefficient follows directly), we can set $h(y_i(a), y_j(b)) = |y_i(a) - y_j(b)|$ and estimate

$$\theta_0(\pi) = \mathbb{E} \left[\sum_{(a,b) \in \{0,1\}^2} |Y_i(a) - Y_j(b)| 1(\pi(X_i) = a, \pi(X_j) = b) \right],$$

hence we can evaluate what would be the resulting inequality whenever certain groups of the population are treated and others are not. We can also estimate what the intergenerational mobility, as measured by the Kendall- τ coefficient, would be if only certain groups were treated by setting

$$g(y_i(a), x_i, y_j(b), x_j, \gamma^{(a)}, \gamma^{(b)}) = \text{sgn}(Y_i(a) - Y_j(b)) \text{sgn}(X_{1i} - X_{1j}),$$

where $Y_i(j)$ for $j = 0, 1$ are the potential permanent incomes of individual i and X_{1i} is the permanent income of the parent of individual i . As a final example, we can also estimate

causal effects on inequality of opportunity by setting

$$g(y_i(a), x_i, y_j(b), x_j, \gamma^{(a)}, \gamma^{(b)}) = |\gamma^{(a)}(x_i) - \gamma^{(b)}(x_j)|,$$

where $\gamma^{(k)}(x_i) = \mathbb{E}[Y_i(k)|X_i = x_i]$ for $k = 0, 1$. Also, using the results in this paper this family of causal estimands can be extended to dyadic settings. For instance, one could analyze the impacts of trade or migration agreements when only certain countries enter into the agreements.

Under typical selection on observables and common support assumptions, [Terschuur \(2023\)](#) provides locally robust moments for such a family of causal U-statistics. While the methodological innovation in [Terschuur \(2023\)](#) is in the estimation of the optimal allocation rule π and its statistical properties, the results on estimating the causal estimands for a given π_{ab} rely completely on the results in this paper.

J Simulations

In the tables below we see the results of the simulations for $\sigma \in \{0.2, 0.3\}$ for such lower StN ratios noisier estimates such as Ridge start malfunctioning and well-performing estimators such as lasso or CIF need more observations before getting good results.

$\sigma = 0.2$			Gini of the FVs					
Plug in Lasso			Debiased Lasso		Plug in Ridge		Debiased Ridge	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.012	0.611	-0.050	0.409	-0.080	0.000	-0.031	0.605
n = 500	-0.016	0.292	-0.009	0.730	-0.074	0.000	-0.010	0.687
n = 1000	-0.016	0.083	-0.003	0.909	-0.020	0.042	-0.009	0.611
n = 3000	-0.009	0.060	-0.001	0.935	-0.005	0.427	-0.005	0.554
Plug in RF			Debiased RF		Plug in CIF		Debiased CIF	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.026	0.280	-0.023	0.730	-0.070	0.000	-0.023	0.692
n = 500	-0.020	0.060	-0.005	0.865	-0.033	0.000	-0.004	0.901
n = 1000	-0.020	0.002	-0.003	0.897	-0.026	0.000	-0.002	0.925
n = 3000	-0.022	0.000	-0.001	0.913	-0.022	0.000	-0.001	0.923
Plug in XGBoost			Debiased XGBoost		Plug in Catboost		Debiased Catboost	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	0.030	0.427	-0.060	0.252	-0.014	0.520	-0.048	0.337
n = 500	0.015	0.292	-0.019	0.335	0.002	0.685	-0.007	0.794
n = 1000	0.009	0.361	-0.009	0.593	0.002	0.700	-0.003	0.897
n = 3000	0.005	0.367	-0.002	0.907	0.002	0.687	-0.001	0.946

Table 6: Simulation based on 500 Monte Carlo iterations, true value for the Gini of the fitted values is 0.18.

$\sigma = 0.3$			Gini of the FVs					
Plug in Lasso			Debiased Lasso		Plug in Ridge		Debiased Ridge	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.041	0.395	-0.085	0.264	-0.073	0.000	-0.044	0.621
n = 500	-0.028	0.115	-0.019	0.508	-0.066	0.000	-0.017	0.563
n = 1000	-0.027	0.022	-0.009	0.722	-0.035	0.002	-0.015	0.427
n = 3000	-0.017	0.006	-0.002	0.913	-0.025	0.000	-0.009	0.355
Plug in RF			Debiased RF		Plug in CIF		Debiased CIF	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	-0.017	0.552	-0.035	0.716	-0.067	0.002	-0.034	0.659
n = 500	-0.016	0.212	-0.011	0.788	-0.032	0.004	-0.008	0.839
n = 1000	-0.017	0.034	-0.007	0.806	-0.024	0.000	-0.004	0.879
n = 3000	-0.020	0.000	-0.002	0.889	-0.021	0.000	-0.002	0.917
Plug in XGBoost			Debiased XGBoost		Plug in Catboost		Debiased Catboost	
	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
n = 100	0.059	0.042	-0.082	0.298	-0.015	0.466	-0.068	0.317
n = 500	0.028	0.022	-0.036	0.095	0.002	0.563	-0.015	0.657
n = 1000	0.017	0.067	-0.020	0.206	0.002	0.560	-0.007	0.794
n = 3000	0.008	0.127	-0.006	0.673	0.002	0.565	-0.001	0.946

Table 7: Simulation based on 500 Monte Carlo iterations, true value for the Gini of the fitted values is 0.18.

K Cross-Fitting

We create a partition $\mathcal{C} = \{C_1, \dots, C_K\}$ of the set $\mathcal{N} = \{1, \dots, n\}$. Then we create partition $\mathcal{I} = \{I_1, \dots, I_L\}$ of the set $\{(i, j) \in \mathcal{N}^2 : i < j\}$ as depicted in Figure 7 for $n = 21$ and $K = 3$. This partition results in $L = K(K + 1)/2$ blocks. For example, in Figure 7 when pairs in I_1 are used to compute the sample orthogonal moment, the nuisance parameters are estimated with observations in $(C2, C3)$.

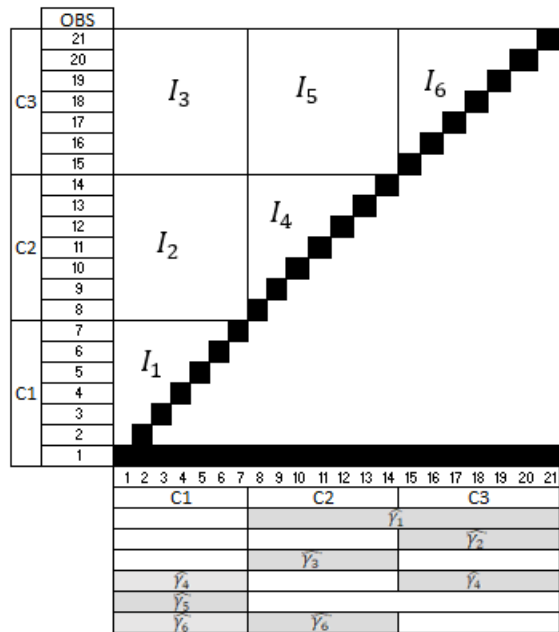


Figure 7: $n = 21$, $L = 3$

This is nothing else than a partition of $\mathcal{C}^2$ in squares intersected with the set $\{i < j : i, j \in \mathcal{N}\}$.

References

- AIYAR, S. AND C. EBEKE (2020): “Inequality of opportunity, inequality of income and economic growth,” *World Development*, 136, 105115.
- ALDOUS, D. J. (1981): “Representations for partially exchangeable arrays of random variables,” *Journal of Multivariate Analysis*, 11, 581–598.
- ALESINA, A. AND E. LA FERRARA (2005): “Preferences for redistribution in the land of opportunities,” *Journal of public Economics*, 89, 897–931.
- ARCONES, M. A. AND E. GINÉ (1993): “Limit theorems for U-processes,” *The Annals of Probability*, 1494–1542.
- ATHEY, S., J. TIBSHIRANI, AND S. WAGER (2019): “Generalized random forests,” *The Annals of Statistics*, 47, 1148–1178.
- ATKINSON, A. B. ET AL. (1970): “On the measurement of inequality,” *Journal of economic theory*, 2, 244–263.
- BELLONI, A. AND V. CHERNOZHUKOV (2011): “l1-penalized quantile regression in high-dimensional sparse models,” .
- (2013): “Least squares after model selection in high-dimensional sparse models,” .
- BICKEL, P. J. (1982): “On adaptive estimation,” *The Annals of Statistics*, 647–671.
- BLUNDELL, R. W. AND J. L. POWELL (2004): “Endogeneity in semiparametric binary response models,” *The Review of Economic Studies*, 71, 655–679.
- BRUNORI, P., P. HUFÉ, AND D. MAHLER (2021): “The roots of inequality: Estimating inequality of opportunity from regression trees and forests,” *IZA Discussion Paper No. 14689*.
- BRUNORI, P. AND G. NEIDHÖFER (2021): “The evolution of inequality of opportunity in Germany: A machine learning approach,” *Review of Income and Wealth*, 67, 900–927.
- BRUNORI, P., F. PALMISANO, AND V. PERAGINE (2019a): “Inequality of opportunity in sub-Saharan Africa,” *Applied Economics*, 51, 6428–6458.
- BRUNORI, P., V. PERAGINE, AND L. SERLENGA (2019b): “Upward and downward bias when measuring inequality of opportunity,” *Social Choice and Welfare*, 52, 635–661.

- CARRANZA, R. (2020): “Inequality of outcomes, inequality of opportunity, and economic growth,” Tech. rep., ECINEQ, Society for the Study of Economic Inequality.
- (2022): “Upper and Lower Bound Estimates of Inequality of Opportunity: A Cross-National Comparison for Europe,” *Review of Income and Wealth*.
- CHEN, X. (2007): “Large sample sieve estimation of semi-nonparametric models,” *Handbook of econometrics*, 6, 5549–5632.
- CHEN, X. AND H. WHITE (1999): “Improved rates and asymptotic normality for non-parametric neural network estimators,” *IEEE Transactions on Information Theory*, 45, 682–691.
- CHERNOZHUKOV, V., D. CHETVERIKOV, M. DEMIRER, E. DUFLO, C. HANSEN, W. NEWEY, AND J. ROBINS (2018): “Double/debiased machine learning for treatment and structural parameters,” *The Econometrics Journal*, 21, C1–C68.
- CHERNOZHUKOV, V., J. C. ESCANCIANO, H. ICHIMURA, W. K. NEWEY, AND J. M. ROBINS (2022): “Locally robust semiparametric estimation,” *Econometrica*, 90, 1501–1535.
- CHIANG, H. D., K. KATO, Y. MA, AND Y. SASAKI (2021a): “Multiway cluster robust double/debiased machine learning,” *Journal of Business & Economic Statistics*, 1–11.
- CHIANG, H. D., Y. MA, J. RODRIGUE, AND Y. SASAKI (2021b): “Dyadic Double/Debiased Machine Learning for Analyzing Determinants of Free Trade Agreements,” *arXiv preprint arXiv:2110.04365*.
- CLÉMENÇON, S., G. LUGOSI, AND N. VAYATIS (2008): “Ranking and empirical minimization of U-statistics,” *The Annals of Statistics*, 36, 844–874.
- CRANE, H. AND H. TOWNSNER (2018): “Relatively exchangeable structures,” *The Journal of Symbolic Logic*, 83, 416–442.
- DABBS, B., S. ADHIKARI, AND T. SWEET (2020): “Conditionally Independent Dyads (CID) network models: A latent variable approach to statistical social network analysis,” *Social Networks*, 63, 122–133.
- DAVEZIES, L., X. D’HAULTFŒUILLE, AND Y. GUYONVARCH (2021): “Empirical process results for exchangeable arrays,” *The Annals of Statistics*, 49, 845 – 862.
- DOMÍNGUEZ, M. A. AND I. N. LOBATO (2004): “Consistent estimation of models defined by conditional moment restrictions,” *Econometrica*, 72, 1601–1615.

- DUCLOS, J.-Y., J. ESTEBAN, AND D. RAY (2004): “Polarization: concepts, measurement, estimation,” *Econometrica*, 72, 1737–1772.
- FARRELL, M. H., T. LIANG, AND S. MISRA (2021): “Deep neural networks for estimation and inference,” *Econometrica*, 89, 181–213.
- FERREIRA, F. H., C. LAKNER, M. A. LUGO, AND B. ÖZLER (2018): “Inequality of opportunity and economic growth: how much can cross-country regressions really tell us?” *Review of Income and Wealth*, 64, 800–827.
- FERREIRA, F. H. AND V. PERAGINE (2016): “Individual responsibility and equality of opportunity,” in *The Oxford handbook of well-being and public policy*.
- FLEURBAEY, M. (1995): “Equal opportunity or equal social outcome?” *Economics & Philosophy*, 11, 25–55.
- GRAHAM, B. S. (2020a): “Dyadic regression,” *The Econometric Analysis of Network Data*, 23–40.
- (2020b): “Network data,” in *Handbook of econometrics*, Elsevier, vol. 7, 111–218.
- GRAHAM, B. S., F. NIU, AND J. L. POWELL (2021): “Minimax risk and uniform convergence rates for nonparametric dyadic regression,” Tech. rep., National Bureau of Economic Research.
- HEUCHENNE, C. AND A. JACQUEMAIN (2022): “Inference for monotone single-index conditional means: A Lorenz regression approach,” *Computational Statistics & Data Analysis*, 167, 107347.
- HOEFFDING, W. (1948): “A class of statistics with asymptotically normal distributions,” *Annals of Mathematical Statistics*, 19, 293–325.
- HONORÉ, B. AND J. POWELL (2005): “Pairwise difference estimators for nonlinear models,” in *Identification and Inference for Econometric Models W. Andrews and J. H. Stock*, Cambridge University Press, Cambridge.
- HOOVER, D. N. (1979): “Relations on Probability Spaces and Arrays of,” *t*, *Institute for Advanced Study*.
- HUFE, P., R. KANBUR, AND A. PEICHL (2022a): “Measuring Unfair Inequality: Reconciling Equality of Opportunity and Freedom from Poverty,” *The Review of Economic Studies*, 89, 3345–3380.

- HUFE, P., A. PEICHL, P. SCHÜLE, AND J. TODOROVIĆ (2022b): “Fairness in Europe: A Multidimensional Comparison,” in *CESifo Forum*, München: ifo Institut-Leibniz-Institut für Wirtschaftsforschung an der . . . , vol. 23, 45–51.
- HUFE, P., A. PEICHL, AND D. WEISHAAR (2022c): “Lower and upper bound estimates of inequality of opportunity for emerging economies,” *Social Choice and Welfare*, 58, 395–427.
- ICHIMURA, H. AND W. K. NEWEY (2022): “The influence function of semiparametric estimators,” *Quantitative Economics*, 13, 29–61.
- JOCHMANS, K. (2017): “Two-way models for gravity,” *Review of Economics and Statistics*, 99, 478–485.
- KALLENBERG, O. (1989): “On the representation theorem for exchangeable arrays,” *Journal of Multivariate Analysis*, 30, 137–154.
- KLAASSEN, C. A. (1987): “Consistent estimation of the influence function of locally asymptotically linear estimators,” *The Annals of Statistics*, 15, 1548–1562.
- KUECK, J., Y. LUO, M. SPINDLER, AND Z. WANG (2023): “Estimation and inference of treatment effects with L2-boosting in high-dimensional settings,” *Journal of Econometrics*, 234, 714–731.
- LEE, A. J. (2019): *U-statistics: Theory and Practice*, Routledge.
- LOU, Z., X. ZHANG, AND W. B. WU (2023): “High-dimensional analysis of variance in multivariate linear regression,” *Biometrika*, 110, 777–797.
- MAO, L. (2018): “On causal estimation using U-statistics,” *Biometrika*, 105, 215–220.
- MARRERO, G. A. AND J. G. RODRÍGUEZ (2012): “Inequality of opportunity in Europe,” *Review of Income and Wealth*, 58, 597–621.
- (2013): “Inequality of opportunity and growth,” *Journal of development Economics*, 104, 107–122.
- NEWHEY, W. K. (1991): “Uniform convergence in probability and stochastic equicontinuity,” *Econometrica: Journal of the Econometric Society*, 1161–1167.
- (1994): “The asymptotic variance of semiparametric estimators,” *Econometrica: Journal of the Econometric Society*, 1349–1382.

- NEWHEY, W. K. AND D. MCFADDEN (1994): “Large sample estimation and hypothesis testing,” *Handbook of econometrics*, 4, 2111–2245.
- NOLAN, D. AND D. POLLARD (1988): “Functional limit theorems for U -processes,” *The Annals of Probability*, 16, 1291–1298.
- POWELL, J. L. (1987): “Semiparametric estimation of bivariate latent variable models,” *SSRI Workshop Series*.
- RAMOS, X. AND D. VAN DE GAER (2016): “Approaches to inequality of opportunity: Principles, measures and evidence,” *Journal of Economic Surveys*, 30, 855–883.
- ROBINS, J. M., A. ROTNITZKY, AND L. P. ZHAO (1995): “Analysis of semiparametric regression models for repeated outcomes in the presence of missing data,” *Journal of the american statistical association*, 90, 106–121.
- ROBINSON, P. M. (1988): “Root-N-consistent semiparametric regression,” *Econometrica: Journal of the Econometric Society*, 931–954.
- ROEMER, J. E. (1998): *Equality of opportunity*, Harvard University Press.
- ROEMER, J. E. AND A. TRANNOY (2016): “Equality of opportunity: Theory and measurement,” *Journal of Economic Literature*, 54, 1288–1332.
- RYCHENER, Y., B. TASKESEN, AND D. KUHN (2022): “Metrizing Fairness,” *arXiv preprint arXiv:2205.15049*.
- SAKAMOTO, K. (2024): “Dyadic Regression with Sample Selection,” *arXiv preprint arXiv:2405.17787*.
- SALAS-ROJO, P. AND J. G. RODRÍGUEZ (2022): “Inheritances and wealth inequality: a machine learning approach,” *The Journal of Economic Inequality*, 1–25.
- SASAKI, Y. AND T. URA (2021): “Estimation and inference for policy relevant treatment effects,” *Journal of Econometrics*.
- SCHICK, A. (1986): “On asymptotically efficient estimation in semiparametric models,” *The Annals of Statistics*, 1139–1151.
- SHERMAN, R. P. (1994): “U-processes in the analysis of a generalized semiparametric regression estimator,” *Econometric theory*, 10, 372–395.
- SILVA, J. S. AND S. TENREYRO (2006): “The log of gravity,” *The Review of Economics and statistics*, 88, 641–658.

- STUTE, W. (1991): “Conditional U-statistics,” *The Annals of Probability*, 812–825.
- TERSCHUUR, J. (2022): “Debiased Machine Learning Inequality in Europe,” *arXiv preprint arXiv:2212.02407*.
- (2023): “Locally Robust Policy Learning: Inequality, Inequality of Opportunity and Intergenerational mobility,” *Working Paper*.
- TINBERGEN, J. (1962): “Shaping the world economy; suggestions for an international economic policy,” (*New York: The Twentieth Century Fund*).
- VAN DE GAER (1993): “Equality of opportunity and investment in human capital,” Phd thesis, Katholieke Universiteit Leuven.
- VAN DER VAART, A. W. (2000): *Asymptotic statistics*, vol. 3, Cambridge university press.
- WAGER, S. AND G. WALTHER (2015): “Adaptive concentration of regression trees, with application to random forests,” *arXiv preprint arXiv:1503.06388*.
- WERNER, T. (2021): “A review on instance ranking problems in statistical learning,” *Machine Learning*, 1–49.
- YAMAMURO, S. (1974): *Differential calculus in topological linear spaces*, vol. 374, Springer.
- YIN, A., A. YUAN, AND M. T. TAN (2024): “Highly robust causal semiparametric U-statistic with applications in biomedical studies,” *The international journal of biostatistics*, 20, 69–91.
- ZHANG, T. AND B. YU (2005): “Boosting with early stopping: Convergence and consistency,” .